



Equipment

- 0102H [GCs+HWJ*]
- 0106D [Box Key Reset holder]
- 0108A [Add+USB]
- 0109A [Network customer RJ45]
- 0202H [AX5 X/Xa Safety OCT 2x6A]
- 0302A [Exhausting, Remote control]
- 0303A [underpressure feedback]
- 0407E [2xCabinet W600H800D400+MP]
- 0410A [ECO mounting bundle kit]
- 0411B [Basic Parts TSc CE]
- 0605M [LightBarrier SICK*]
- 1102B [Main Cabi supply S+N switched]
- 1107A [Transform 36V4x100VA]
- 1110A [EP PowerBox 4x4A]
- 1204A [Foot switch, endtruck, 2x2 FrontRear]
- 1501A [Cabinet cooling, Door, GCs, 75W]
- 1504A [GCs fan cooling box]
- 0202G [L1-AX5 Y OCT Safety 3A]
- 0406A [L1-Plasma Interface Cabinet]
- 0507B [L1-SL 200*]
- 0909B [L1-KJB Q1500]
- 0910J [L1-No CutOffBox*]
- 1001A [L1-Laser Pointer]
- 1301A [L1-Havarie, Torchholder, Magnetic]
- 1301B [L1-Havarie tactil lift up]

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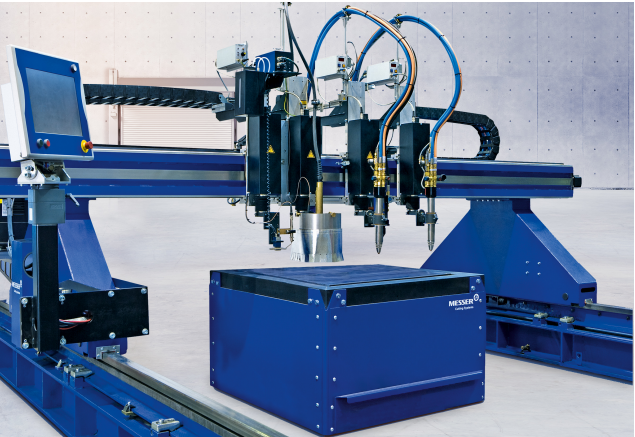
Customer:
SEFAMAR

project 289001295
machine MultiTherm Eco
length / width 10m / 4m
power 9 kVA
Toolbox Toolbox_105
drawing No. 289001295_SEFAMAR

editor Johann.Rais
building year 2024
regulation EN60204-1

Number of pages 185
Eplan-version 2.9.4
15117

machines number: 289001295



shop floor: EU

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Doku	09.01.2025	Johann.Rais					customer
Index Revision	Date	Name	Date	Name	Date	name	Remarks
	Created on		reviewed		Approved		

directory to Modul Overview location aware

module		description		group box	page
[=]	1B	table of contents		Module Overview	2...4
	4A	topology EtherCAT		EtherCAT cable;SPS-Kartenübersicht	30...39
	Z6A			cable overview	144...151
	Z6B			cable connection plan	152...185
[+]	B (gantry)				
[=]	3C	grounding bar	grounding bar	connection overview;connection points	23;24
[=]	BI	DC power supply interface	EP power - distributor	24VDC, EtherCat;rack overview	72;73
[=]	IB	interface box	Fieldbus module	power supply, EtherCat;rack overview	98;99
[=]	IBB	interface box	Fieldbus module	power supply, EtherCat;rack overview	100;101
[+]	E (end truck left)				
[=]	EA	emergency stop additions	emergency stop	connection	88
[=]	EF	emergency stop foot protection		sensor, safety EP box	90
[=]	EL	emergency stop light barrier	receiver front	Safety circuit	94
[=]	ELB	emergency stop light barrier	receiver rear	Safety circuit	95
[=]	MX	motor X	AM8531 OCT X	connection motor encoder	115
[+]	G (end truck right)				
[=]	EAB	emergency stop additions	emergency stop	connection	89
[=]	EFB	emergency stop foot protection		sensor, safety EP box	91
[=]	MXa	motor Xa	AM8531 OCT Xa	connection motor encoder	116
[=]	OB	operation panel Keyboard		EtherCAT;rack overview	119;120
[=]	OP	operation panel CP7202	Operation interface	EtherCAT;emergency stop, operating elements;operating elements;rack overview;supply, Akku	122...126
[=]	PG1	plasma gas console	type Q-gas 4500	power supply gas;rack overview	130...132
[=]	UT	unit temperature	climatisation	backplane operation panel GCs	142
[+]	HB (cabinet main)				
[=]	1A	cover sheet	cover sheet	Messer	1

directory to Modul Overview location aware

module	description		group box	page
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+ HB (cabinet main)

=	1C	diagram conventions	Cable Symbols;Cover page;General Description;Local Identifiers;Overview Device;Overview Terminal strip;Structure and Overview Locations	5...11
=	3B	grounding cabinet	cabinet;mounting plate +HB drive controller	grounding points in cabinet;structure cabinet
=	3BB	grounding cabinet	Cabinet;cabinet	grounding points in cabinet;structure cabinet
=	3D	grounding machine	grounding machine	overview
=	AA	AC power supply cabinet	AC operating voltage	main switch, PE/N;overview terminal strip
=	AB	AC distribution	AC operating voltage	operating voltage;overview terminal strip
=	AL	AC levels	transformer	rack overview;supply
=	AS	AC service socket	service socket	feed-in
=	BA	DC power supply	operating voltage	connection overview;electronical fuse channel 1 / 2;electronical fuse channel 3 / 4;power supply;rack overview
=	BU	DC USV	control voltage buffered	connection overview;feed-in;rack overview
=	CXXa	controller X,Xa	AX5206 X,Xa	power supply 400 VAC;rack overview
=	CY1	controller Y1	AX5103 Y1	entrances;power supply 400 VAC;rack overview
=	EP	emergency stop process	emergency stop process	Safety circuit interface;rack overview
=	IC	interface control	star hub	power supply, EtherCat;rack overview
=	IT	interface terminal	bus coupler;coupler, 8 x DI, 8 x DO;power supply terminal	EtherCAT, power supply;EtherCAT, refreshment
=	UTB	unit temperature	cooling cabinet	supply

+ HOK (operation panel key switch)

=	OK	operation panel key switch	key switch box	elements	121
---	----	----------------------------	----------------	----------	-----

+ HP1 (interface cabinet plasma)

=	3B1	grounding cabinet	cabinet;mounting plate plasma Interlock	grounding points in cabinet;structure cabinet	18...20
=	AA1	AC power supply cabinet	feed-in	overview terminal strip;X0 , PE/N/1~	46;47
=	AB1	AC distribution	AC operating voltage	operating voltage;overview terminal strip	53...55
=	BA1	DC power supply	operating voltage	connection overview;electronical fuse channel 1 / 2;electronical fuse channel 3 / 4;power supply	66...71
=	EI1	emergency stop interface	cabinet external	control voltage, safety circuit;rack overview	92;93
=	IT1	interface terminal	bus coupler;coupler, 8 x DI, 8 x DO	EtherCAT, power supply	111;112
=	PS1	plasma source		rack overview	133

directory to Modul Overview location aware

module	description		group box	page
--------	-------------	--	-----------	------

+

HP1 (interface cabinet plasma)

=

UE

unit exhausti

rack overview

139

+

L1 (Lifter)

=

3E1

grounding elements

grounding Lifter

connection point

29

=

DC1

detection collision

torch magnetic holder

sensor, input

82

=

DC1B

detection collision

torch dig switch tactile

sensor, input

83

=

DL1

detection limiting

limit Y+;limiting Y+

cabinet wiring;connection overview

84;85

=

DL1B

detection limiting

limit Y-

cabinet wiring;connection overview

86;87

=

LI1

lifter

SL 200

AC/DC power supply;rack overview

113;114

=

MY1

motor Y1

AM8531 OCT Y1

connection motor encoder

117

=

PB1

plasma torch connection unit

type Q-Port 4500

power/gas supply;rack overview

128;129

=

PT1

plasma torch

supply torch

137

=

UP1

unit pointer

laser diode

connection

141

+

P (periphery)

=

PA1

plasma power switch box

Q1500

power supply

127

=

PS1

plasma source

Q1500

PLC interface, EtherCAT;grounding;rack overview

134...136

=

SP

sensor pressure

differential pressure monitor

exhaustion

138

=

UE

unit exhausti

external control

filter fan switch on

140

+

PZ (customer)

=

1S

supply connection plan

electrical connection plan

connection plan;connection plan exhaust Donaldson;connection plan machine

12...14

=

NC

network connectivity

network

customer

118

+

T (table)

=

3CB

grounding bar

grounding bar

connection overview;connection points

25;26

=

3DB

grounding machine

grounding machine

excerpt machine, track

28

General explanations on Electrical Documentation

This electrical documentation contains the necessary technical information required to understand the basic idea of the global electrical schematics with the New Electrical Modular Concept.

This includes: **Cover page**

General Description

Structure and Overview locations

Local identifiers

Overview Terminal strip

Overview Device

General Description

Module : An electrical design constructed with a group of standardized parts for a specific machine functionality in an electrical schematic that can be used on many different machine lines.

- + Symbol : The plus symbol is used as a prefix to designate a location
- = Symbol : The equal symbol is used as a prefix to designate a module
- Symbol : The minus symbol is used as a prefix to designate a device tag

Potential distribution terminal : A group of wire terminals that distribute power or a signal from 1 source to multiple locations

Feed-through terminals : Wire terminals that are used as a connection point for cables

Device overview : An overview page at the begining of a module that shows all of the connections to a device

Connection overview : A page that shows all of the connections to a potential distribution terminal

Revision Number

Option Name

Customer Name

Commission Number

Function Name DE/EN

Module Description in Detail

Sector

Location

Module Description

Module Name

Page Number

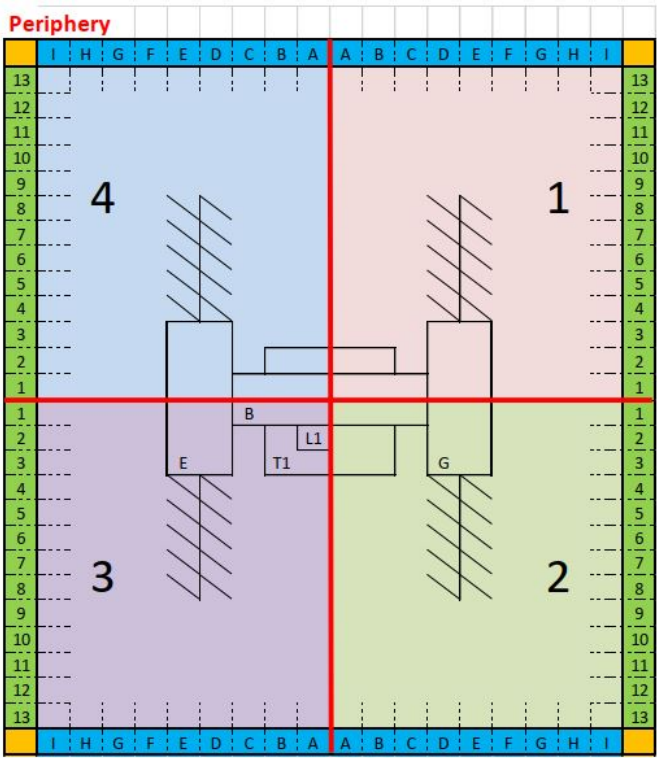
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			General Description	module	diagram conventions	Page	
			Toolbox version	customer:	SEFAMAR				sector	4B3E location	+HB	=1C
			1C1 - ABI - 33.61.00 - CC: EU	commission:	289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. by:	marcelo.azevedo 26.09.2024						6	

Structure and Overview Locations

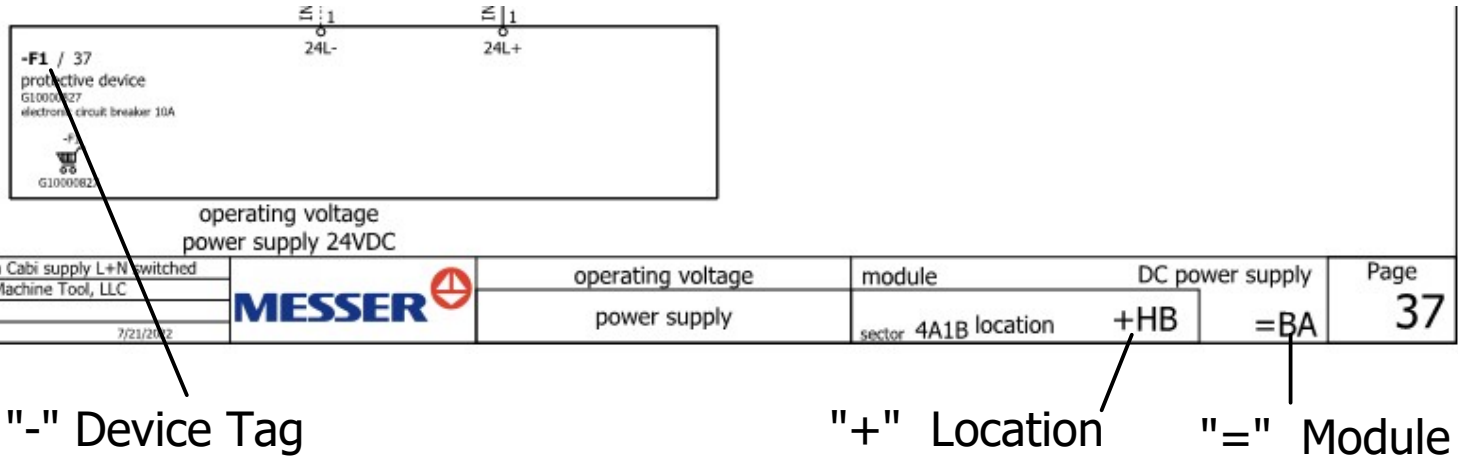
Structure consists of total 3 levels.

"+" Location "=" Module "-" Device Tag

Overview grid with X and Y coordinates:



Eg: 1D3G
1 Sector
D X-Coordinate
3 Y-Coordinate
G Endtruck Right



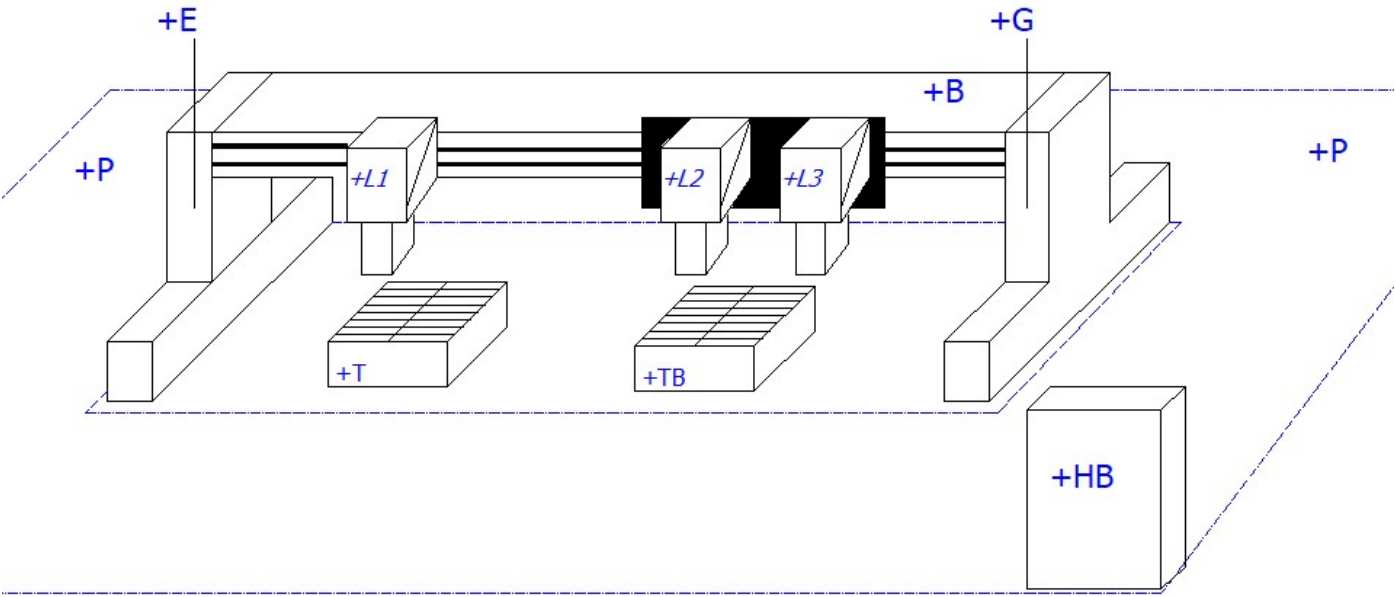
Overview Locations:

"+" - Location

- B Beam
- L Lifter
- E Endtruck Left
- G Endtruck Right
- P Periphery
- T Table
- C Cover / Housing

"+" - Cabinet

- HB Main Cabinet
- HP Cabinet Plasma Interface
- HL Cabinet Laser Interface
- HD Cabinet Drilling
- HC Cabinet Controller
- HO Cabinet Global Pedestal
- HQ Cabinet Laser Head
- HT Cutting Table Interface



Local Identifiers

Location box

Terminal

Connection definition

Cable length calculated
by ABIGAIL (Invisible)

Possible cables found
by ABIGAIL (Invisible)

Cable

Cable Shield

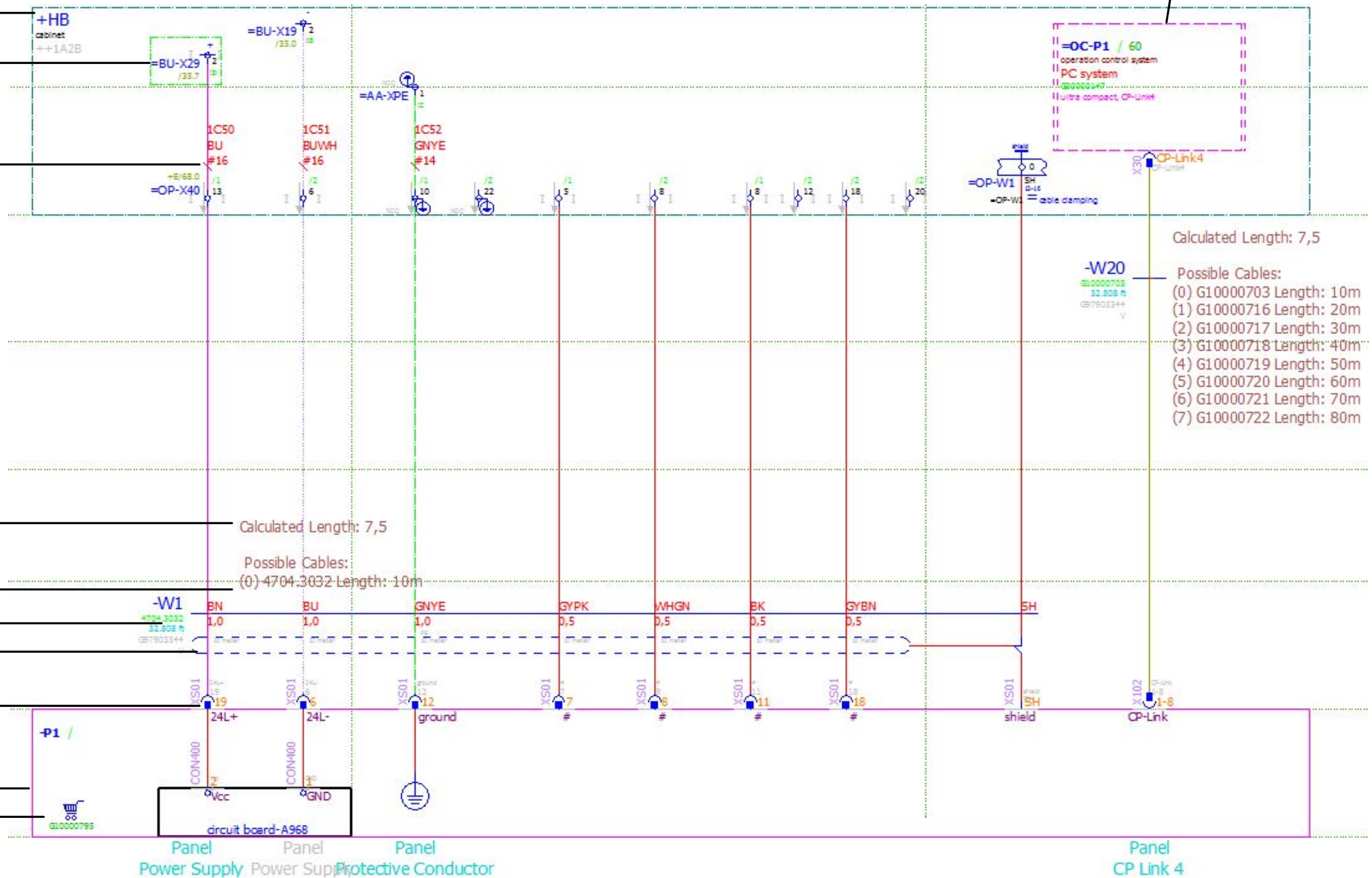
PLC Connection Point

PLC Device Box (Internal)

Part Definition Cart

Function Path text

PLC Device Box (External)



Overview Terminal strip

-X3 = L3 Anschluss

X3

G10000470

Klemme

Brücke

Seite

1	a	Einspeisung Phase	/19.7
	b	Höhenverstellung Trafoschutz	=AL/20.1
	c	Regler Versorgung AC	=CXB+G/34.1
	d		
2	a		
	b		
	c		
	d		

How to populate potential distribution terminals

Terminals are numbered ascending from left to right
Connection points from top to down with a,b,c,d...

G10000485

G10000431

Klemme

Brücke

Etage

Seite

1	,	1	-Absaugung aktiv	/129.8
2	,	2	-Absaugung aktiv	/129.9
3	,	1	-Absaugung aktiv	/129.9
4	,	2	-Absaugung aktiv	/129.9

How to populate feed-through terminals

Represent One to One connection of a Signal
Terminal has a number from left to right per level

Rack Overview Device



		G10001010	
		controller	
address:	-K1	AX5206, 2 channel, AC 2x 6A, FW >=2.10	
		SteckerBMK	
GND			
kanal	grounding		
M6	ground	controller grounding	/28.9
		SteckerBMK	
Safety			
kanal	PCI		
PCI	Bus	controller Safety Card	/28.5
		SteckerBMK	
X01			
kanal	supply grid		
1	L1	controller power supply AC	/28.1
2	L2	=	/28.2
3	L3	=	/28.2
PE	GND	controller ground	/28.2
		SteckerBMK	
X02			
kanal	DC Link		
1	DC+		
2	DC-		
		SteckerBMK	
X03			
kanal	supply DC		
1	Up+	controller power supply DC	/28.3
2	Us+	controller supply DC	/28.4
3	GND	controller power supply DC	/28.4
		SteckerBMK	
X04			
kanal	Bus		
1-4	Bus-IN	controller EtherCAT	/28.8
		SteckerBMK	
X05			
1-4	Bus-OUT	controller EtherCAT	/28.8

All connections going to a Device will be shown in the Rack Overview page in the corresponding module.

The Rack Overview page is only used when there are connections on multiple pages

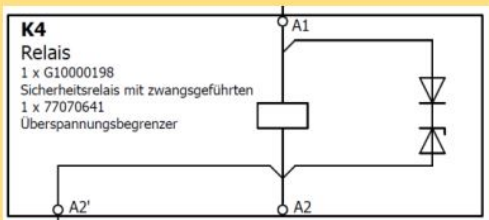
A box drawn with solid lines indicates that device is included in the module on that page.

A box drawn with dashed lines indicates that device is not included in that module.

A dashed box does not indicate a device location. You can have multiple module devices in one location.

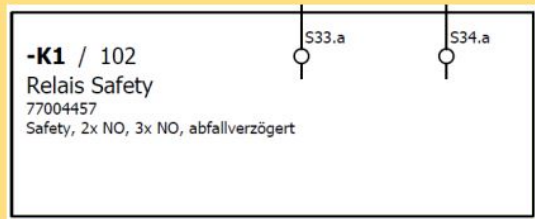
Without Overview
Devices

Inside a module

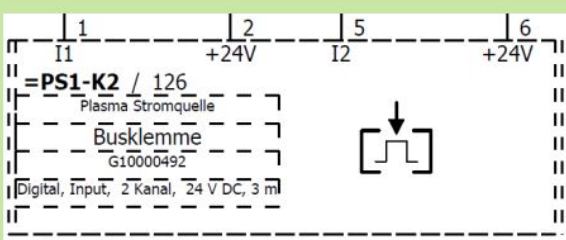
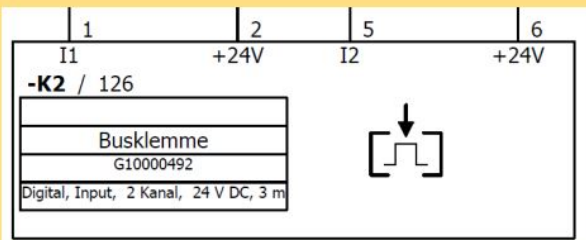


Outside a module

With Overview
Devices

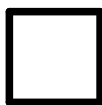
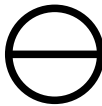
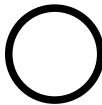


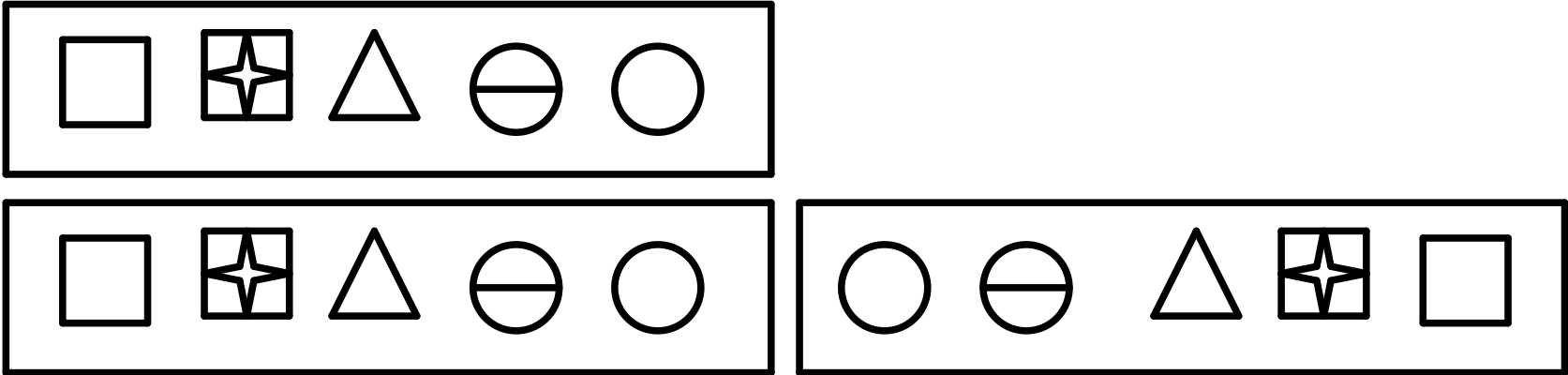
PLC-Boxes



Cable Symbols

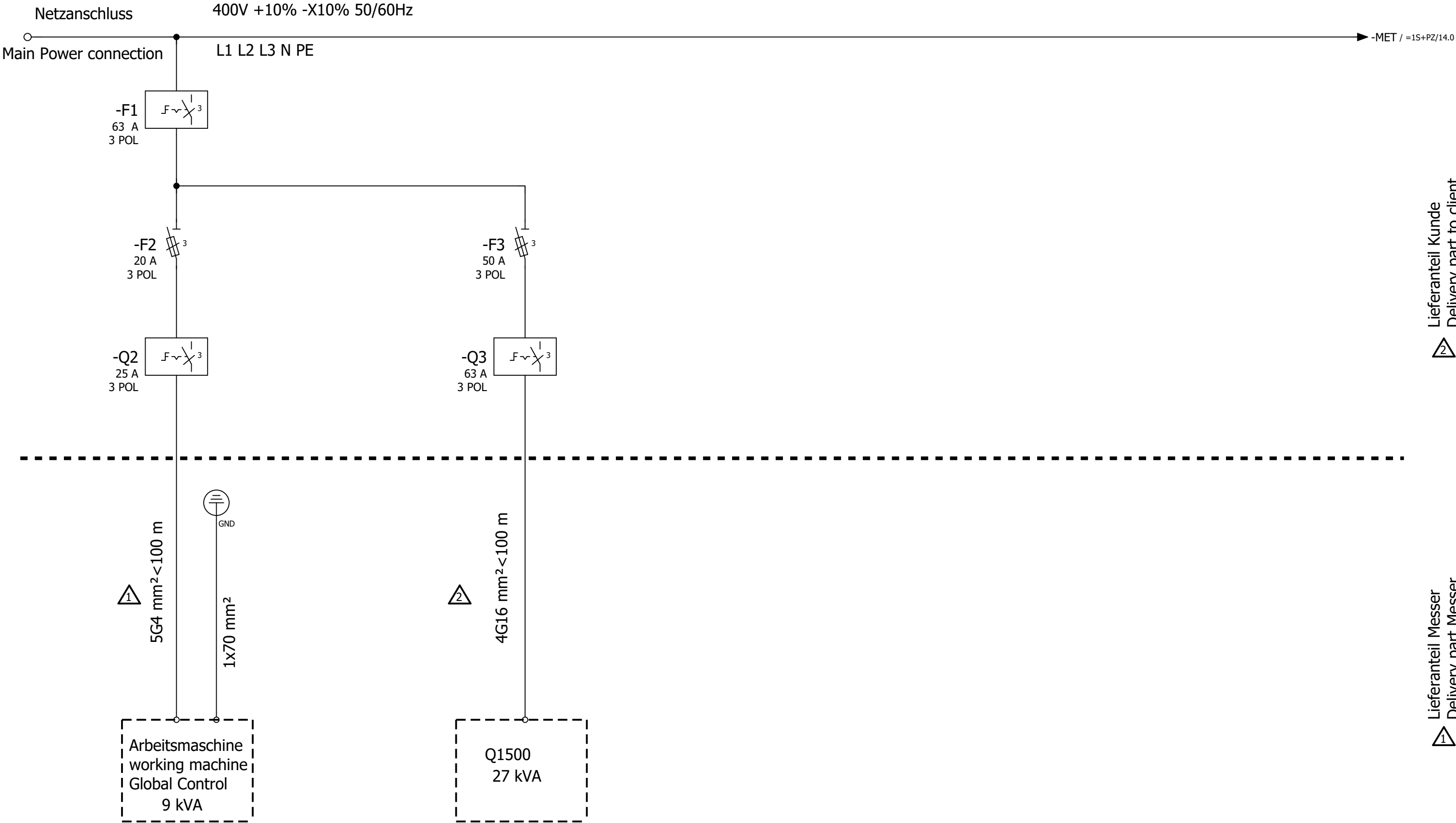
See the Z6A section of the electrical schematics to find the cable symbols associated with each cable.

-  = Noisy
-  = Medium Impact
-  = Low Impact
-  = Sensitive Signal
-  = Isolating

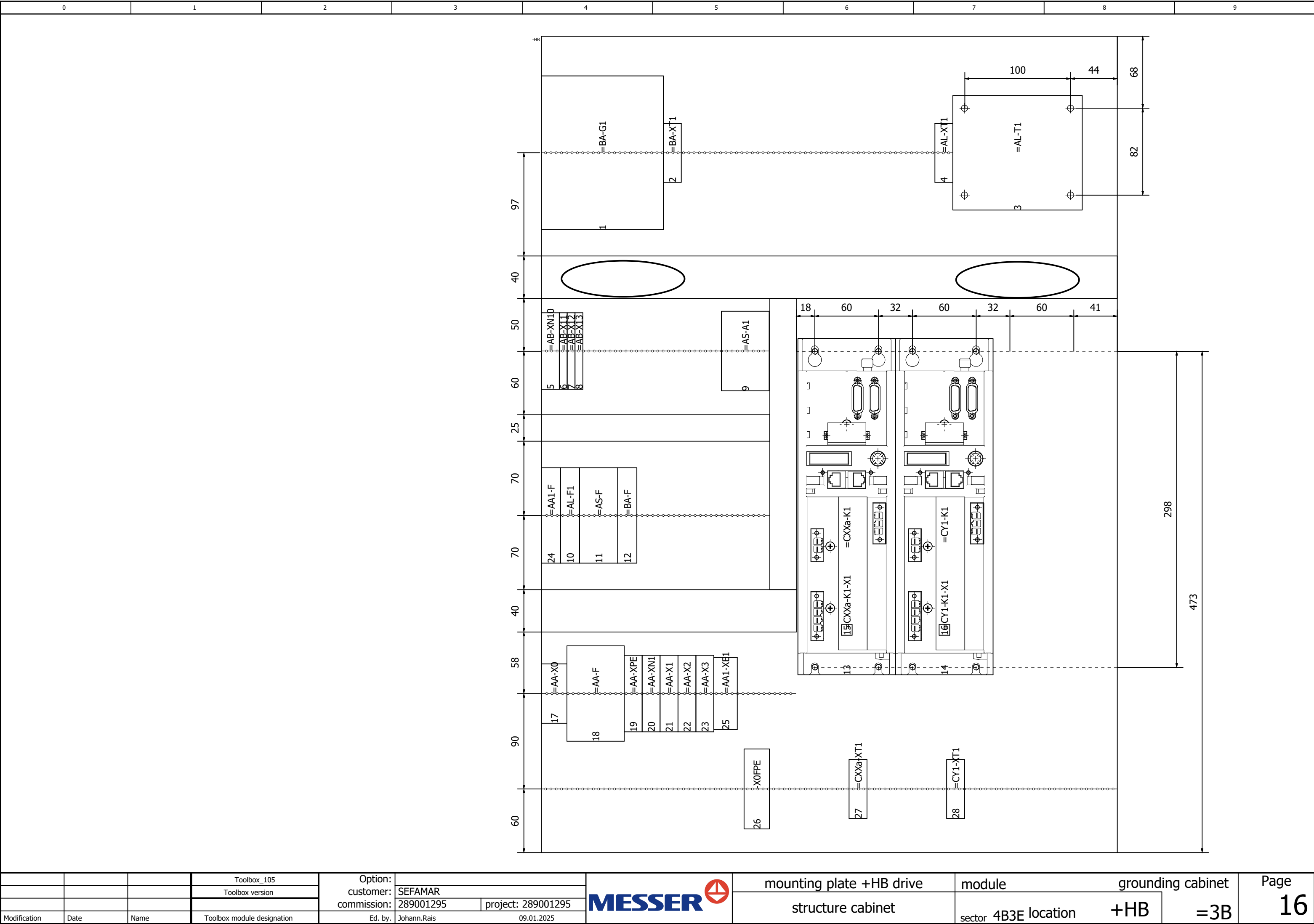


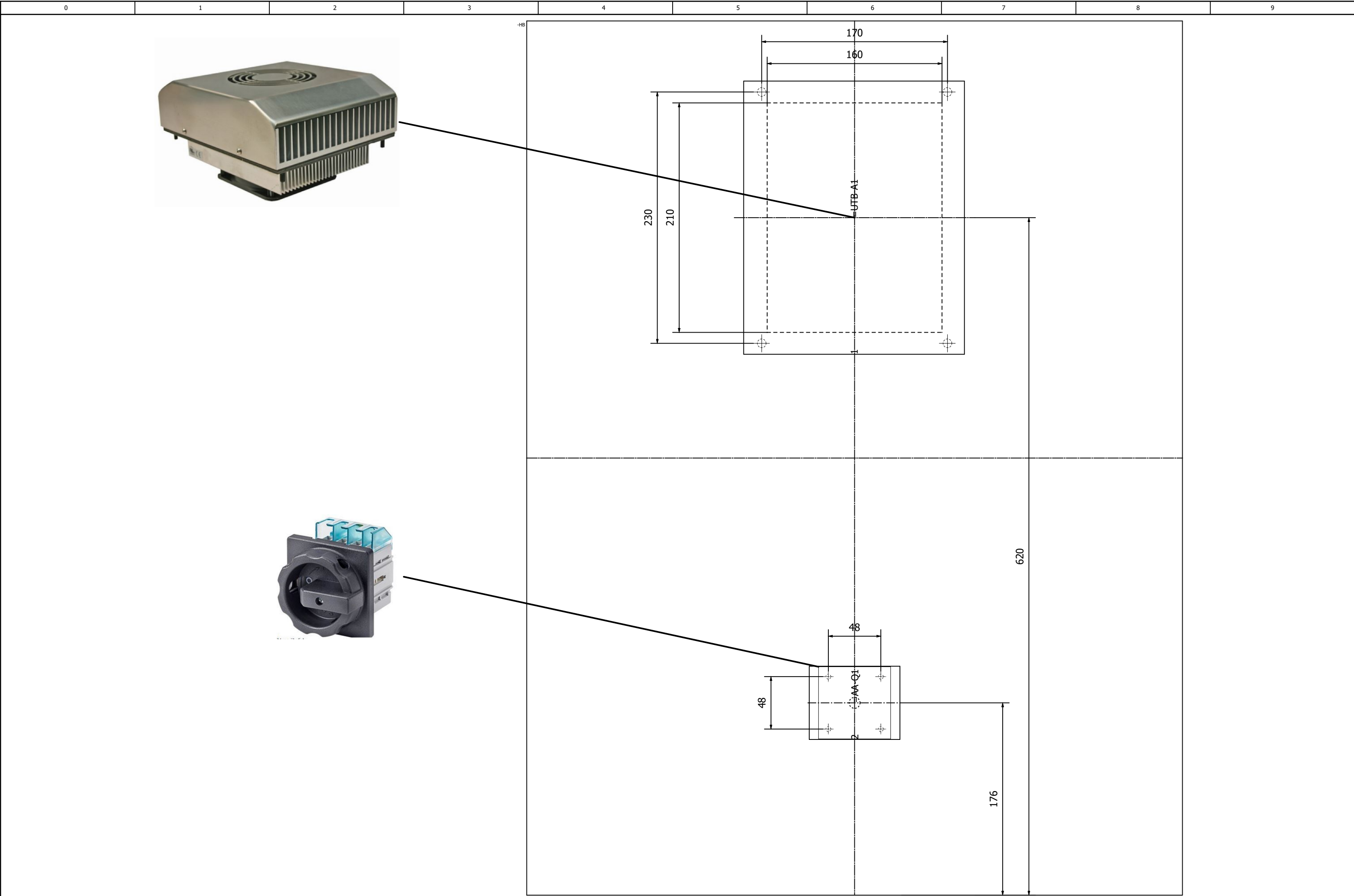
0	1	2	3	4	5	6	7	8	9	
Die Absicherungs- und Schalterbaugruppen sind entsprechend den örtlichen Bestimmungen der Energieversorgungsunternehmen auszulegen und stellen im Anschlussplan eine Empfehlung dar.					The main switchgear and circuit protection have to be arranged according to the local power supply authority regulations. The connection diagram represents a recommendation					
Spannung: Maximale zulässige Abweichung +10-10% von der Sollspannung und Spannungsspitzen/Einbrüche von max. 5 ms Dauer und +/- 30V.					Line voltage: max. admissible fluctuations +6 -10% and spikes/cuts of 5 ms duration and 30V + or -					
NC Arbeitstemperaturbereich + 5 bis +45 Grad C Lagerungstemperaturbereich -10 bis +50 Grad C Rel. Luftfeuchtigkeit bis 90% (nichtkondensierend)					NC Working temperature range + 5 to +45 degree centigrade Storage temperature range -10 to +50 degree centigrade Relative humidity max. 90% (condensation-free)					
72387975										
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			electrical connection plan	module supply connection plan	Page
			Toolbox version	customer:	SEFAMAR			connection plan	sector 1E6P location +PZ =1S	12
			1S - ABI - 34.102.00 - CC: EU	commission:	289001295	project: 289001295				
Modification	Date	Name	Toolbox module designation	Ed. by.	marcelo.azevedo 01.10.2024					

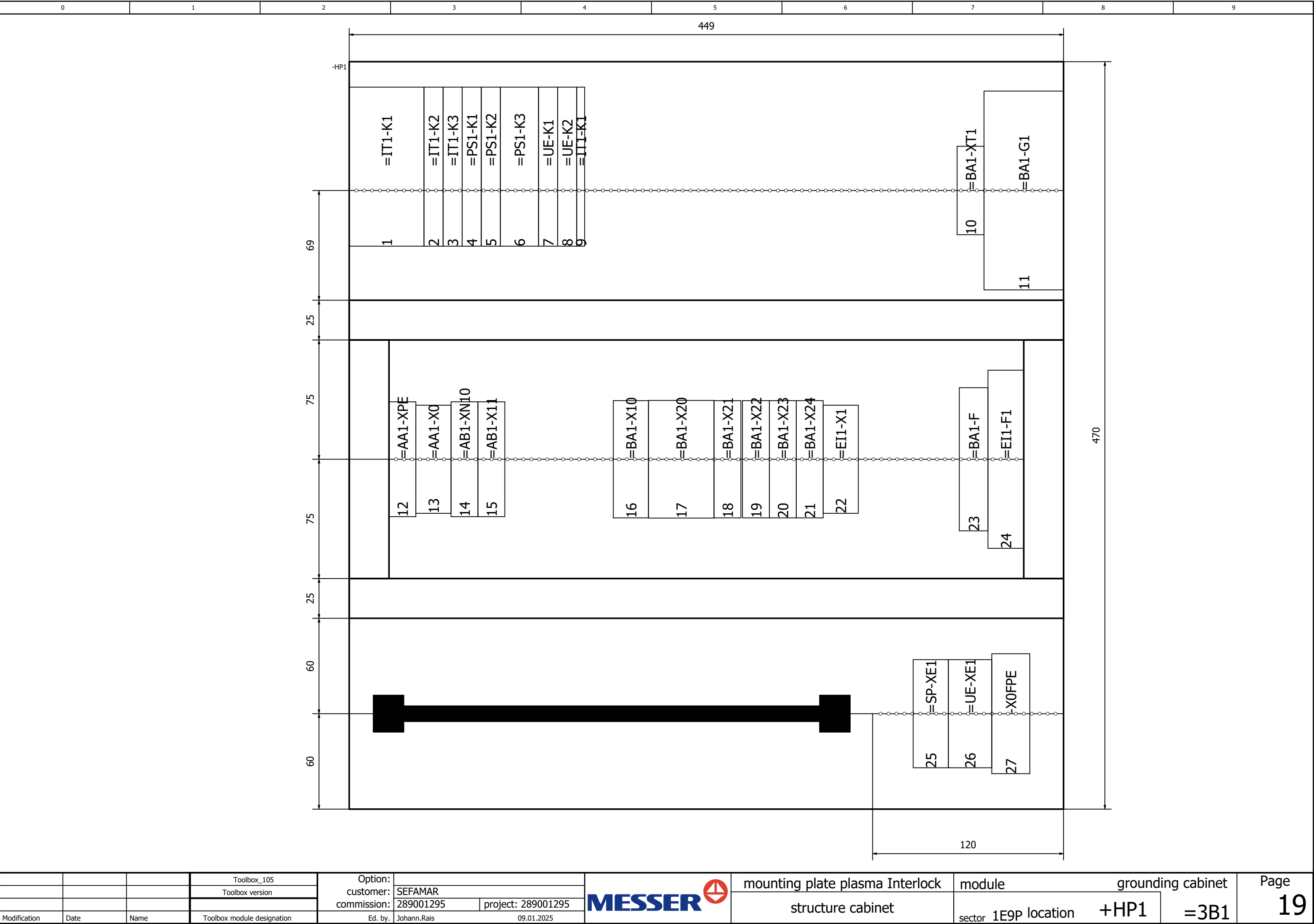
Bitte beachten: Rechtsdrehfeld
Please note: clockwise rotating field

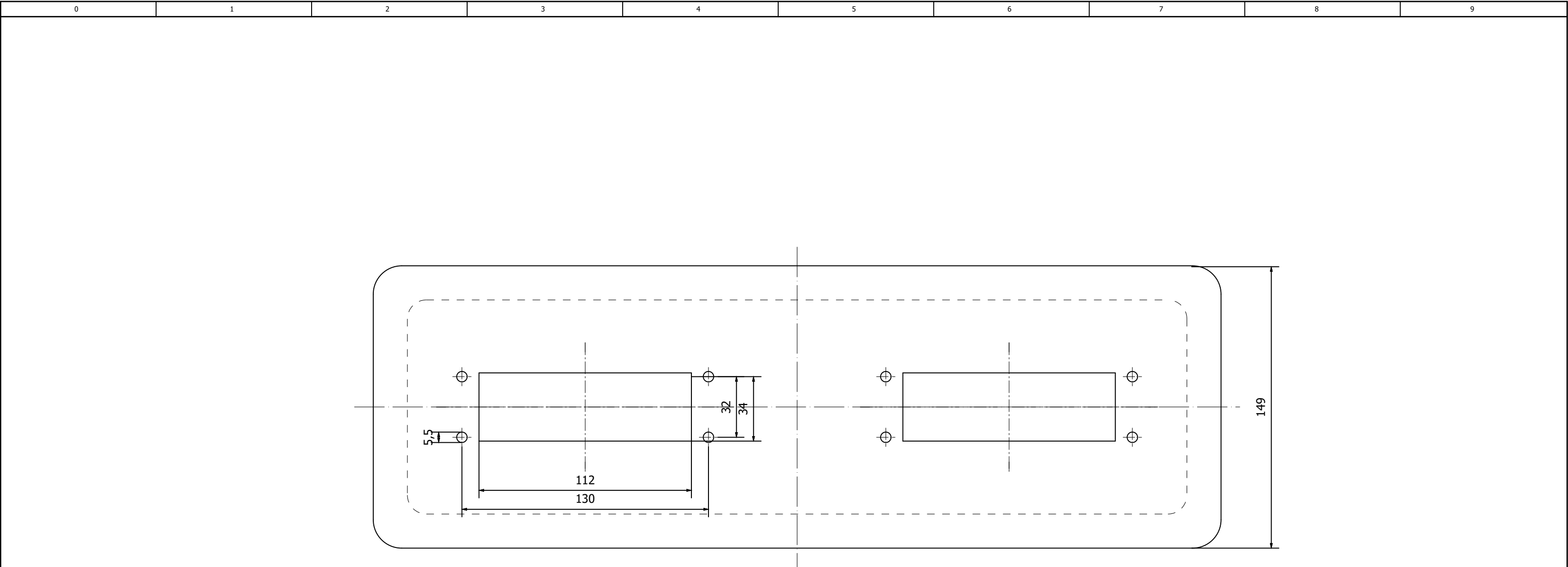


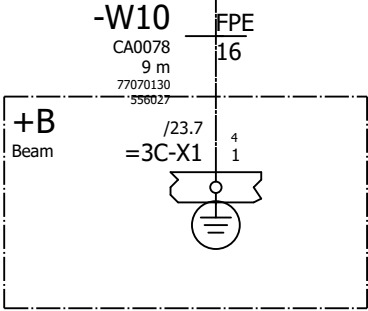
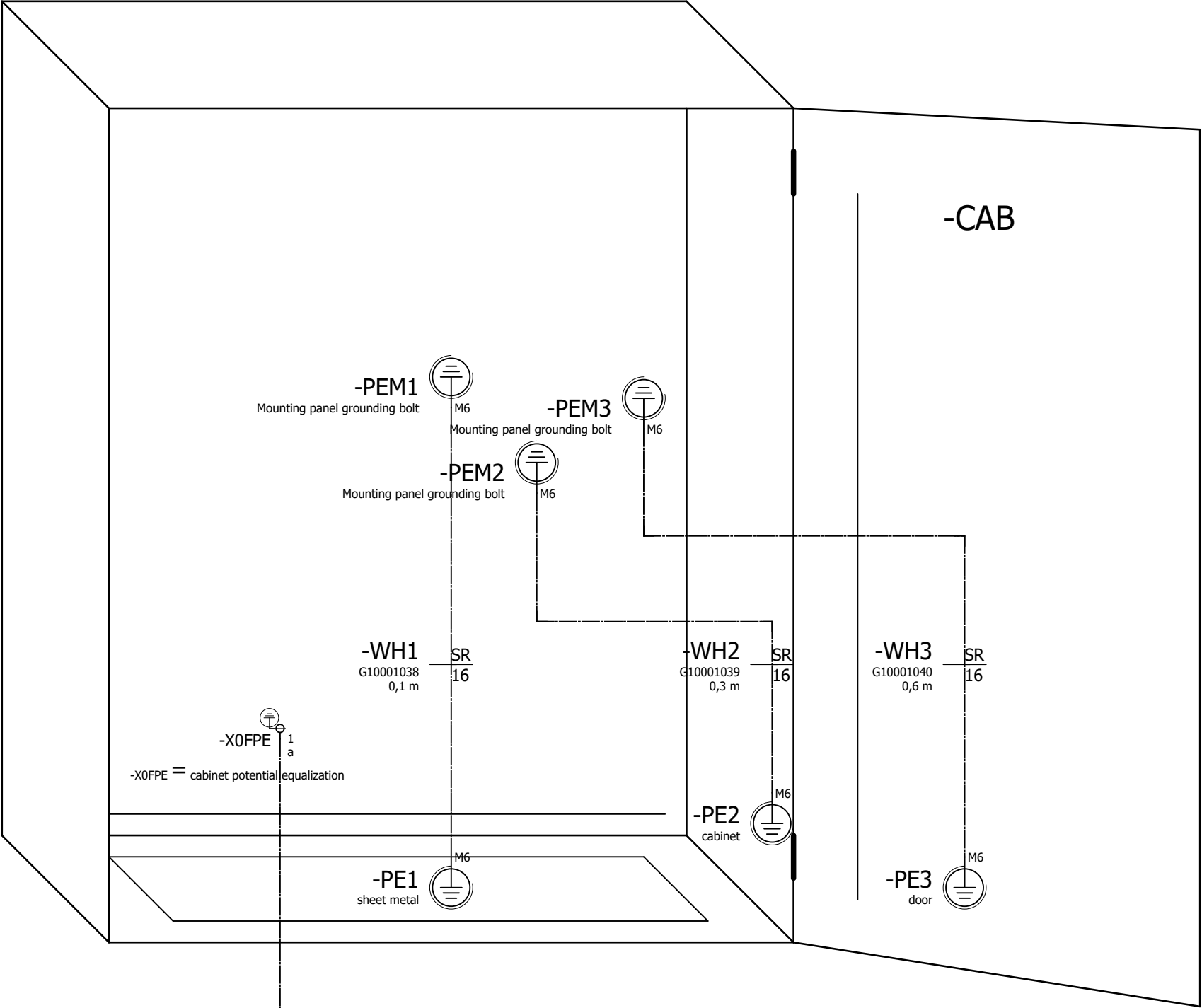
72387975



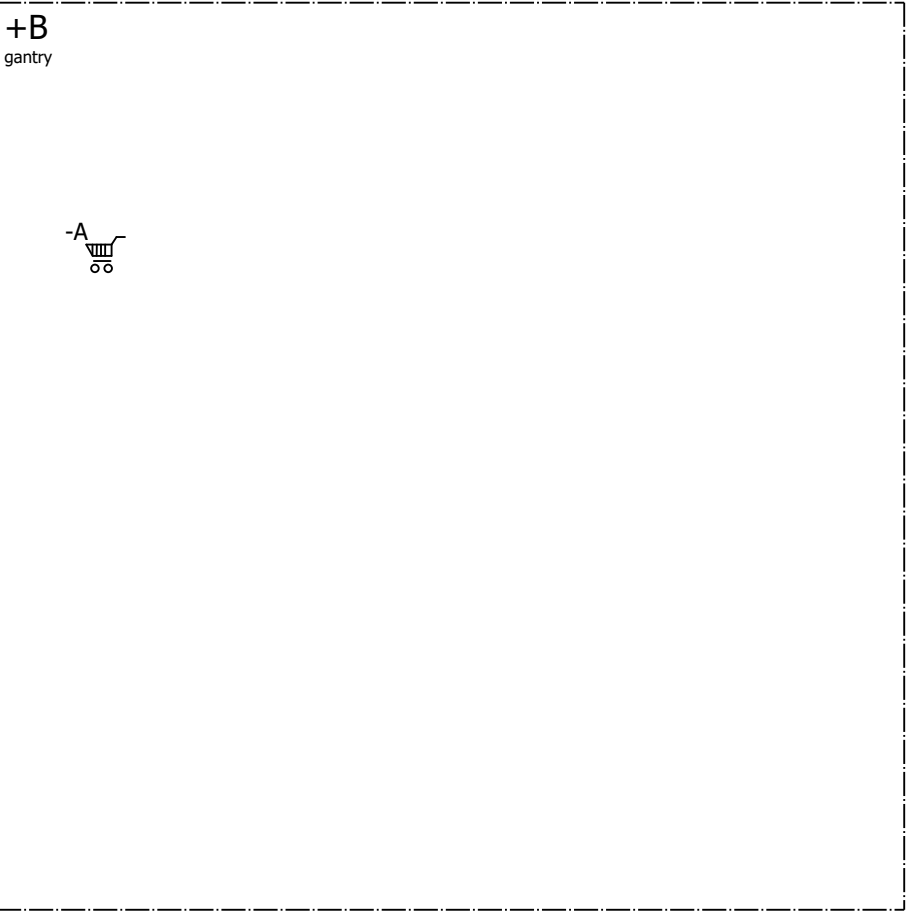








-CAB			
parts list // LN folder 72881342			
piece	article	ID	
1	77071588	V	corpus, single door, 600 x 800 x 400, mounting plate



cabinet

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

X1

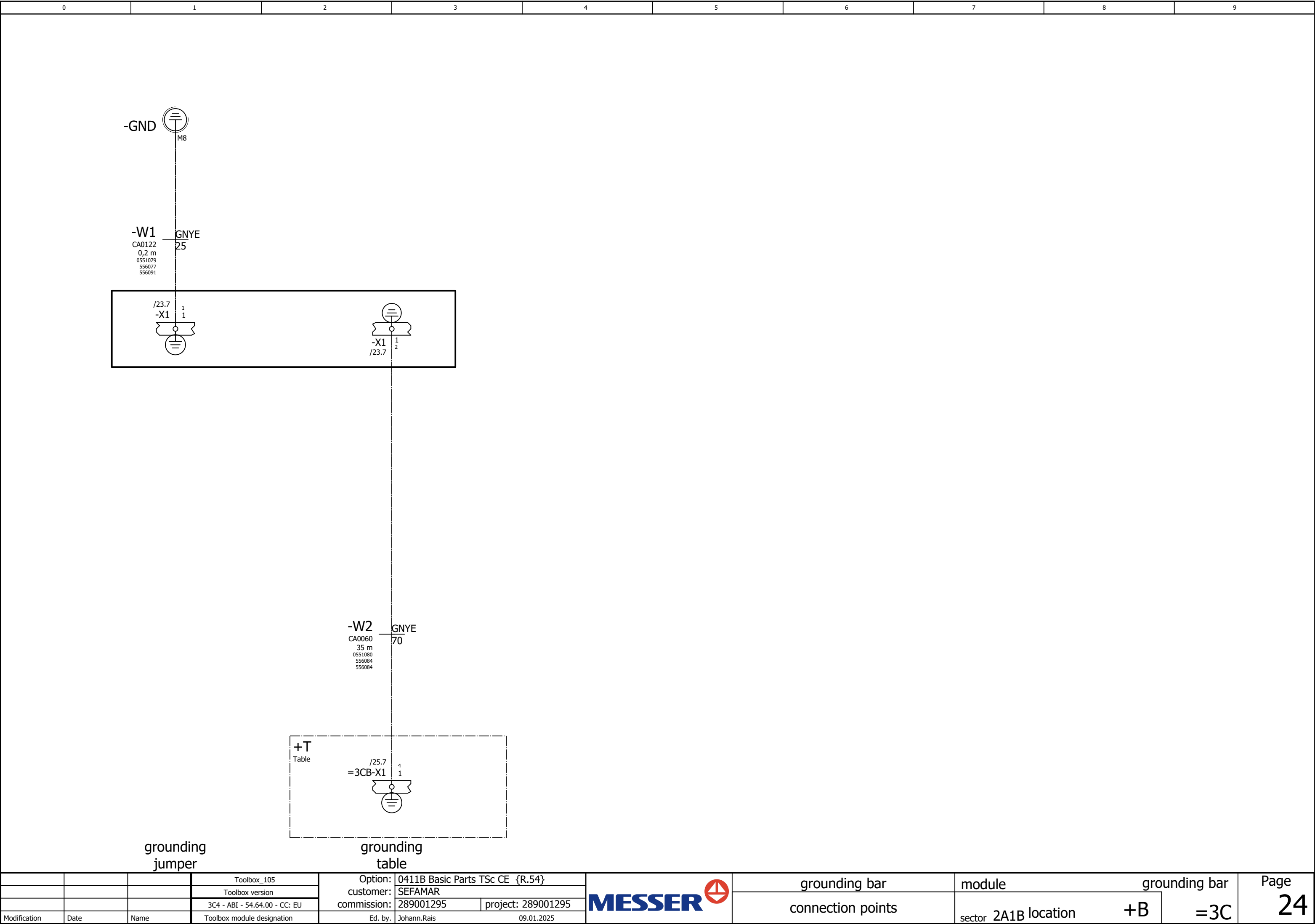
-X1 = potential equalization machine



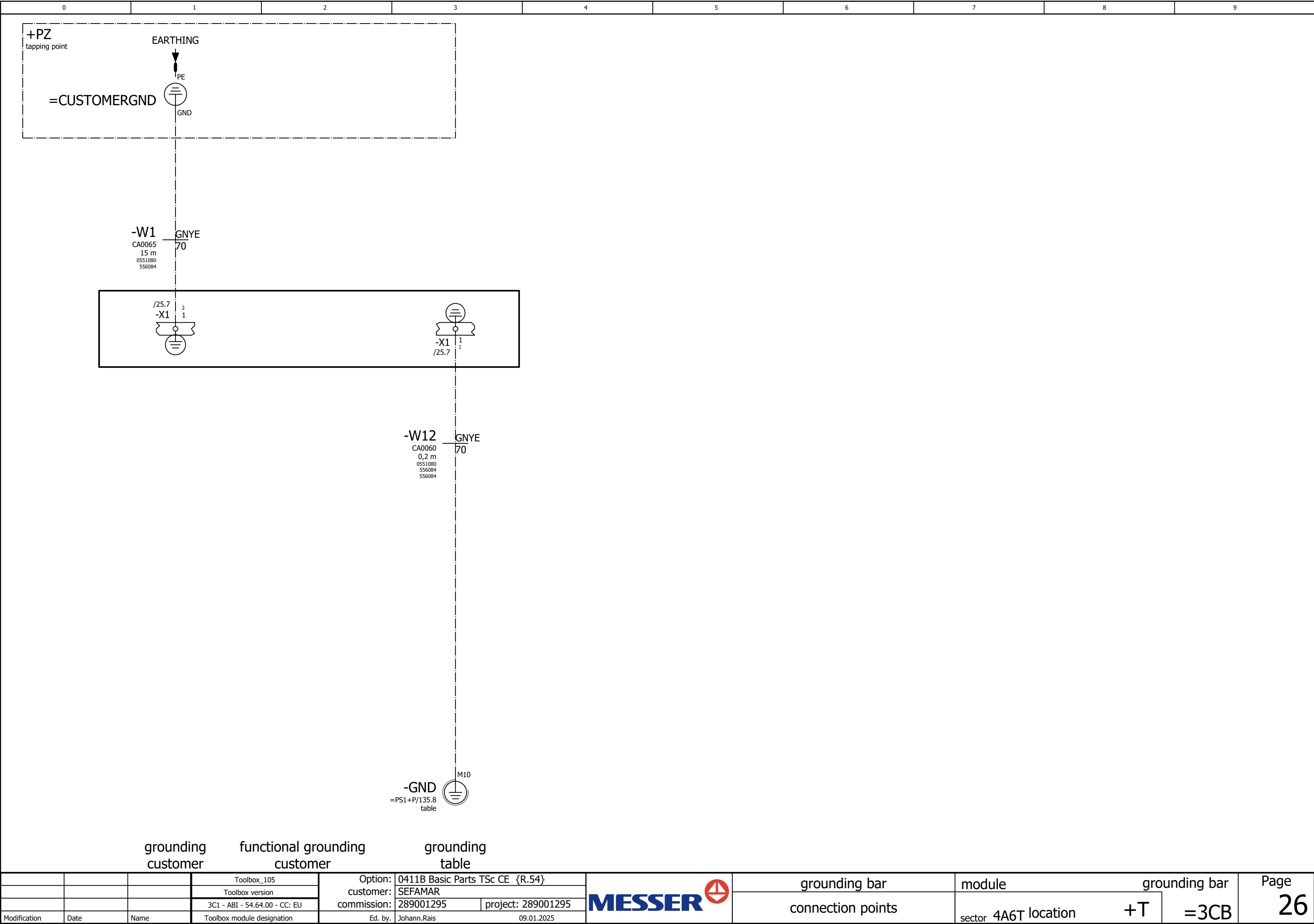
jumper

•	1 - grounding jumper / /24.1	G00000775
•	2 - grounding table / /24.3	
•	3 - cabinet / =3B+HB/15.1	
•	4 - cabinet / =3BB+HB/21.1	
•	5 - grounding carriage plate / =3E1+L1/29.1	
•	6 - panel grounding / =OP+G/124.9	
•	7 - torch connection console grounding / =PB1+L1/129.9	
•	8 - gas console grounding / =PG1+G/131.9	
•	9 - /	
•	10 - /	
•	11 - /	
•	12 - /	

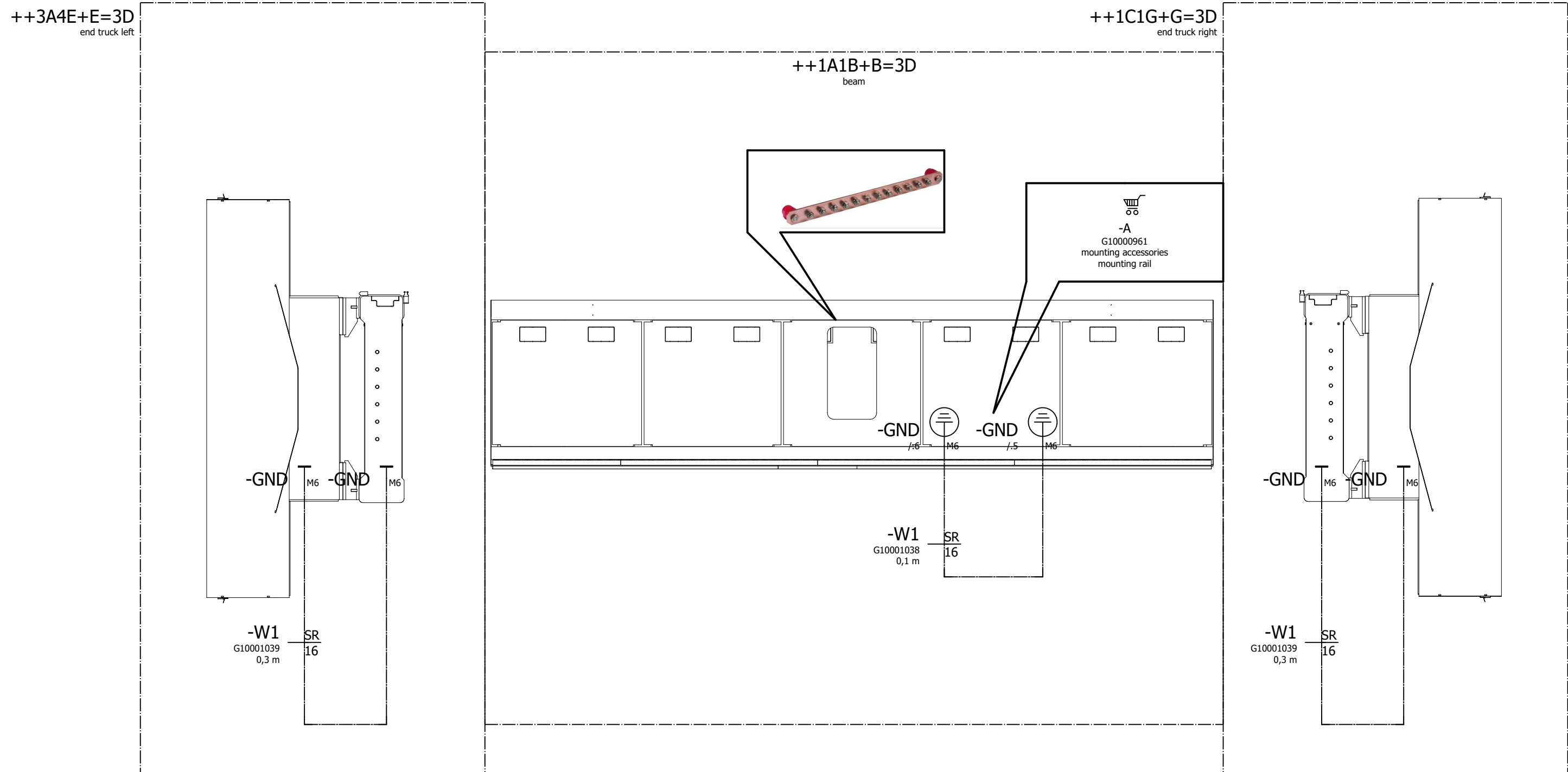
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			grounding bar	module	grounding bar	Page 23
			Toolbox version	customer:	SEFAMAR			connection overview			
			3C4 - ABI - No Rev - CC: EU	commission:	289001295	project: 289001295					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025			sector	2A1B location	



0	1	2	3	4	5	6	7	8	9			
X1 -X1 = potential equalization machine jumper 1 - grounding table / /26.3G00000775 2 - grounding customer / /26.1 3 - cabinet / =3B1+HP1/18.1 4 - grounding table / =3C+B/24.3 5 - track 2 / =3DB/28.9 6 - track 2 / =3DB/28.9 7 - plasma power source functional ground / =PS1+P/135.6 8 - / 9 - / 10 - / 11 - / 12 - / 												
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}		MESSER 	grounding bar	module	grounding bar	Page	
			Toolbox version	customer:	SEFAMAR						25	
			3C1 - ABI - No Rev - CC: EU	commission:	289001295	project: 289001295		connection overview	sector 4A6T location	+T	=3CB	
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025						



0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---



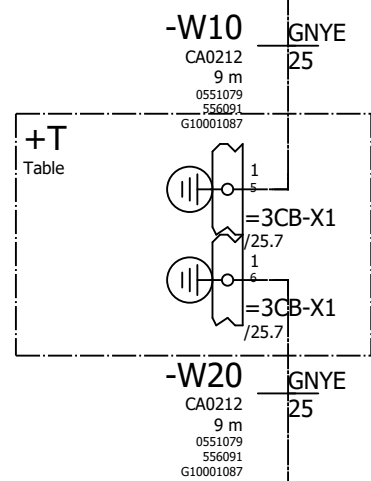
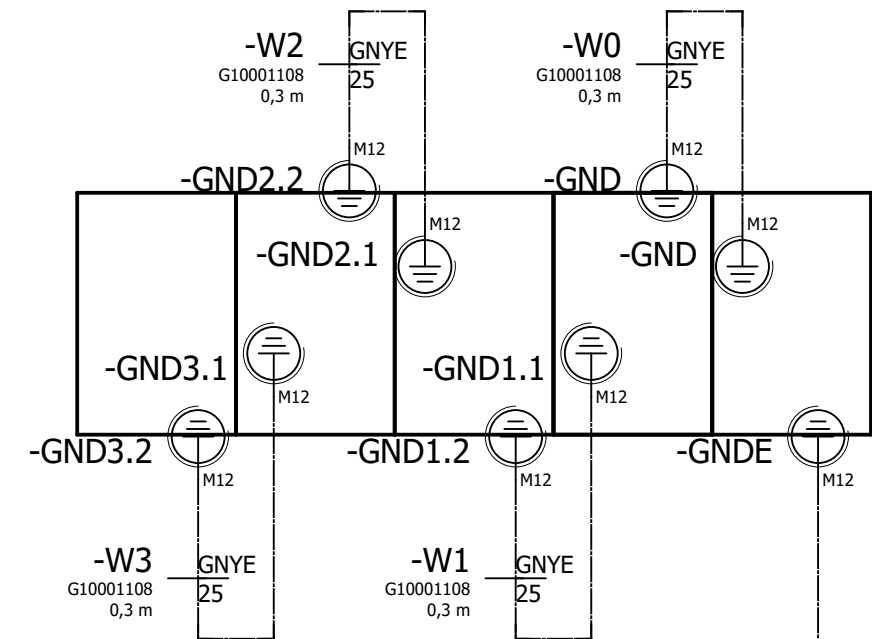
potential
connection

potential
connection

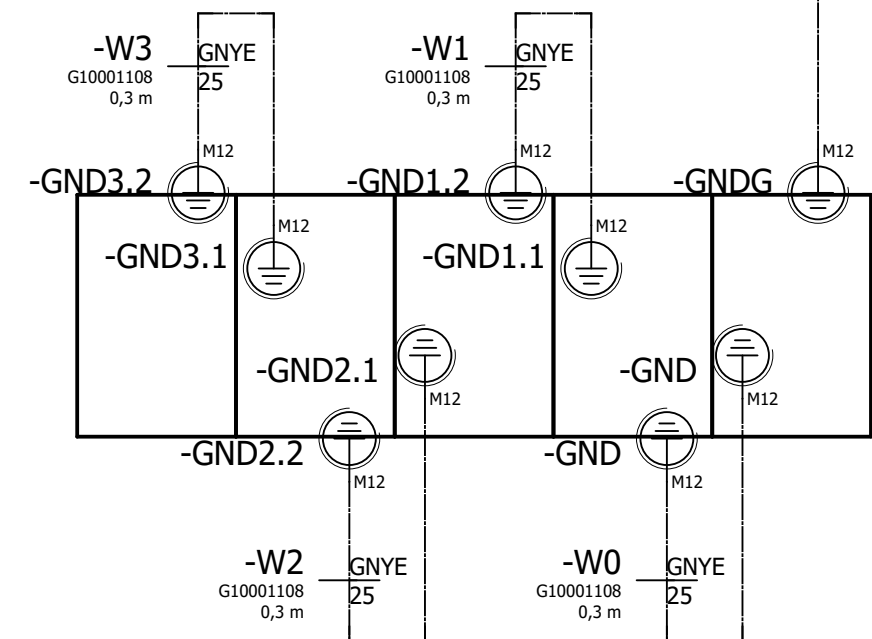
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			grounding machine	module	grounding machine	Page 27
			Toolbox version	customer:	SEFAMAR			overview	sector 4B3E location	+HB =3D	
			3D5 - ABI - 31.60.01 - CC: EU	commission:	289001295	project: 289001295					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025					

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

+PRE
track -X

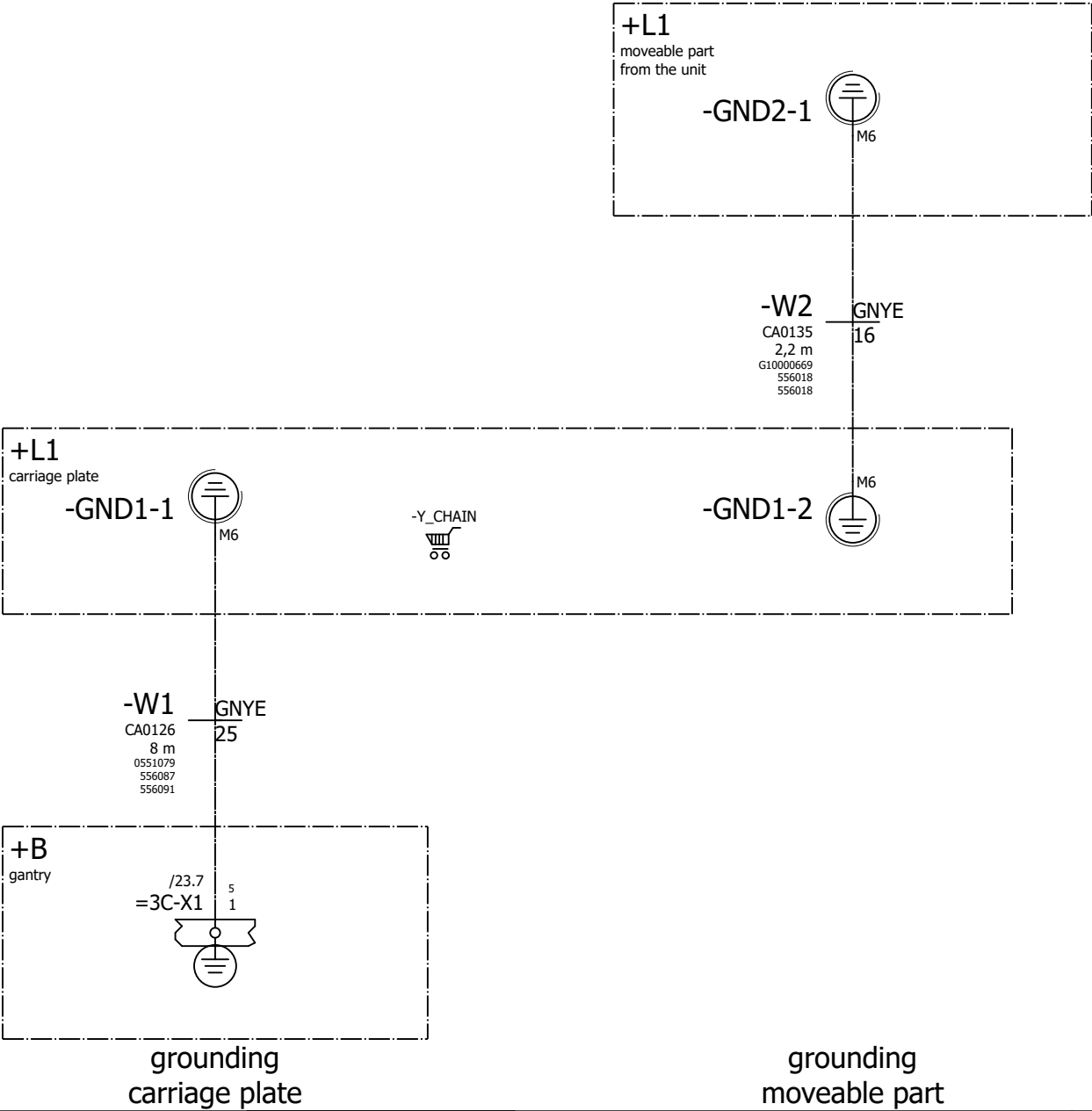


+PRG
track -X



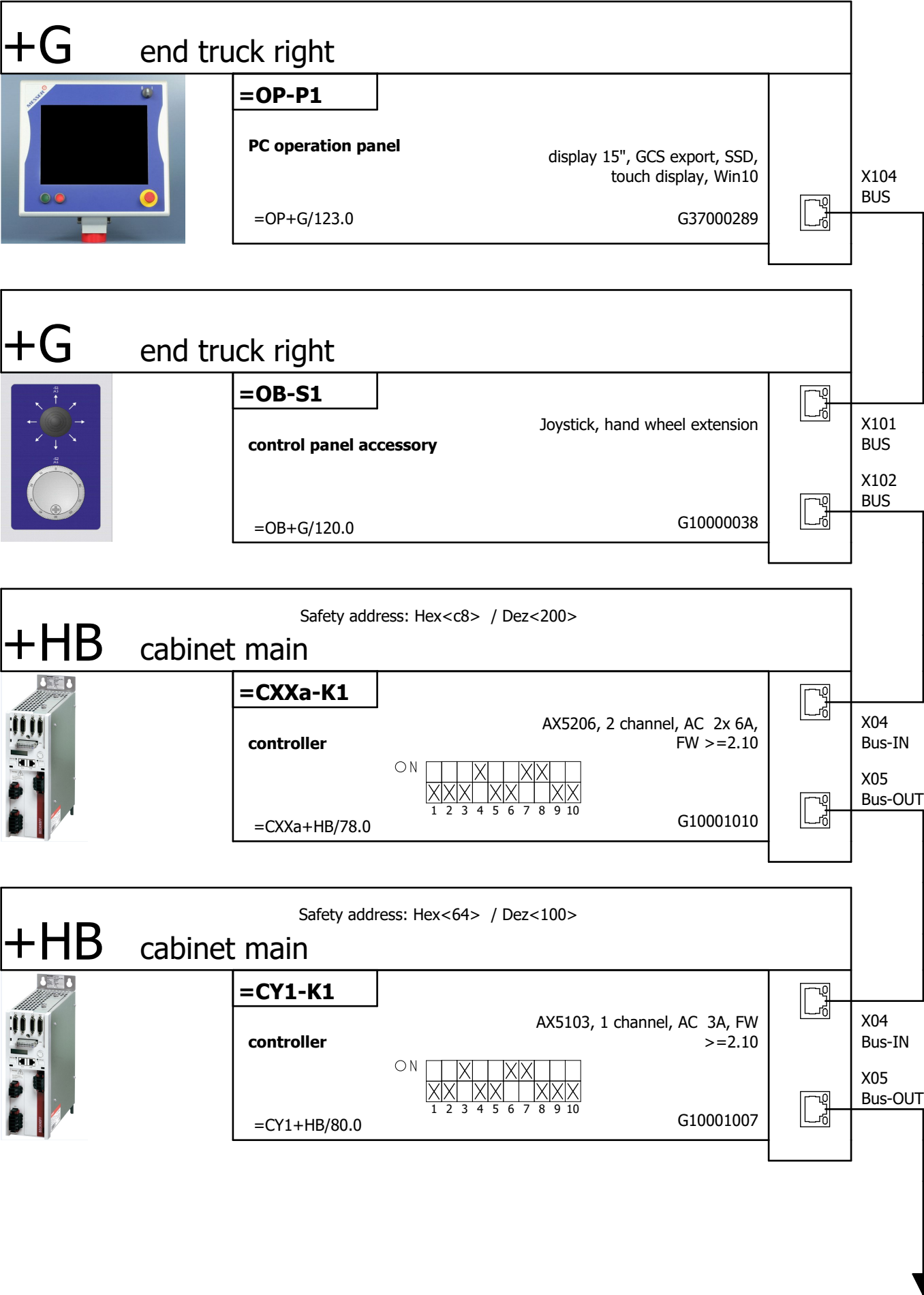
track	track	track	track	track
5	4	3	2	1

			Toolbox_105	Option:	2505C Global Rail Grounding M12 {R.54}			grounding machine	module	grounding machine	Page	
			Toolbox version	customer:	SEFAMAR			excerpt machine, track	sector 2A3T location	+T	=3DB	28
			3D4 - ABI - 54.54.00 - CC: EU	commission:	289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais 09.01.2025							



EtherCAT -Topology

\$-FOS-F20-4A

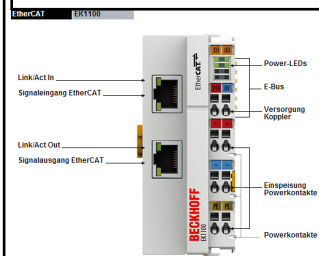


ECAT

+HB=CY1-K1:X05:1-4

$$=IT++4B3E+HB/104$$

+HB cabinet main



=IT-K1

bus terminal

bus coupler

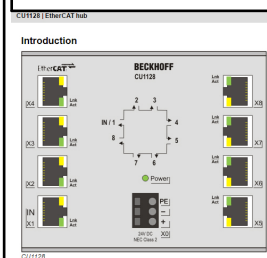
$$=IT+HB/104.1$$

G13000078

X1

x2

+HB cabinet main



=IC-K1

distributor active

EtherCAT star hub, 8x RJ45

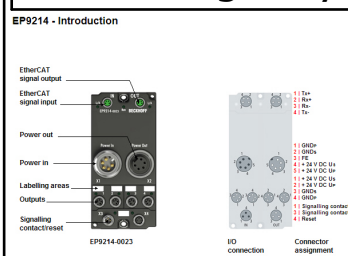
$$=IC+HB/103.0$$

G10000188

X1

X8

+B gantry



=BI-K1

distributor power

EtherCAT-box, power distribution,
M8, 4 x 4A

$$=BI+B/73.0$$

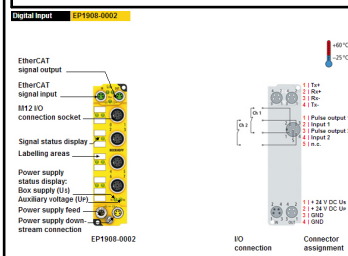
G10000103

L/A
BUS

L/A

+B gantry

Safety address: Hex<9> / Dez<9>



=IB-K1

distributor active

EtherCAT Safety box, IN, 4 channel, M12

$$=IB+B/99.0$$

G10000100

Bus

L/A

EtherCAT -Topologie

ECAT

+B=IB-K1:L/A:OUT

=IBB++4C1B+B/101

+B

gantry

Safety address: Hex<8> / Dez<8>

Digital input

EP1908-0002

EtherCAT signal output

EtherCAT signal input

BM2 IO connection socket

Signal status display

Labelling area

Power supply status display

Box supply (V1)

Auxiliary voltage (V2)

Power supply feed

Power supply down

screen connection

EP1908-0002

IO connection

Connector assignment

=IBB-K1

distributor active

=IBB+B/101.0

EtherCAT Safety box, IN, 4 channel, M12

G10000100

L/A Bus

L/A Bus

			Toolbox_105	Option: customer: commission:		MESSER		module	topology EtherCAT	Page 32	
			Toolbox version		SEFAMAR						
			automated		289001295		project: 289001295				
Modification	Date	Name	evaluation has been generated	Ed. by.	Johann.Rais	09.01.2025	EtherCAT cable	sector	location	+A	=4A

Johann.Rais

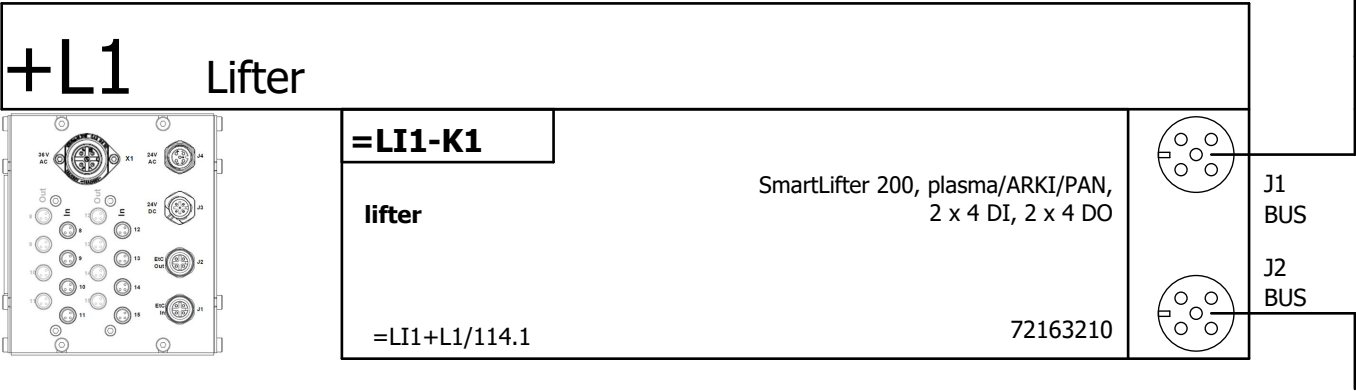
09.01.2025

EtherCAT -Topologie

ECAT_2

+HB=IC-K1:X2:OUT

=LI1++CA1L+L1/114



ECAT_3

=IT1++1E9P+HP1/111

Diagram illustrating the EtherCAT interface on a Beckhoff terminal block. The diagram shows the following connections:

- Link/Act In** and **Signalleingang EtherCAT** connected to the top RJ45 port.
- Link/Act Out** and **Signalleingang EtherCAT** connected to the bottom RJ45 port.
- Power-LEDs** connected to the 15-pin DIN connector.
- E-Bus** connected to the 15-pin DIN connector.
- Versorgung Koppler** connected to the 15-pin DIN connector.
- Einspeisung Powerkontakte** connected to the 15-pin DIN connector.
- Powerkontakte** connected to the 15-pin DIN connector.

bus terminal

G13000078

G13000078

X2	
BUS	

EtherCAT -Topologie

ECAT_PS1
+HP1=PS1-K3:X1:OUT1
=PS1++1E8P+P/136

+Pperiphery



=PS1-E1

plasma power source

KJB, Q1500, 400V, 50/60 Hz, 37 kVA

=PS1+P/135.0

G72100001

X308
Bus

X309
Bus

+HB=IT-K1	+HB=IT-K2	+HB=IT-K3	+HB=IT-K4	+HB=IT-K5	+HB=IT-K6	+HB=IT-K7	+HB=IT-K8	+HB=IT-K9	+HB=IT-K10																																																																																																																																																					
G13000078 EK1100	G10001248 EL1904	G10001248 EL1904	G10000441 EL1859	G10001248 EL1904	G13000079 EL9410	G10001248 EL1904	G10001248 EL1904	G10000699 EL9184	G10001255 EL2904																																																																																																																																																					
	Twin Safe ON <table><tr><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>	X	X	X								X	X	X	X	X	X	1	2	3	4	5	6	7	8	Twin Safe ON <table><tr><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>X</td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>	X	X	X							X			X	X	X	X	1	2	3	4	5	6	7	8		Twin Safe ON <table><tr><td></td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>X</td><td>X</td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>		X							X	X		X	X	X	X	X	1	2	3	4	5	6	7	8		Twin Safe ON <table><tr><td>X</td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td></tr><tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>	X	X			X				X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	Twin Safe ON <table><tr><td></td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td></tr><tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>		X			X				X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	Twin Safe ON <table><tr><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>	X	X	X										X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10
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=IT+HB/104.1	=IT+HB/106.1	=IT+HB/106.6	=IT+HB/107.1	=IT+HB/107.6	=IT+HB/105.0	=IT+HB/108.1	=IT+HB/108.6	=IT+HB/109.1	=IT+HB/109.6																																																																																																																																																					
Safety address: Dez. <11>		Safety address: Dez. <13>		Safety address: Dez. <4>		Safety address: Dez. <37>		Safety address: Dez. <36>																																																																																																																																																						
								Safety address: Dez. <7>																																																																																																																																																						

+HB=IT-K11

G10001262
EL6900

Twin Safe

ON

	X								
X		X	X	X	X	X	X	X	X
1	2	3	4	5	6	7	8	9	10

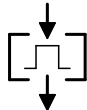
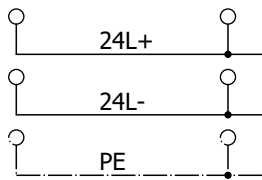
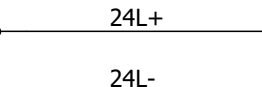
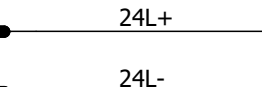
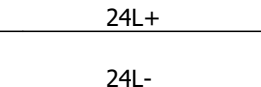
24L+

24L-

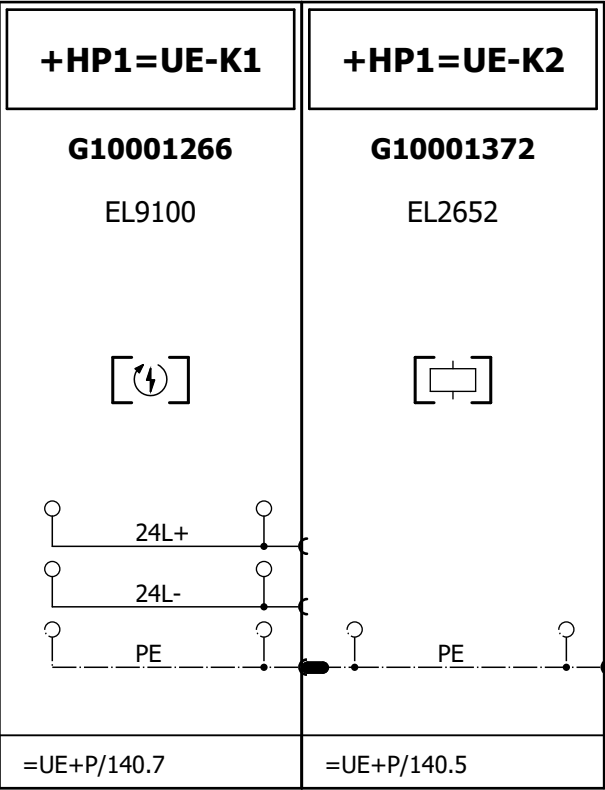
=IT+HB/110.1

Safety address: Dez. <2>

			Toolbox_105	Option: customer: commission:					module	topology EtherCAT	Page 37	
			Toolbox version		SEFAMAR							
			automated		289001295	project: 289001295						
Modification	Date	Name	evaluation has been generated	Ed. by.	Johann.Rais	09.01.2025		SPS-Kartenübersicht	sector	location	+A	=4A

+HP1=IT1-K1	+HP1=IT1-K2	+HP1=IT1-K3	+HP1=PS1-K1	+HP1=PS1-K2	+HP1=PS1-K3																								
G13000078 EK1100	G10001248 EL1904	G10000441 EL1859	G13000079 EL9410	G10000910 EL9182	G10000781 EK1122																								
	Twin Safe ON <table><tr><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td></td><td></td></tr><tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>					X	X			X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8				
				X	X																								
X	X	X	X	X	X	X	X																						
1	2	3	4	5	6	7	8																						
																													
=IT1+HP1/111.1	=IT1+HP1/112.1	=IT1+HP1/112.6	=PS1+P/136.1	=PS1+P/136.6	=PS1+P/136.8																								

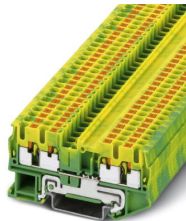
Safety address: Dez. <80>

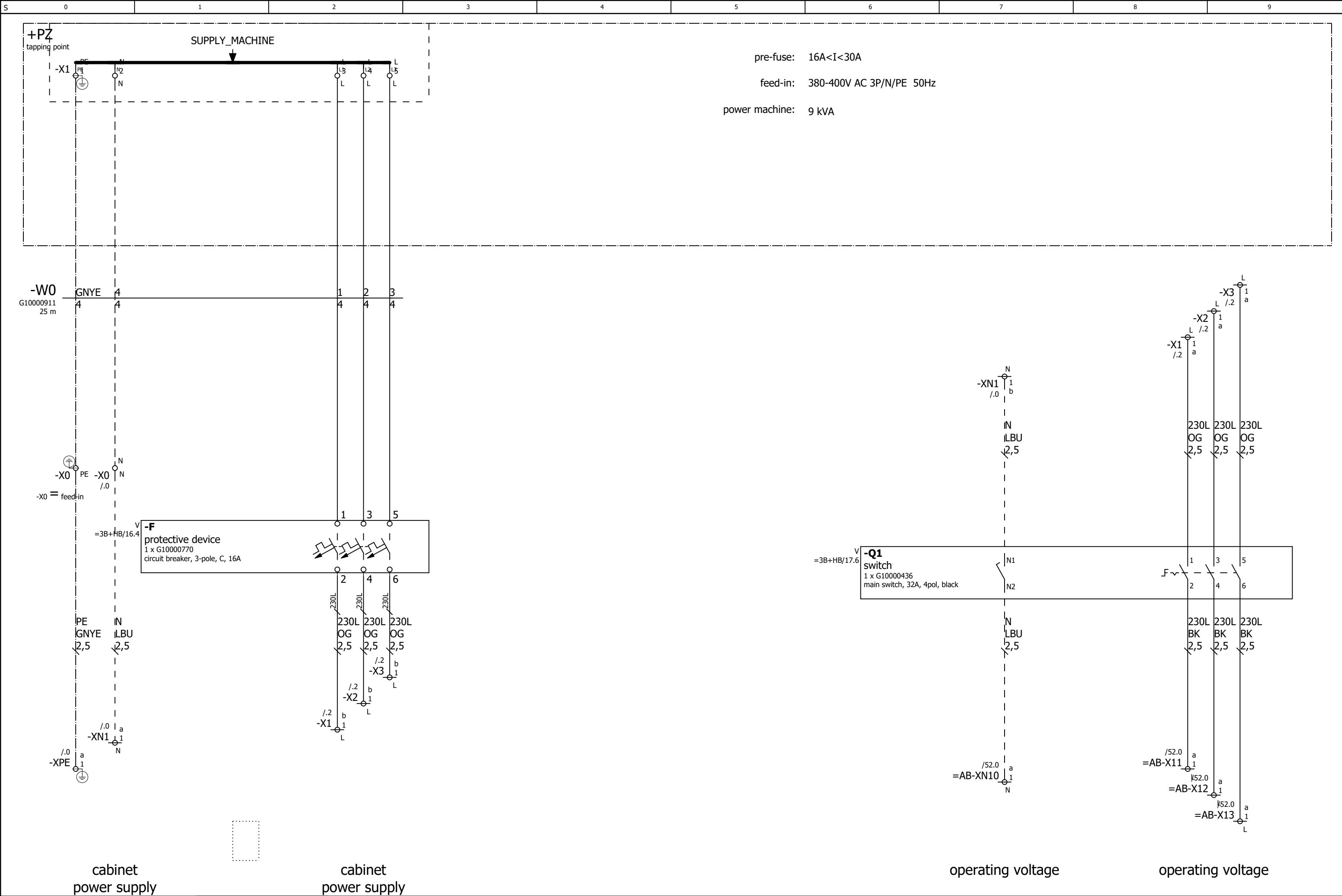


0	1	2	3	4	5	6	7	8	9												
X1  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td colspan="3">a - operating voltage /45.8</td></tr><tr><td colspan="3">b - cabinet power supply /45.2</td></tr><tr><td colspan="3">c - service power socket =AS/58.3</td></tr></table>										•	1	77070157	a - operating voltage /45.8			b - cabinet power supply /45.2			c - service power socket =AS/58.3		
•	1	77070157																			
a - operating voltage /45.8																					
b - cabinet power supply /45.2																					
c - service power socket =AS/58.3																					
			Option: 1		MESSER	overview terminal strip		module AC power supply cabinet													
			customer: SEFAMAR																		
			commission: 289001295 project: 289001295																		
Modification	Date	Name	Toolbox module designation		Ed. by.	Johann.Rais	09.01.2025	sector location +HB =AA													
									Page 40												

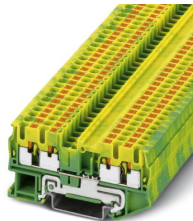
0	1	2	3	4	5	6	7	8	9									
X3  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td></td><td>a - operating voltage</td><td>/45.8</td></tr><tr><td></td><td>b - cabinet power supply</td><td>/45.2</td></tr></table>										•	1	77070157		a - operating voltage	/45.8		b - cabinet power supply	/45.2
•	1	77070157																
	a - operating voltage	/45.8																
	b - cabinet power supply	/45.2																
			Toolbox_105	Option:	3		module		AC power supply cabinet		Page							
			Toolbox version	customer:	SEFAMAR		overview terminal strip	sector		location	+HB	=AA	42					
				commission:	289001295									project: 289001295				
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025												

0	1	2	3	4	5	6	7	8	9												
XN1  jumper <table><tr><td>•</td><td>1</td><td>G10000679</td></tr><tr><td colspan="2">a - cabinet power supply</td><td>/45.0</td></tr><tr><td colspan="2">b - operating voltage</td><td>/45.7</td></tr><tr><td colspan="2">c - service power socket</td><td>=AS/58.3</td></tr></table>										•	1	G10000679	a - cabinet power supply		/45.0	b - operating voltage		/45.7	c - service power socket		=AS/58.3
•	1	G10000679																			
a - cabinet power supply		/45.0																			
b - operating voltage		/45.7																			
c - service power socket		=AS/58.3																			
			Toolbox_105	Option: 4			module	AC power supply cabinet		Page											
			Toolbox version	customer: SEFAMAR			overview terminal strip			43											
				commission: 289001295							project: 289001295										
Modification	Date	Name	Toolbox module designation	Ed. by. Johann.Rais	09.01.2025		sector	location	+HB	=AA											

0	1	2	3	4	5	6	7	8	9												
XPE  jumper <table><tr><td>•</td><td>1</td><td>77070158</td></tr><tr><td>a - cabinet power supply</td><td>/45.0</td><td></td></tr><tr><td>b - star hub supply / PE</td><td>=IC/103.2</td><td></td></tr><tr><td>c - service power socket</td><td>=AS/58.5</td><td></td></tr></table>										•	1	77070158	a - cabinet power supply	/45.0		b - star hub supply / PE	=IC/103.2		c - service power socket	=AS/58.5	
•	1	77070158																			
a - cabinet power supply	/45.0																				
b - star hub supply / PE	=IC/103.2																				
c - service power socket	=AS/58.5																				
			Toolbox_105	Option: 5			module	AC power supply cabinet		Page											
			Toolbox version	customer: SEFAMAR						44											
				commission: 289001295		project: 289001295	overview terminal strip														
Modification	Date	Name	Toolbox module designation	Ed. by. Johann.Rais	09.01.2025		sector	location	+HB	=AA											

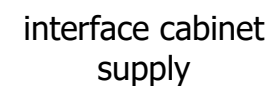


XPE



jumper

•	1	77070158
a - interface cabinet supply /47.2		
b - exhaust grounding / PE =UE+P/140.6		



X11



jumper

•	1	77070157
a - operating voltage =AA/45.8		
b - interface cabinet supply =AA1+HP1/47.5		
c - controller power supply AC / L1 =CXXa/78.1		
d - controller power supply AC / L1 =CY1/80.1		

0	1	2	3	4	5	6	7	8	9															
X12  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td colspan="3">a - operating voltage =AA/45.8</td></tr><tr><td colspan="3">b - transformer supply =AL/57.1</td></tr><tr><td colspan="3">c - controller power supply AC / L2 =CXXa/78.2</td></tr><tr><td colspan="3">d - controller power supply AC / L2 =CY1/80.2</td></tr></table>										•	1	77070157	a - operating voltage =AA/45.8			b - transformer supply =AL/57.1			c - controller power supply AC / L2 =CXXa/78.2			d - controller power supply AC / L2 =CY1/80.2		
•	1	77070157																						
a - operating voltage =AA/45.8																								
b - transformer supply =AL/57.1																								
c - controller power supply AC / L2 =CXXa/78.2																								
d - controller power supply AC / L2 =CY1/80.2																								
			Toolbox_105	Option: 2			module		AC distribution	Page														
			Toolbox version	customer: SEFAMAR		overview terminal strip				49														
				commission: 289001295 project: 289001295																				
Modification	Date	Name	Toolbox module designation	Ed. by. Johann.Rais 09.01.2025			sector	location	+HB =AB															

0	1	2	3	4	5	6	7	8	9															
X13  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td colspan="3">a - operating voltage =AA/45.8</td></tr><tr><td colspan="3">b - operating voltage power supply 24VDC =BA/63.6</td></tr><tr><td colspan="3">c - controller power supply AC / L3 =CXXa/78.2</td></tr><tr><td colspan="3">d - controller power supply AC / L3 =CY1/80.2</td></tr></table>										•	1	77070157	a - operating voltage =AA/45.8			b - operating voltage power supply 24VDC =BA/63.6			c - controller power supply AC / L3 =CXXa/78.2			d - controller power supply AC / L3 =CY1/80.2		
•	1	77070157																						
a - operating voltage =AA/45.8																								
b - operating voltage power supply 24VDC =BA/63.6																								
c - controller power supply AC / L3 =CXXa/78.2																								
d - controller power supply AC / L3 =CY1/80.2																								
			Toolbox_105	Option: 3	MESSER 	module		AC distribution		Page														
			Toolbox version	customer: SEFAMAR		overview terminal strip			50															
				commission: 289001295 project: 289001295		sector	location	+HB =AB																
Modification	Date	Name	Toolbox module designation	Ed. by. Johann.Rais 09.01.2025																				

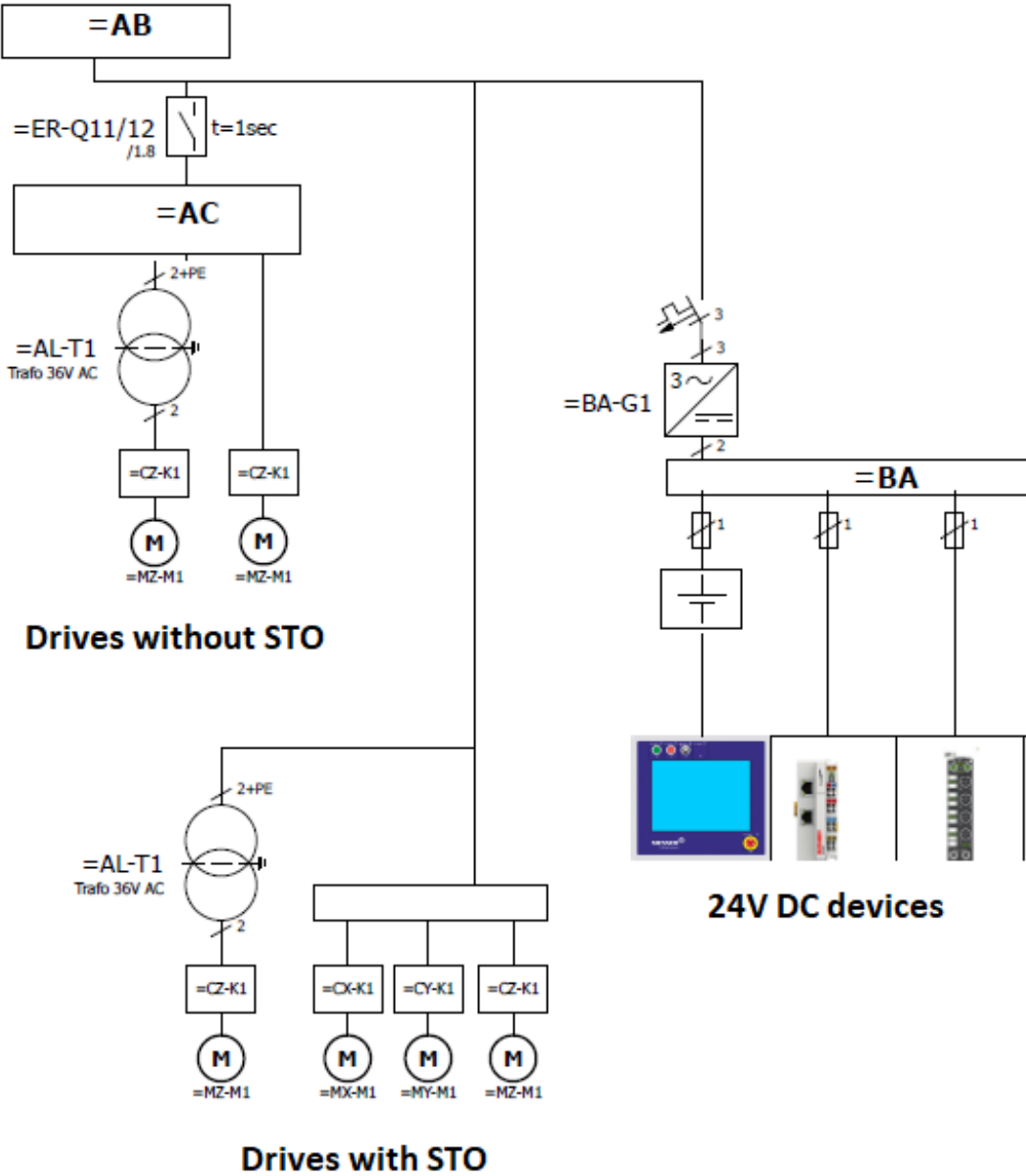
XN10



jumper

•	1	G10000679
a - operating voltage =AA/45.7		
b - operating voltage power supply 24VDC / 230N =BA/63.4		
c - interface cabinet supply =AA1+HP1/47.3		
d - transformer supply / 0 =AL/57.2		

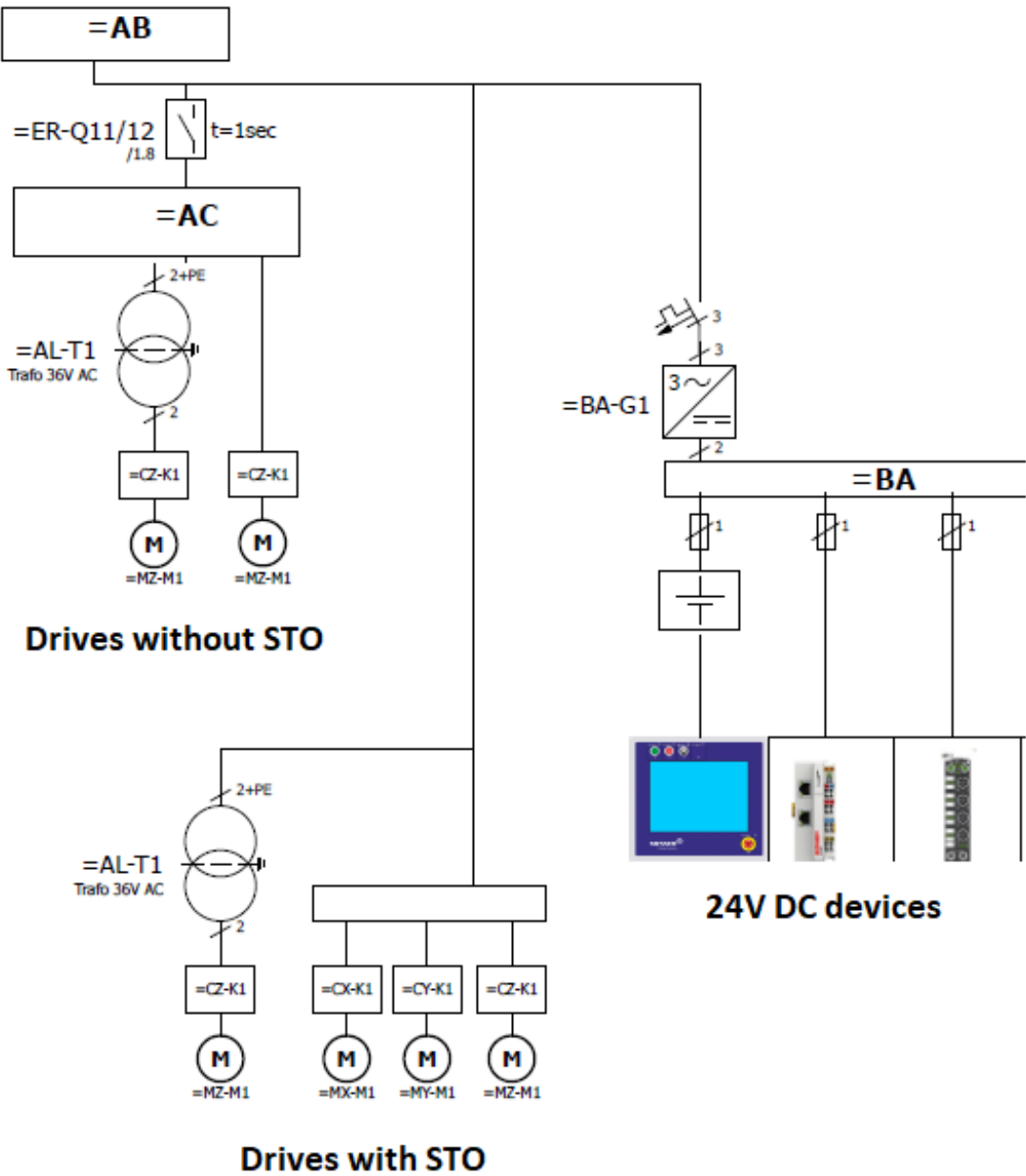
AC supply principle



0	1	2	3	4	5	6	7	8	9

0	1	2	3	4	5	6	7	8	9

AC supply principle





77070862

transformer

-T1

1 phase, 4 x 100 VA, 230V/36V

X1

L	230	transformer supply	/57.1
N	0	=	/57.2
PE	ground	=	/57.2

X2.1

L	36	SmartLifter AC power supply	=LI1+L1/114.3
N	0		=LI1+L1/114.2

X2.2

L	36
N	0

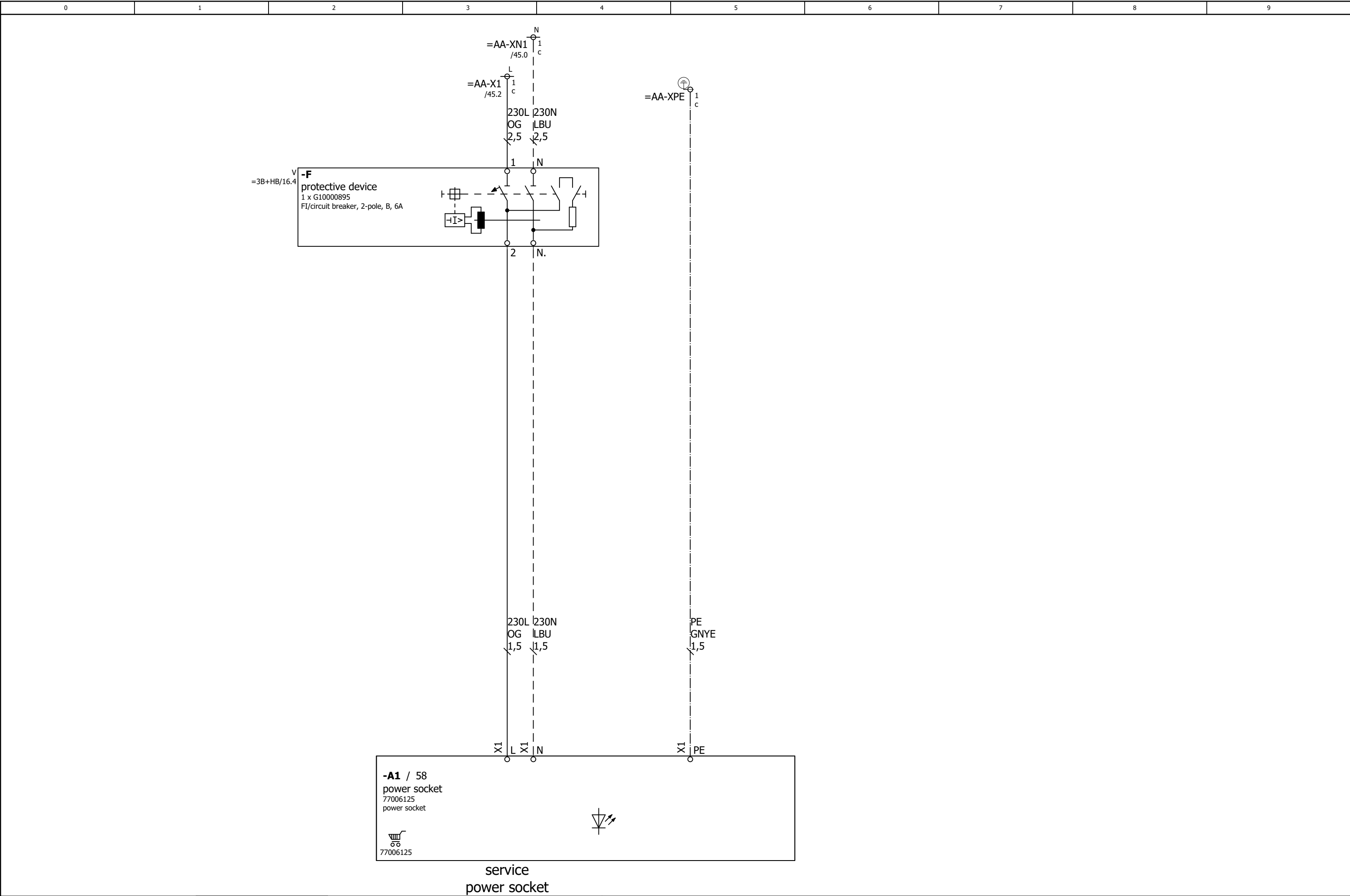
X2.3

L	36
N	0

X2.4

L	36
N	0







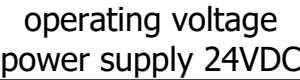
G10000827

protective device

-F1

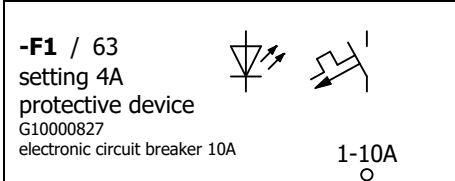
electronic circuit breaker 10A

Contact		
13	NO	
14	NO	
IN+		
1	24L+	operating voltage power supply 2
2	24L+	
IN-		
1	24L-	operating voltage power supply 2
2	24L-	
OUT+		
1	1-10A	operating voltage 24L+ channel
2	1-10A	operating voltage 24L+ channel
3	1-10A	operating voltage 24L+ channel
4	1-10A	operating voltage 24L+ channel



			Toolbox_105	Option:	1102B Main Cabi supply S+N switched			operating voltage		module		DC power supply		Page
			Toolbox version	customer:	SEFAMAR				power supply	sector	4B3E location	+HB	=BA	63
			BA1 - ABI - 34.80.00 - CC: EU	commission:	289001295	project: 289001295								
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025								

0	1	2	3	4	5	6	7	8	9
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1-10A

+	1
---	---

24L+

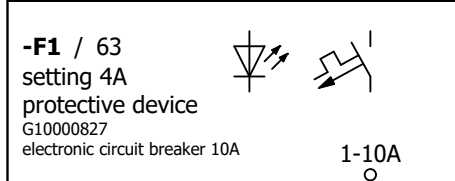
BU

1

/61.0	a
-------	---

a

1



1-10A

 $1 + 2$ $24L +$

BU

1

/61.7 | a

a

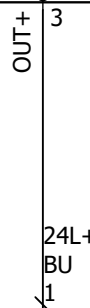
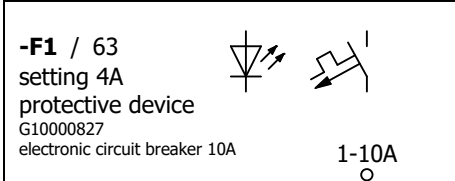
 $\ominus^1$

operating voltage 24L+

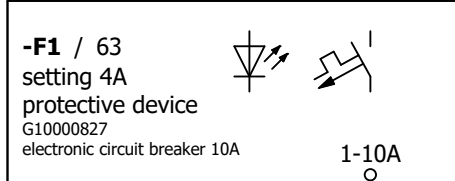
operating voltage 24L+

			Toolbox_105	Option: customer: commission:	1102B Main Cabi supply S+N switched			operating voltage	module	DC power supply	Page 64	
			Toolbox version		SEFAMAR			electronical fuse channel 1 / 2	sector 4B3E location	+HB		=BA
			BA1 - ABI - 34.80.00 - CC: EU		289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	09.01.2025						

0	1	2	3	4	5	6	7	8	9
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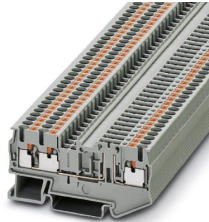


operating voltage 24L+
channel 3

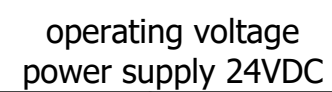


operating voltage 24L+
channel 4

			Toolbox_105	Option: customer: commission:	1102B Main Cabi supply S+N switched			operating voltage	module	DC power supply	Page 65	
			Toolbox version		SEFAMAR			electronical fuse channel 3 / 4	sector 4B3E location	+HB		=BA
			BA1 - ABI - 34.80.00 - CC: EU		289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	09.01.2025						

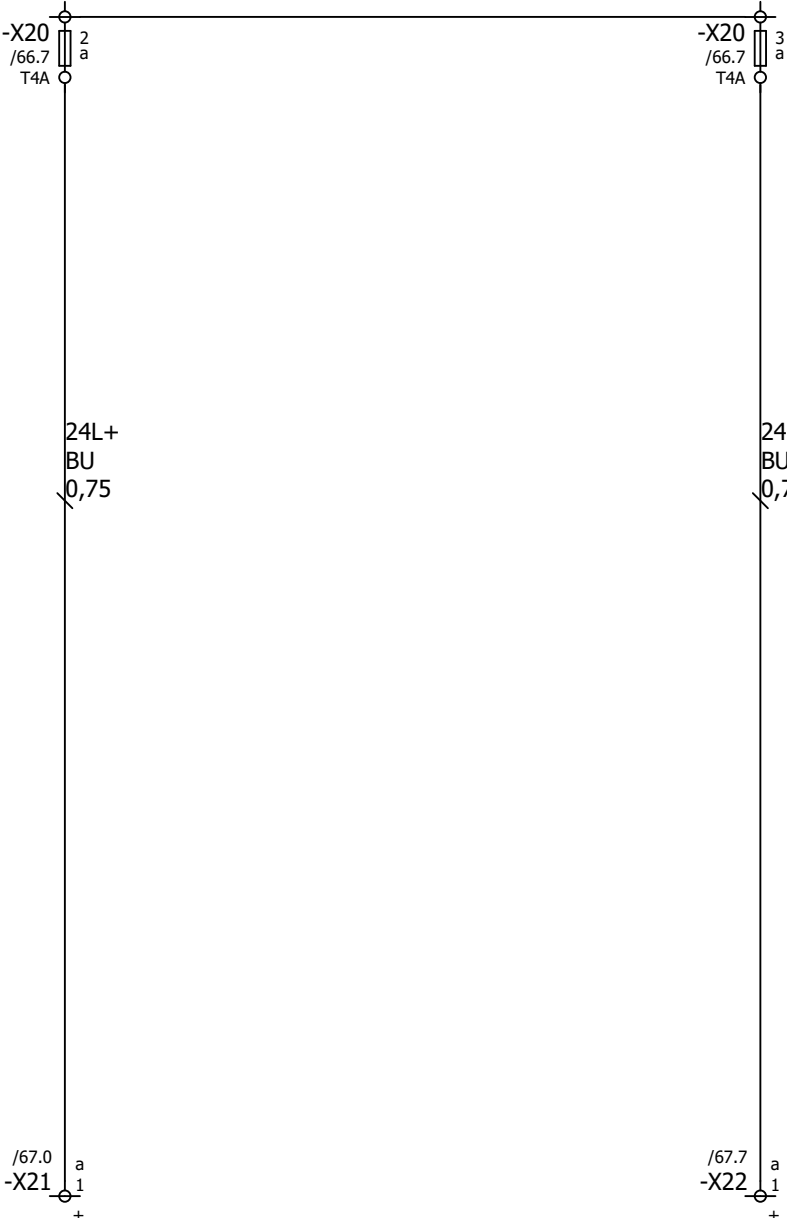
0	1	2	3	4	5	6	7	8	9					
X10  -X10 = operating voltage 24L- jumper 1 77070157 a - operating voltage power supply 24VDC / 24L- /69.4 b - Safety interface external =EI1/93.1 c - EtherCAT coupler operating supply / 24L- =IT1/111.6 d - EtherCAT coupler powerbus / - =IT1/111.7 2 77070157 a - feed-in / 24L- =PS1+P/136.1 b - plasma power source standby / 5-13 =PS1+P/136.2 X20  -X20 = potential distribution operating voltage 24L+ jumper 1 G10000602;G10000506 a - operating voltage power supply 24VDC / 24L+ /69.5 2 G10000602;G10000459;541021 a - operating voltage 24L+ channel 1 /70.4 3 G10000602;G10000459;541021 a - operating voltage 24L+ channel 2 /70.6 4 G10000602;G10000459;541021 a - operating voltage 24L+ channel 3 /71.4 5 G10000602;G10000459;541021 a - operating voltage 24L+ channel 4 /71.6														
			Toolbox_105	Option:	0406A L1-Plasma Interface Cabinet		MESSER 	operating voltage	module	DC power supply		Page		
			Toolbox version	customer:	SEFAMAR			connection overview	sector	1E9P location	+HP1	=BA1	66	
			BA2 - ABI - No Rev - CC: EU	commission:	289001295	project: 289001295								
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025								

0	1	2	3	4	5	6	7	8	9							
X23 -X23 = operating voltage 24L+ channel 3 jumper • 1 77070157 a - operating voltage 24L+ channel 3 /71.4 b - EtherCAT coupler powerbus / + =IT1/111.6 X24 -X24 = operating voltage 24L+ channel 4 jumper • 1 77070157 a - operating voltage 24L+ channel 4 /71.6 b - plasma interface supply / 24L+ =PS1+P/136.0																
			Toolbox_105		Option: 0406A L1-Plasma Interface Cabinet customer: SEFAMAR commission: 289001295	project: 289001295	operating voltage connection overview	module		DC power supply		Page				
			Toolbox version													
			BA2 - ABI - No Rev - CC: EU													
Modification	Date	Name	Toolbox module designation		Ed. by.	Johann.Rais	09.01.2025	sector 1E9p location		+HP1	=BA1	68				



MESSER 

0	1	2	3	4	5	6	7	8	9
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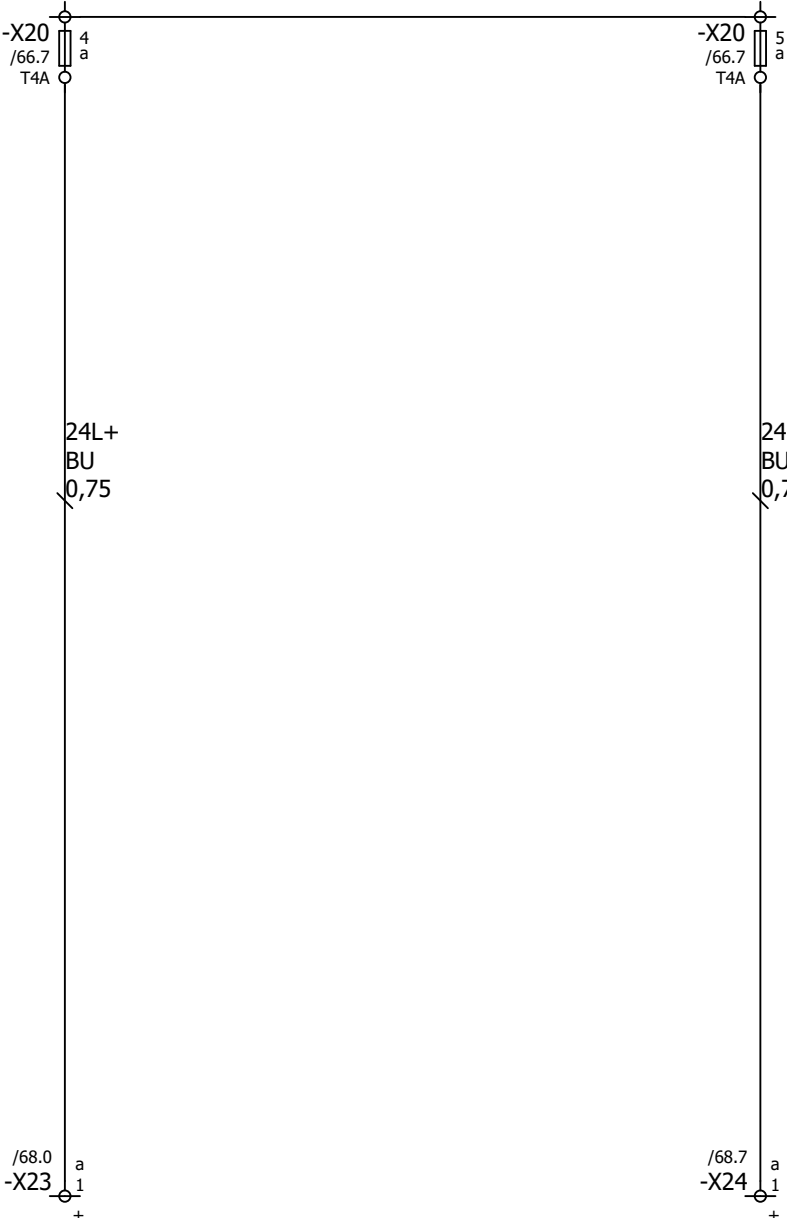


operating voltage 24L+
channel 1

operating voltage 24L+
channel 2

			Toolbox_105	Option: customer: commission:	0406A L1-Plasma Interface Cabinet			operating voltage	module	DC power supply	Page 70	
			Toolbox version		SEFAMAR			electronical fuse channel 1 / 2	sector 1E9p location	+HP1		=BA1
			BA2 - ABI - 34.101.00 - CC: EU		289001295 project: 289001295							
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais 09.01.2025							

0	1	2	3	4	5	6	7	8	9
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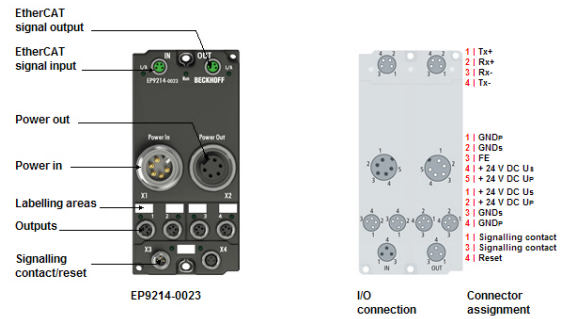


operating voltage 24L+
channel 3

operating voltage 24L+
channel 4

			Toolbox_105	Option: customer: commission:	0406A L1-Plasma Interface Cabinet			operating voltage		module DC power supply		Page 71
			Toolbox version		SEFAMAR			electrical fuse channel 3 / 4	sector 1	E9p location +HP1	=BA1	
			BA2 - ABI - 34.101.00 - CC: EU		289001295 project: 289001295							
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais 09.01.2025							

EP9214 - Introduction



G10000103

distributor power

-K1

EtherCAT-box, power distribution, M8, 4 x 4A

1

OUT 1

1	24L+s	=IB/99.1
2	24L+p	EP module supply =IB/99.2
3	24L-s	= =IB/99.3
4	24L-p	= =IB/99.3

2

OUT 2

1	24L+s	SmartLifter DC operating voltage =LI1+L1/114.5
2	24L+p	SmartLifter DC control voltage =LI1+L1/114.6
3	24L-s	SmartLifter DC operating voltage =LI1+L1/114.7
4	24L-p	= =LI1+L1/114.7

3

OUT 3

1	24L+s
2	24L+p
3	24L-s
4	24L-p

4

OUT 4

1	24L+s
2	24L+p
3	24L-s
4	24L-p

L/A

	Bus	
IN	Bus	EtherCAT /73.8
OUT	Bus	= /73.8

X1

	Power IN	
1	24L-s	distribution 24 VDC external /73.1
2	24L-p	= /73.2
3	PE	= /73.3
4	24L+s	= /73.3
5	24L+p	= /73.4

X2

	Power OUT	
1	24L-s	
2	24L-p	
3	PE	
4	24L+s	
5	24L+p	

X3

	diagnosis IN	
1	NO	
3	COM	
4	reset	

X4

	diagnosis OUT	
1	NO	
3	COM	
4	reset	



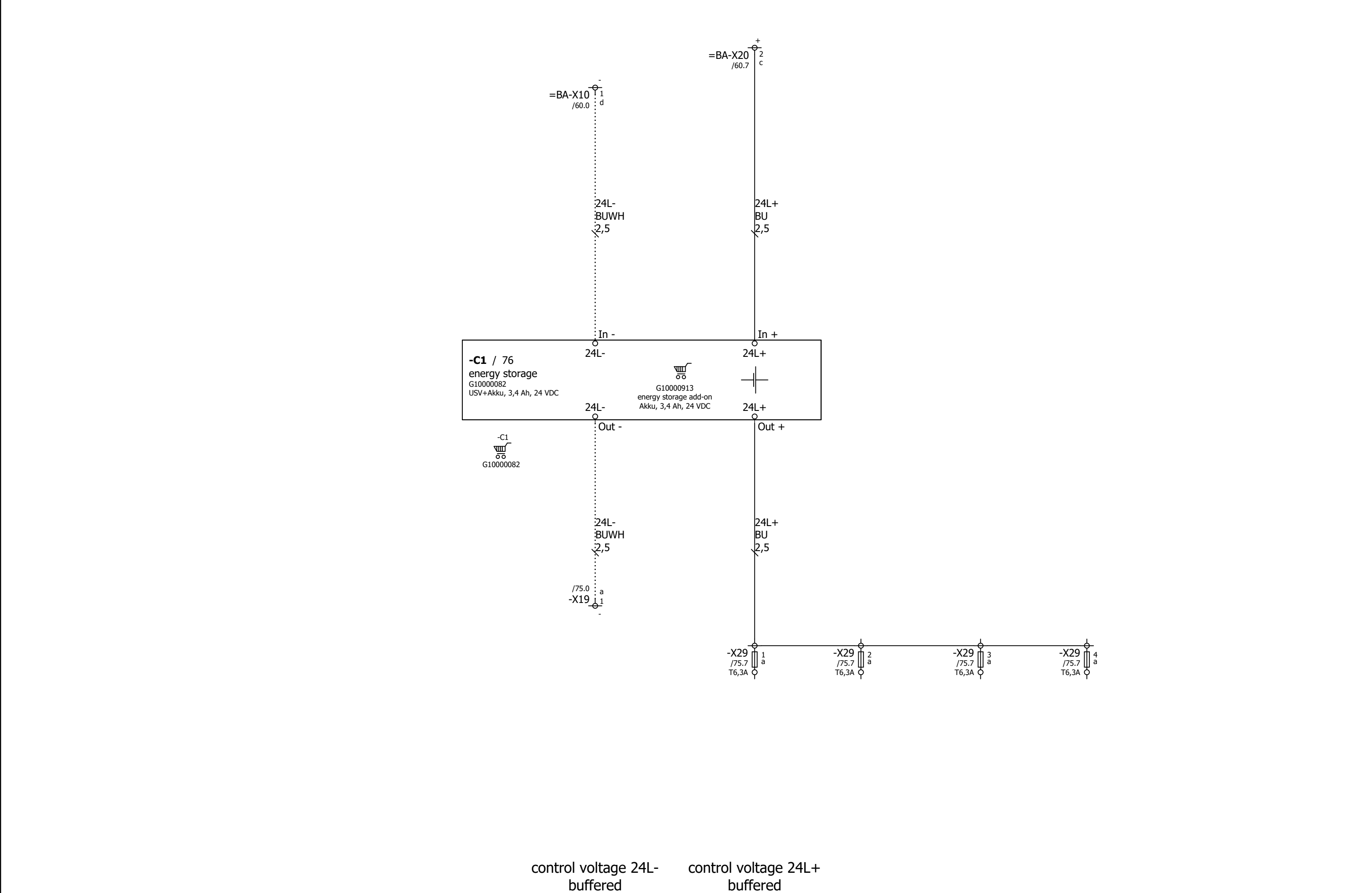
G10000082

energy storage

-C1

USV+Akkus, 3,4 Ah, 24 VDC

In +	24L+	control voltage 24L+ buffered	/76.5
In -	24L-	control voltage 24Lbuffered	/76.4
Out +	24L+	control voltage 24L+ buffered	/76.5
Out +	24L+ 0,2A		
Out -	24L-	control voltage 24Lbuffered	/76.4
Alarm			
13	Alarm		
14	Alarm		
Bat. Charge			
33	Bat. Charge		
34	Bat. Charge		
Bat. Mode			
23	Bat. Mode		
24	Bat. Mode	PC turn off	=OP+G/124.6
Remote			
R1	Remote		
R2	Remote in	USV turn off	=OP+G/124.7





G10001010

controller

-K1

AX5206, 2 channel, AC 2x 6A,
FW >=2.10

GND

grounding		
M6	ground	controller grounding /78.9

Safety

PCI		
PCI	Bus	controller Safety Card /78.5

X01

supply grid		
1	L1	controller power supply AC /78.1
2	L2	= /78.2
3	L3	= /78.2
PE	GND	controller ground /78.2

X02

DC Link		
1	DC+	
2	DC-	

X03

supply DC		
1	Up+	controller power supply DC /78.3
2	Us+	controller supply DC /78.4
3	GND	controller power supply DC /78.4

X04

Bus		
1-4	Bus-IN	controller EtherCAT /78.8

X05

1-4	Bus-OUT	controller EtherCAT /78.8
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X06

digital I/O's	
0	input
0V	24L-
1	input
2	input
3	input
4	input
5	input
6	input
7	in/out
24	24L+

X11

Hiperface	
1	SIN+
2	GND-9V
3	COS+
4	n.c.
5	Data+
6	Us-9V
7	n.c.
8	n.c.
9	REFSIN
10	n.c.
11	REF COS
12	n.c.
13	Data-
14	n.c.
15	n.c.
SH	shield

X12

Resolver	
1	Temp.
2	AGND
3	COS-
4	SIN-
5	REF-
6	n.c.
7	n.c.
8	n.c.
9	Temp.GND
10	COS+

X12

11	SIN+
12	REF+
13	n.c.
14	n.c.
15	n.c.
SH	shield

X13

Motor	
PE	GND motor coil =MX+E/115.2
U	Motor = =MX+E/115.1
V	Motor = =MX+E/115.2
W	Motor = =MX+E/115.2

X13/X14

SH	shield connection shield =MX+E/115.6
----	--------------------------------------

X14

Temp./brake	
B+	brake + connection # =MX+E/115.3
B-	brake GND = =MX+E/115.4
PE	shield
T+	Temp.+ connection encoder =MX+E/115.5
T-	Temp.- = =MX+E/115.5

X21

Hiperface	
1	SIN+
2	GND-9V
3	COS+
4	n.c.
5	Data+
6	Us-9V
7	n.c.
8	n.c.
9	REFSIN
10	n.c.
11	REF COS
12	n.c.
13	Data-
14	n.c.
15	n.c.
SH	shield

X22

Resolver	
1	Temp.
2	AGND
3	COS-
4	SIN-
5	REF-
6	n.c.
7	n.c.
8	n.c.
9	Temp.GND
10	COS+
11	SIN+
12	REF+
13	n.c.
14	n.c.
15	n.c.
SH	shield

X23

Motor	
PE	GND motor coil =MXa+G/116.2
U	Motor = =MXa+G/116.1
V	Motor = =MXa+G/116.2
W	Motor = =MXa+G/116.2

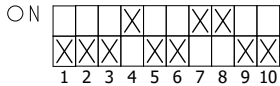
X23/X24

SH	shield connection shield =MXa+G/116.6
----	---------------------------------------

X24

Temp./brake	
B+	brake + connection # =MXa+G/116.3
B-	brake GND = =MXa+G/116.4
PE	shield
T+	Temp.+ connection encoder =MXa+G/116.5
T-	Temp.- = =MXa+G/116.5

Safety adress: HEX -<c8> / Dez: <200>





G10001007

controller

AX5103, 1 channel, AC 3A, FW
>=2.10

-K1

GND

grounding	
5 ground	controller grounding /80.9

Safety

	PCI	
PCI	Bus	controller Safety Card /80.5

X01

	supply grid		
1	L1	controller power supply AC	/80.1
2	L2	=	/80.2
3	L3	=	/80.2
PE	GND	controller ground	/80.2

X02

	DC Link
1	DC+
2	DC-

X03

	supply DC	
1	Up+	controller power supply DC /80.3
2	Us+	controller supply DC /80.4
3	GND	controller power supply DC /80.4

X04

	Bus	
1-4	Bus-IN	controller EtherCAT /80.8

X05

1-4	Bus-OUT	controller EtherCAT	/80.8
-----	---------	---------------------	-------

X06

	digital I/O's	
0	input	
0V	24L-	
1	input	working area beam Y + /81.2
2	input	working area beam Y - /81.3
3	input	
4	input	
5	input	
6	input	
7	in/out	
24	24L+	

X11

	Hiperface
1	SIN+
2	GND-9V
3	COS+
4	n.c.
5	Data+
6	Us-9V
7	n.c.
8	n.c.
9	REFSIN
10	n.c.
11	REF COS
12	n.c.
13	Data-
14	n.c.
15	n.c.
SH	shield

X12

	Resolver
1	Temp.
2	AGND
3	COS-
4	SIN-
5	REF-
6	n.c.
7	n.c.
8	n.c.
9	Temp.GND
10	COS+

X12

11	SIN+
12	REF+
13	n.c.
14	n.c.
15	n.c.
SH	shield

X13

	Motor	
PE	GND	motor coil =MY1+L1/117.2
U	Motor	= =MY1+L1/117.1
V	Motor	= =MY1+L1/117.2
W	Motor	= =MY1+L1/117.2

X13/X14

SH	shield	connection shield	=MY1+L1/117.6
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X14

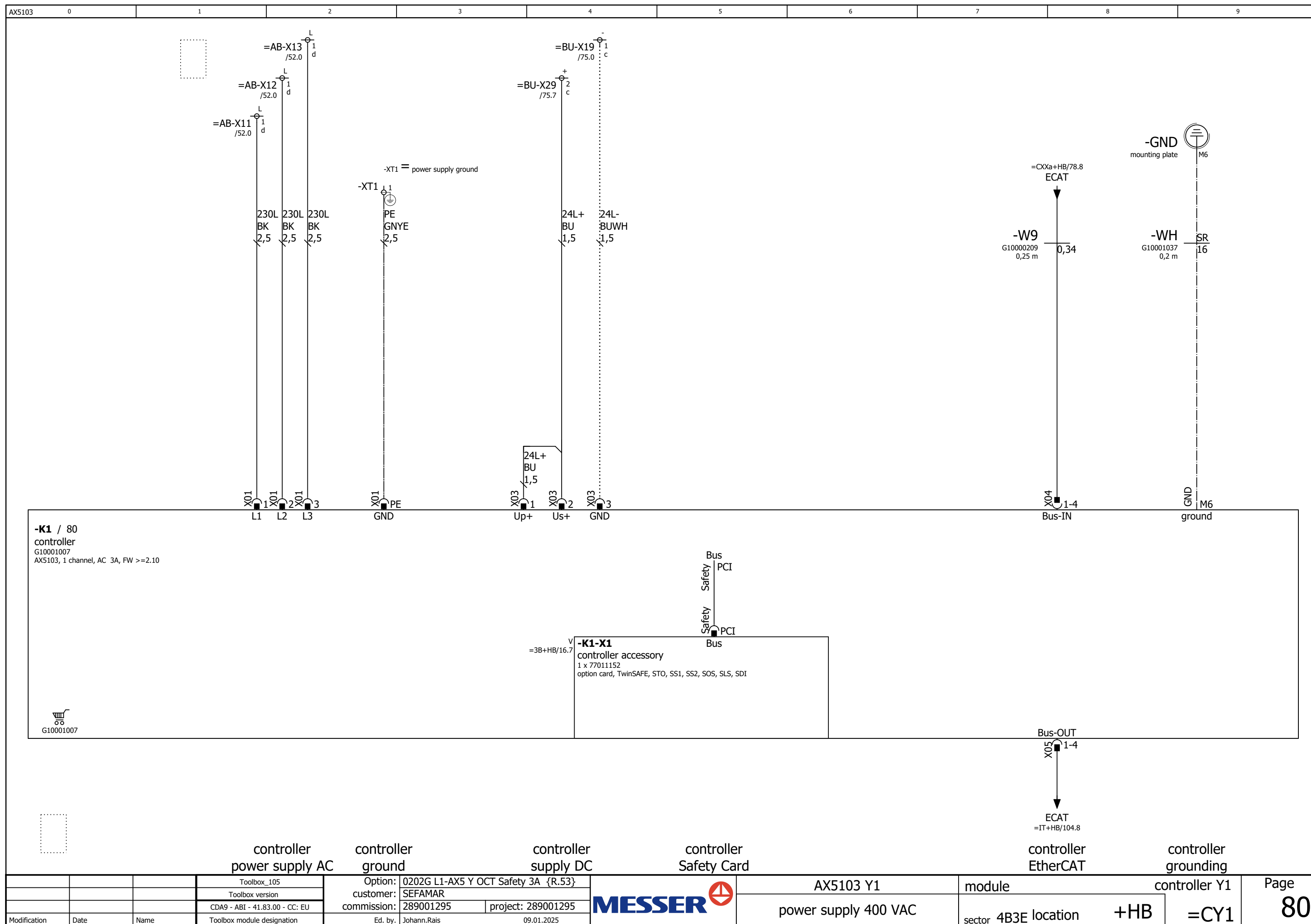
	Temp./brake	
B+	brake + connection #	=MY1+L1/117.3
B-	brake GND	=MY1+L1/117.4
PE	shield	
T+	Temp.+ connection encoder	=MY1+L1/117.5
T-	Temp.-	=MY1+L1/117.5

Safety adress: HEX -<64> / Dez: <100>

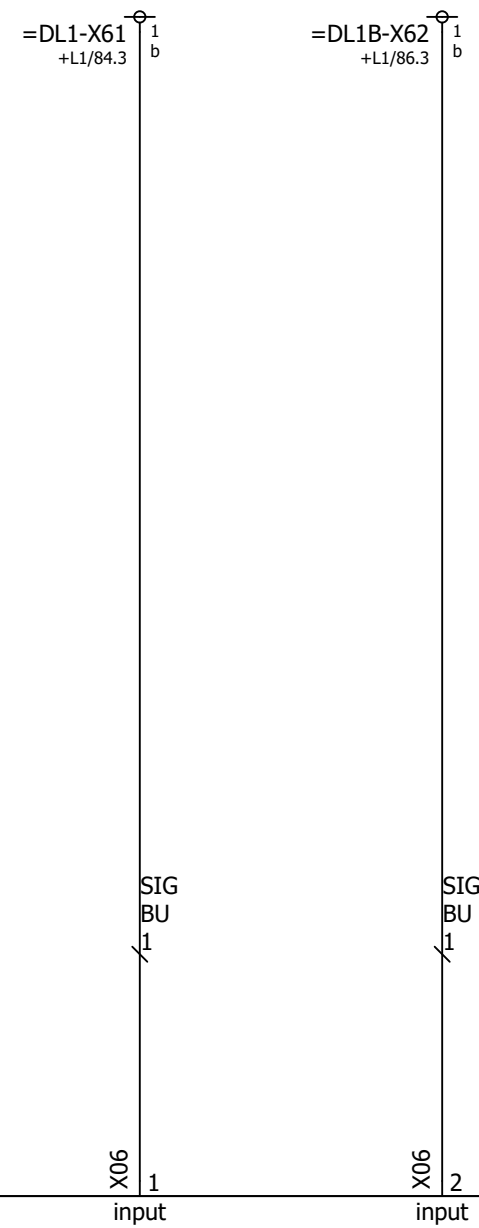
ON

		X			X	X			
X	X		X	X			X	X	X
1	2	3	4	5	6	7	8	9	10

			Toolbox_105	Option: customer: commission:			MESSER		module		controller Y1		Page 79
			Toolbox version		SEFAMAR								
			automated		289001295	project: 289001295							
Modification	Date	Name	evaluation has been generated	Ed. bv.	Johann.Rais	09.01.2025		rack overview	sector	location	+HB	=CY1	



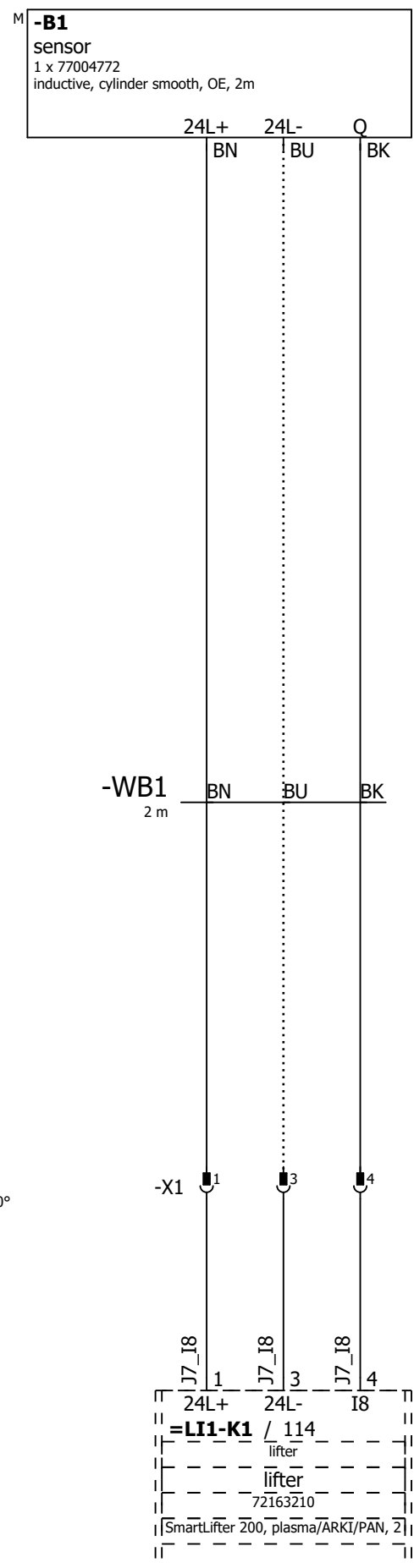
0	1	2	3	4	5	6	7	8	9
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=CY1-K1 / 80
controller
G10001007
AX5103, 1 channel, AC 3A, FW >=2.10

working area beam Y +	working area beam Y -
--------------------------	--------------------------

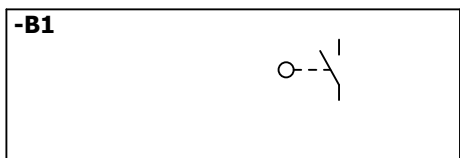
			Toolbox_105	Option: customer: commission:	0202G L1-AX5 Y OCT Safety 3A {R.53}			AX5103 Y1		module		controller Y1		Page 81
			Toolbox version		SEFAMAR			entrances		sector 4B3E location +HB		=CY1		
			CX2* - ABI - 31.32.00 - CC: EU		289001295			project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025								



5	
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			Toolbox_105	Option: customer: commission:	1301A L1-Havarie, Torchholder, Magnetic			torch magnetic holder		module		detection collision		Page 82		
			Toolbox version		SEFAMAR											
			DC1 - ABI - 52.99.00 - CC: EU		289001295			project: 289001295								
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais 09.01.2025			sensor, input		sector CA2L location		+L1 =DC1				

0	1	2	3	4	5	6	7	8	9
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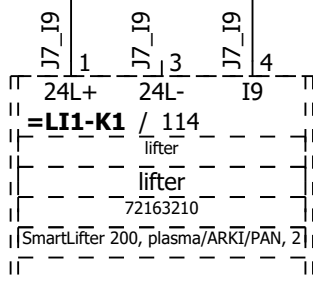
BU

BN

BU

() 1

4

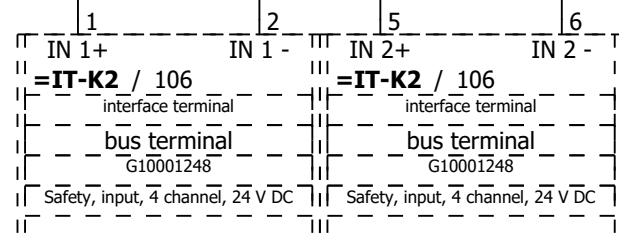
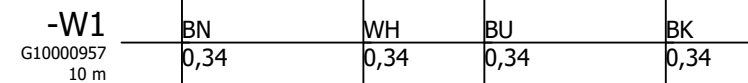
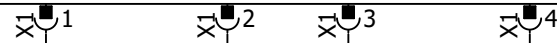
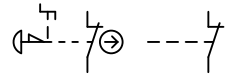
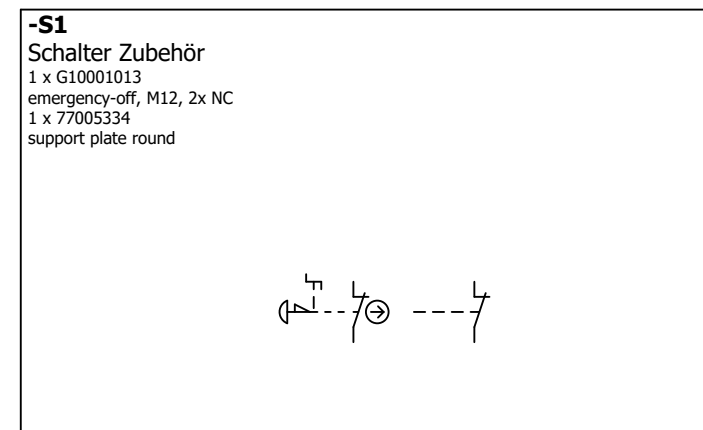


tactile switch

			Toolbox_105	Option: customer: commission:	1301B L1-Havarie tactil lift up {R.34}			torch dig switch tactile	module	detection collision	Page 83	
			Toolbox version		SEFAMAR			sensor, input	sector CA2L location	+L1		=DC1B
			DC2 - ABI - 31.99.00 - CC: EU		289001295 project: 289001295							
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais 09.01.2025							

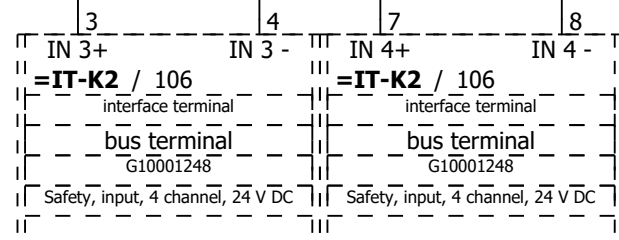
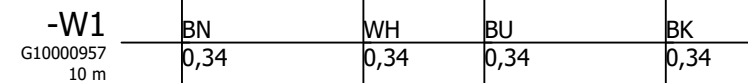
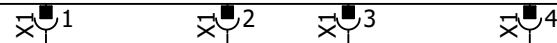
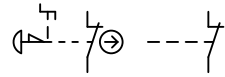
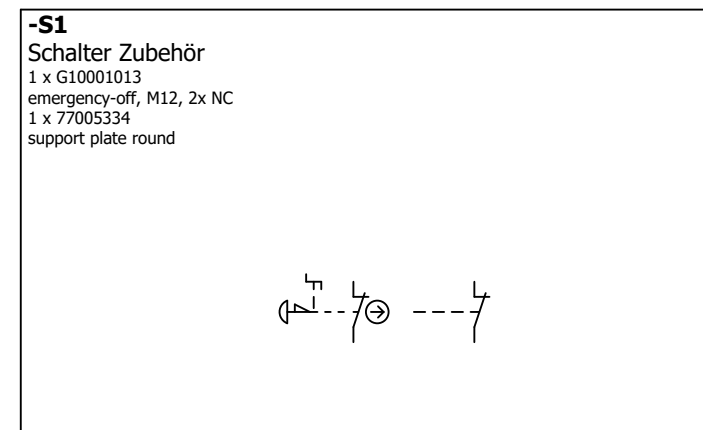
			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			limit Y+	module	detection limiting	Page 84	
			Toolbox version	customer:	SEFAMAR			connection overview	sector CA2L location	+L1		=DL1
			DL12 - ABI - No Rev - CC: EU	commission:	289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025						

			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			limit Y-		module		detection limiting		Page 86
			Toolbox version	customer:	SEFAMAR			connection overview						
			DL12 - ABI - No Rev - CC: EU	commission:	289001295	project: 289001295				sector CA2L location		+L1	=DL1B	
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025								



emergency stop

			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			emergency stop	module	emergency stop additions	Page 88
			Toolbox version	customer:	SEFAMAR			connection	sector 4C1E location	+E =EA	
			EA1 - ABI - 33.98.00 - CC: EU	commission:	289001295	project: 289001295					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025					



emergency stop

			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			emergency stop	module	emergency stop additions	Page 89	
			Toolbox version	customer:	SEFAMAR			connection	sector 1C1G location	+G		=EAB
			EA1 - ABI - 33.98.00 - CC: EU	commission:	289001295	project: 289001295						
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025						

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

-S1
switch
1 x 77005253
position, M12, 2 NC, spring rod

-S2
switch
1 x 77005253
position, M12, 2 NC, spring rod

-S3
switch
1 x 77005253
position, M12, 2 NC, spring rod

-S4
switch
1 x 77005253
position, M12, 2 NC, spring rod

-W1
G10000965
5 m

-W2
G10000965
5 m

+

B=IBB-K1 / 101

interface box

distributor active

G10000100

EtherCAT Safety box, IN, 4 channel,

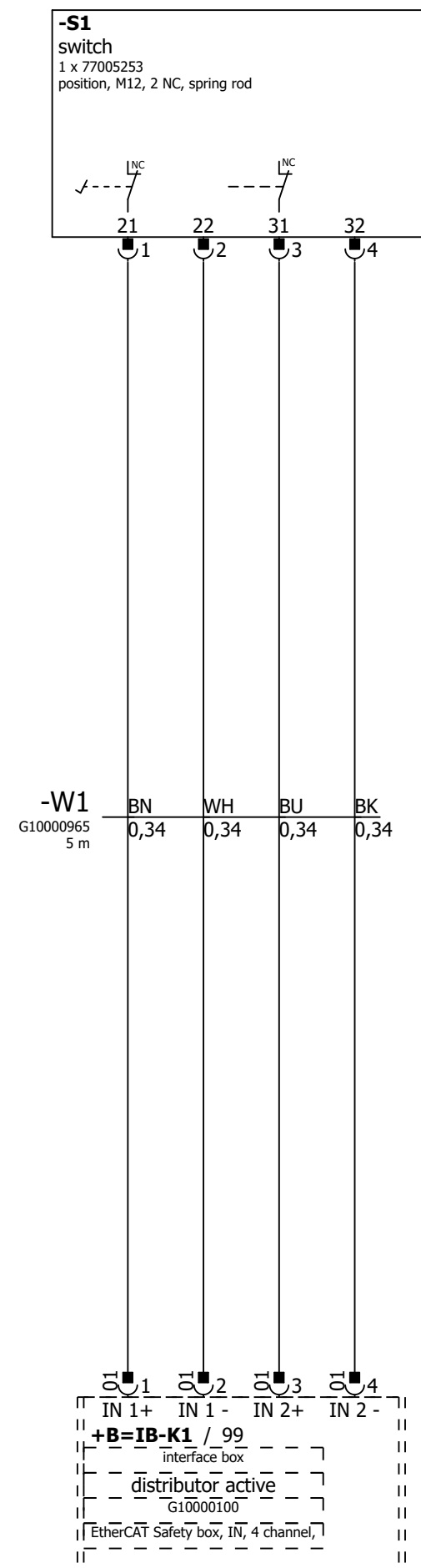
foot safety trigger rear inside

foot safety trigger rear outside

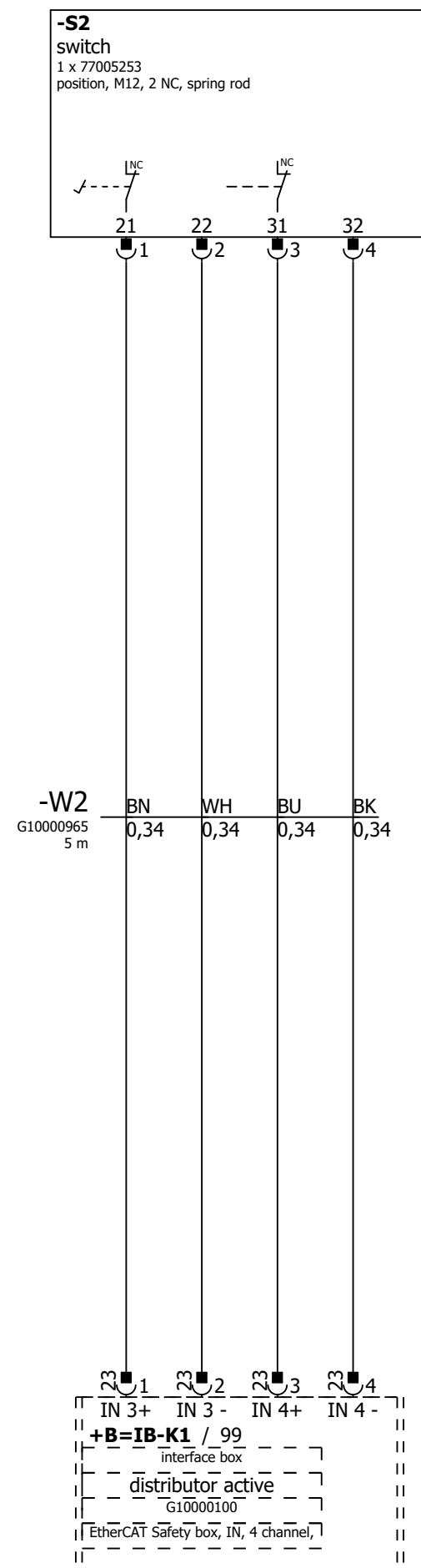
foot safety trigger front outside

foot safety trigger front inside

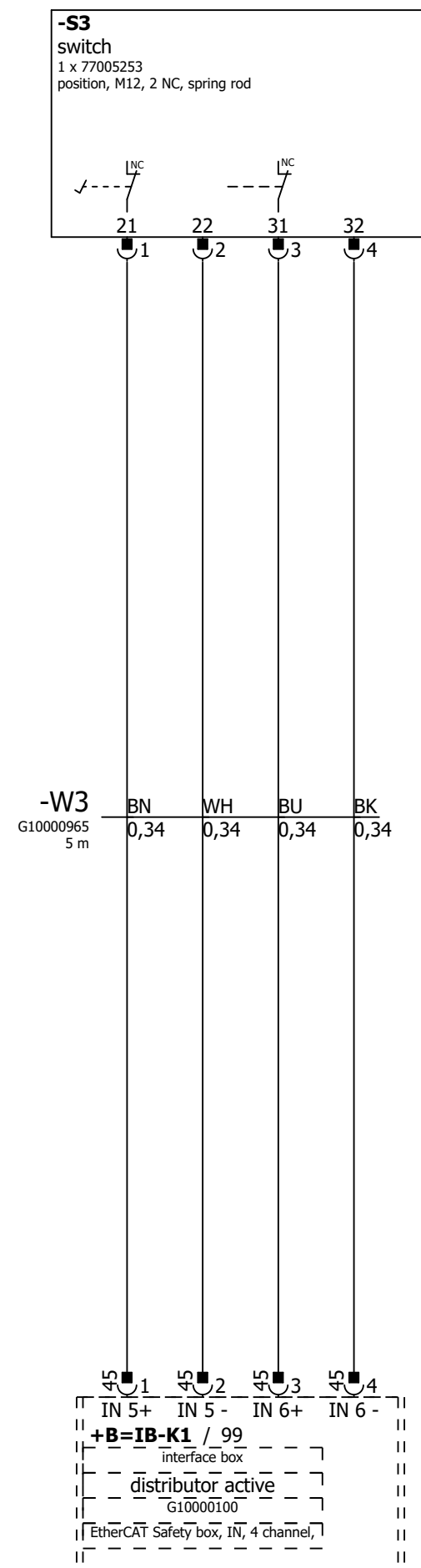
			Toolbox_105		Option: customer:		1204A Foot switch, endtruck, 2x2	MESSER	module emergency stop foot protection		Page
			Toolbox version		commission:		SEFAMAR				90
			EF2 - ABI - 31.33.01 - CC: EU		project:		289001295 289001295				
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	09.01.2025			sensor, safety EP box	sector 3B1E location +E =EF	



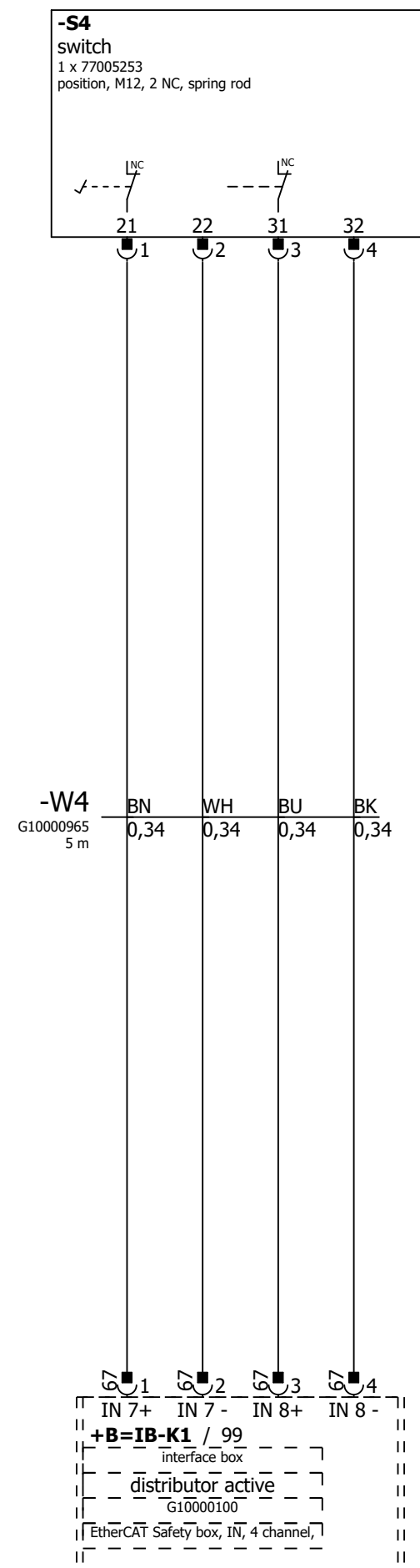
foot safety trigger
rear inside



foot safety trigger
rear outside



foot safety trigger
front outside



foot safety trigger
front inside

			Toolbox_105	Option: customer: commission:	1204A Foot switch, endtruck, 2x2				module emergency stop foot protection		Page 91
			Toolbox version		SEFAMAR						
			EF2 - ABI - 31.33.01 - CC: EU		289001295	project: 289001295					
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	09.01.2025		sensor, safety EP box	sector 2B1G location +G =EFB		



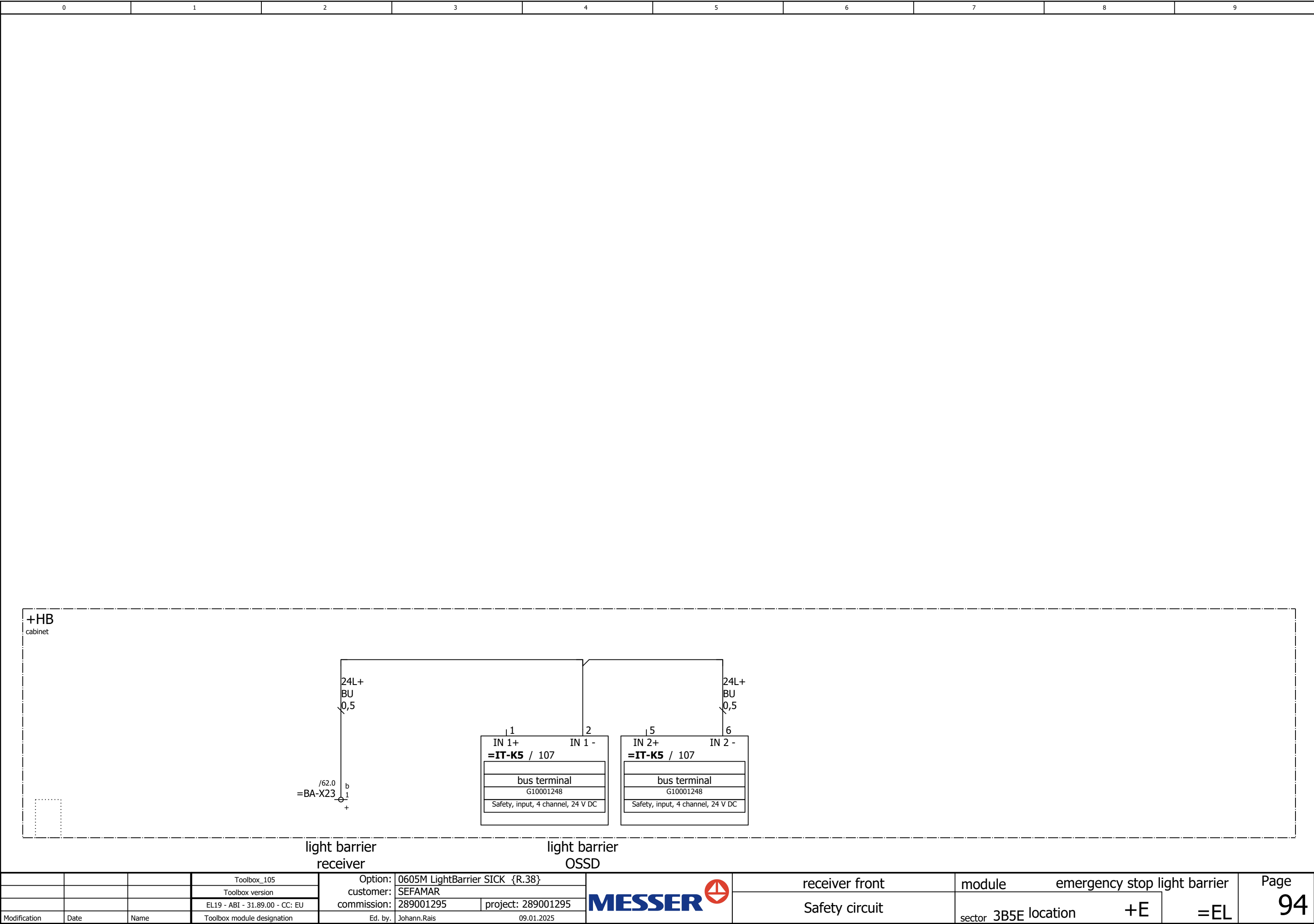
relay

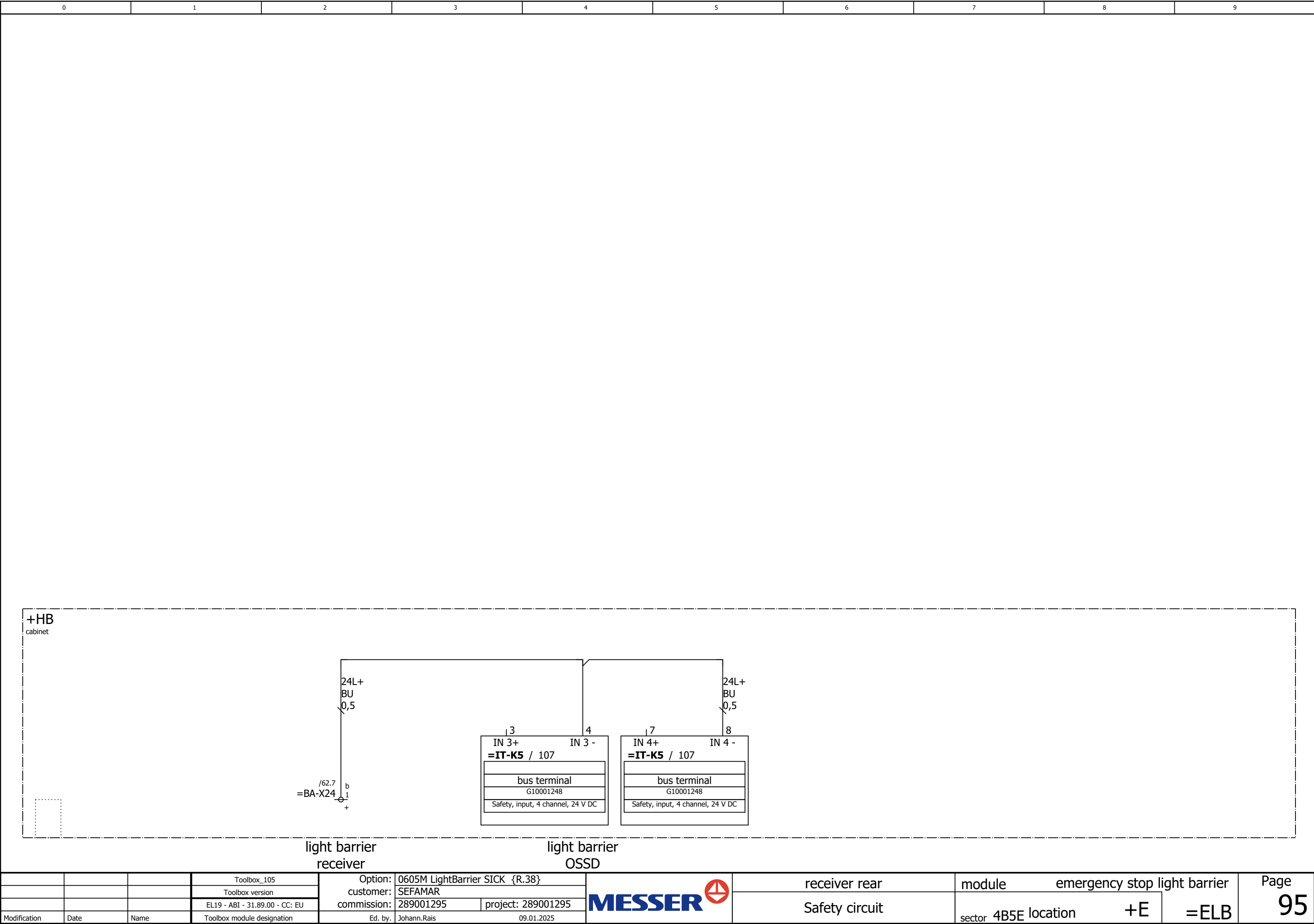
-F1

extension module, 24V DC

72a		
72b		
A1a	Safety interface external	/93.1
A1b		
A2a	Safety interface external	/93.1
A2b		

Logic		
11a	Safety interface input	/93.2
11b		
12a	Safety interface input	/93.2
12b		
23a	plasma emergency stop	=PS1+P/136.4
23b		
24a	plasma emergency stop	=PS1+P/136.6
24b		=PS1+P/136.6
33a		
33b		
34a		
34b		
43a		
43b		
44a		
44b		
53a		
53b		
54a		
54b		
63a		
63b		
64a		
64b		
71a		
71b		







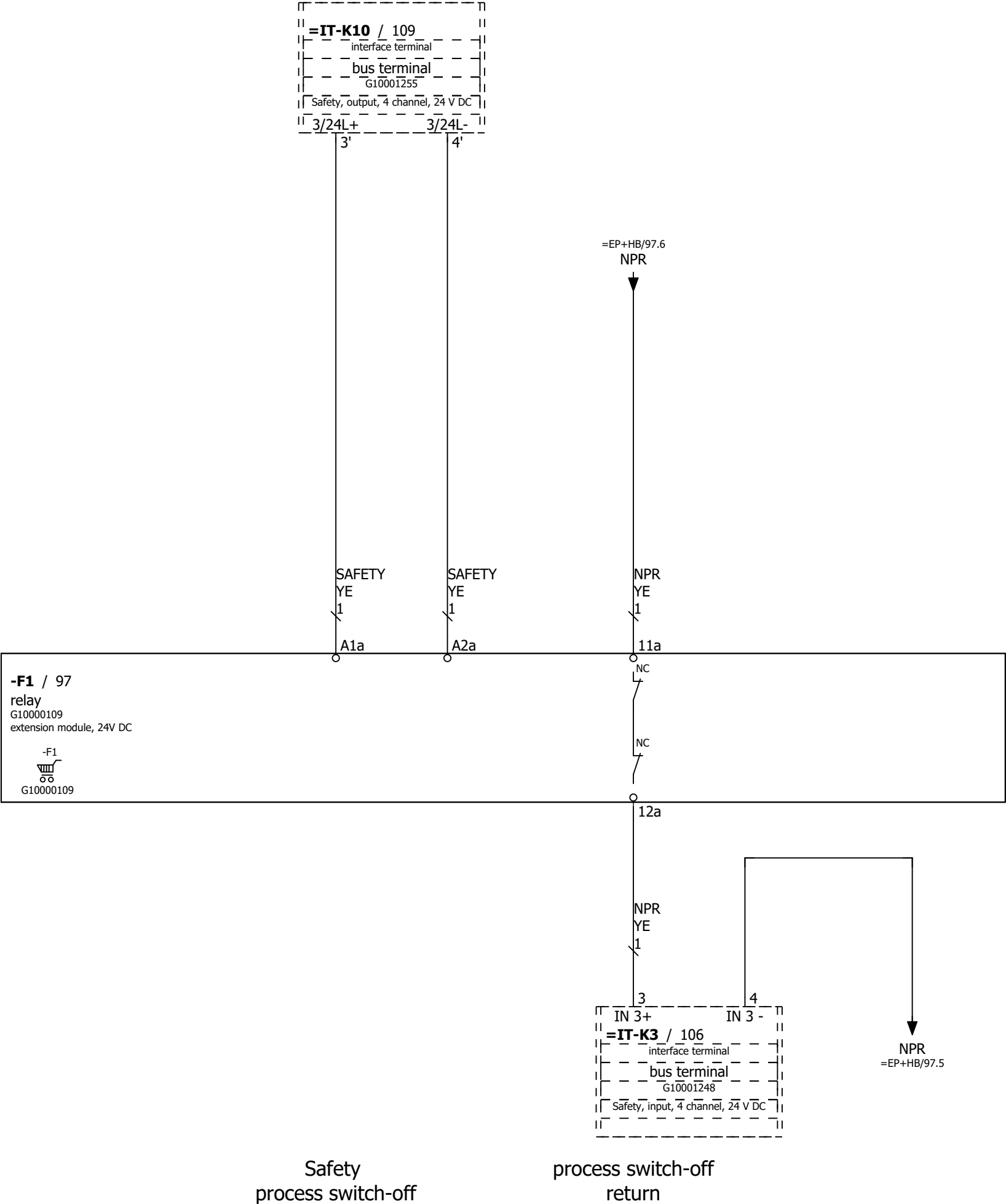
G10000109

relay

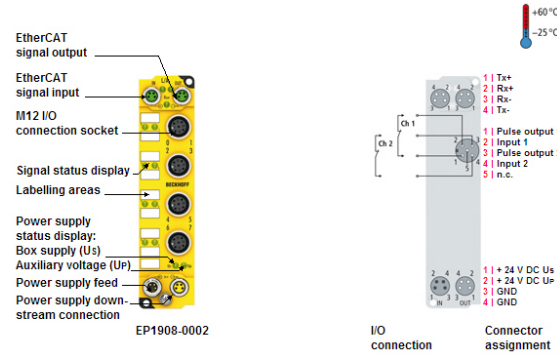
-F1 extension module, 24V DC

72a		
72b		
A1a	Safety process switch-off	/97.3
A1b		
A2a	Safety process switch-off	/97.4
A2b		

Logic		
11a	process switch-off return	/97.5
11b		
12a	process switch-off return	/97.5
12b		
23a	Safety interface external	=EI1+HP1/93.1
23b		
24a	Safety interface external	=EI1+HP1/93.2
24b		
33a		
33b		
34a		
34b		
43a		
43b		
44a		
44b		
53a		
53b		
54a		
54b		
63a		
63b		
64a		
64b		
71a		
71b		



Digital Input EP1908-0002



G10000100
distributor active
-K1 EtherCAT Safety box, IN, 4 channel, M12

01

IN 01		
1	IN 1+	foot safety trigger rear inside =EFB+G/91.1
2	IN 1 -	=EFB+G/91.2
IN 02		
3	IN 2+	foot safety trigger rear inside =EFB+G/91.2
4	IN 2 -	= =EFB+G/91.2

23

IN 03		
1	IN 3+	foot safety trigger rear outside =EFB+G/91.3
2	IN 3 -	=EFB+G/91.4
IN 04		
3	IN 4+	foot safety trigger rear outside =EFB+G/91.4
4	IN 4 -	= =EFB+G/91.4

24V IN

Power IN		
1	24L+s	/99.1
2	24L+p	EP module supply /99.2
3	24L-s	= /99.3
4	24L-p	= /99.3

24V OUT

Power OUT		
1	24L+s	=IBB/101.1
2	24L+p	EP module supply =IBB/101.2
3	24L-s	= =IBB/101.3
4	24L-p	= =IBB/101.3

45

IN 05		
1	IN 5+	foot safety trigger front outside =EFB+G/91.6
2	IN 5 -	=EFB+G/91.6
IN 06		
3	IN 6+	foot safety trigger front outside =EFB+G/91.6
4	IN 6 -	= =EFB+G/91.6

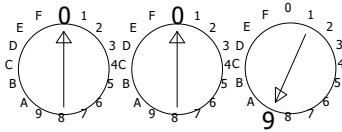
67

IN 07		
1	IN 7+	foot safety trigger front inside =EFB+G/91.8
2	IN 7 -	= =EFB+G/91.8
IN 08		
3	IN 8+	foot safety trigger front inside =EFB+G/91.8
4	IN 8 -	= =EFB+G/91.8

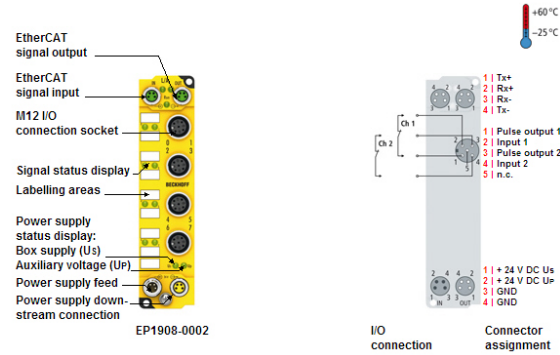
L/A

Bus		
IN	Bus	EtherCAT /99.8
OUT	Bus	= /99.8

Safety adress: HEX -<9> / Dez: <9>



Digital Input EP1908-0002



G10000100

distributor active

-K1

EtherCAT Safety box, IN, 4 channel, M12

01			
IN 01			
1	IN 1+	foot safety trigger rear inside	=EF+E/90.1
2	IN 1 -		=EF+E/90.2
IN 02			
3	IN 2+	foot safety trigger rear inside	=EF+E/90.2
4	IN 2 -		=EF+E/90.2

23			
IN 03			
1	IN 3+	foot safety trigger rear outside	=EF+E/90.3
2	IN 3 -		=EF+E/90.4
IN 04			
3	IN 4+	foot safety trigger rear outside	=EF+E/90.4
4	IN 4 -		=EF+E/90.4

24V IN			
Power IN			
1	24L+s		/101.1
2	24L+p	EP module supply	/101.2
3	24L-s		= /101.3
4	24L-p		= /101.3

24V OUT			
Power OUT			
1	24L+s		
2	24L+p		
3	24L-s		
4	24L-p		

45			
IN 05			
1	IN 5+	foot safety trigger front outside	=EF+E/90.6
2	IN 5 -		=EF+E/90.6
IN 06			
3	IN 6+	foot safety trigger front outside	=EF+E/90.6
4	IN 6 -		=EF+E/90.6

67			
IN 07			
1	IN 7+	foot safety trigger front inside	=EF+E/90.8
2	IN 7 -		=EF+E/90.8
IN 08			
3	IN 8+	foot safety trigger front inside	=EF+E/90.8
4	IN 8 -		=EF+E/90.8

L/A			
Bus			
IN	Bus	EtherCAT	/101.8
OUT	Bus		= /101.8

Safety adress: HEX -<8> / Dez: <8>

0

1

2

3

4

5

6

7

8

9

A

B

C

D

E

F

0

1

2

3

4

5

6

7

8

9

A

B

C

D

E

F

0

1

2

3

4

5

6

7

8

9

A

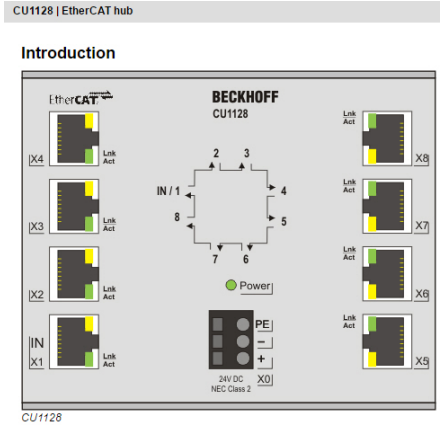
B

C

D

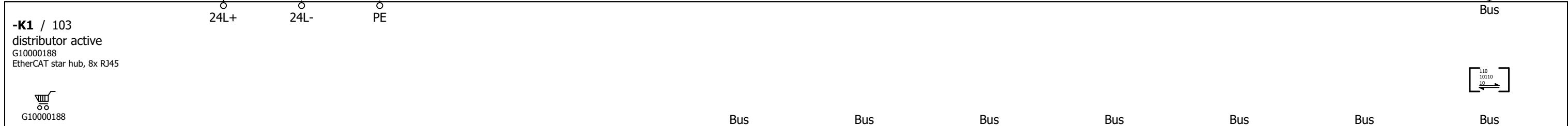
E

F



G10000188
distributor active
-K1 EtherCAT star hub, 8x RJ45

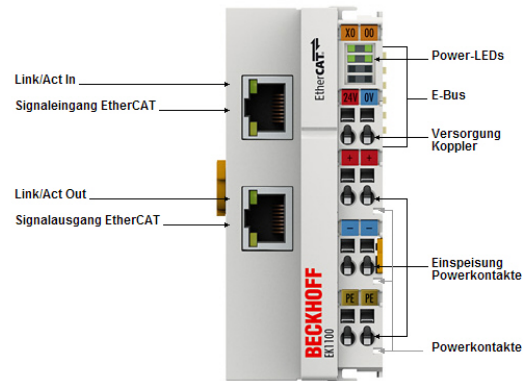
X0			
	supply		
1	24L+	star hub supply	/103.1
2	24L-	=	/103.2
3	PE	=	/103.2
X1			
	Bus		
IN	Bus	EtherCAT	/103.9
X2			
OUT	Bus	EtherCAT distribution	/103.4
X3			
OUT	Bus	EtherCAT distribution	/103.5
X4			
OUT	Bus	EtherCAT distribution	/103.6
X5			
OUT	Bus	EtherCAT distribution	/103.7
X6			
OUT	Bus	EtherCAT distribution	/103.7
X7			
OUT	Bus	EtherCAT distribution	/103.8
X8			
OUT	Bus	EtherCAT	/103.9



EtherCAT

			Toolbox_105	Option:	0411B Basic Parts TSc CE {R.54}			star hub		module		interface control		Page 103
			Toolbox version	customer:	SEFAMAR			power supply, EtherCat		sector 4B3E location		+HB		
			IC1 - ABI - 32.102.00 - CC: EU	commission:	289001295	project: 289001295						=IC		
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025								

EtherCAT EK1100



G13000078

bus terminal

-K1

bus coupler

supply

1 24L+ EtherCAT coupler operating supply /104.5

power contacts

2 + EtherCAT coupler powerbus /104.6

3 - = /104.7

4 PE

supply

5 24L- EtherCAT coupler operating supply /104.6

power contacts

6 +

7 -

8 PE

X1

Bus

IN BUS EtherCAT coupler Bus /104.8

X2

OUT BUS EtherCAT coupler Bus /104.8

Safety Address: HEX -<>/ Dec: <>



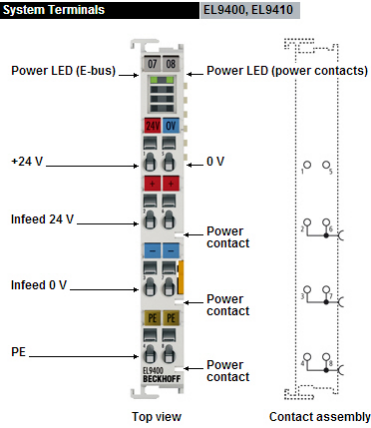
EtherCAT coupler



EtherCAT coupler
operating supply

EtherCAT coupler
powerbus

EtherCAT coupler
Bus



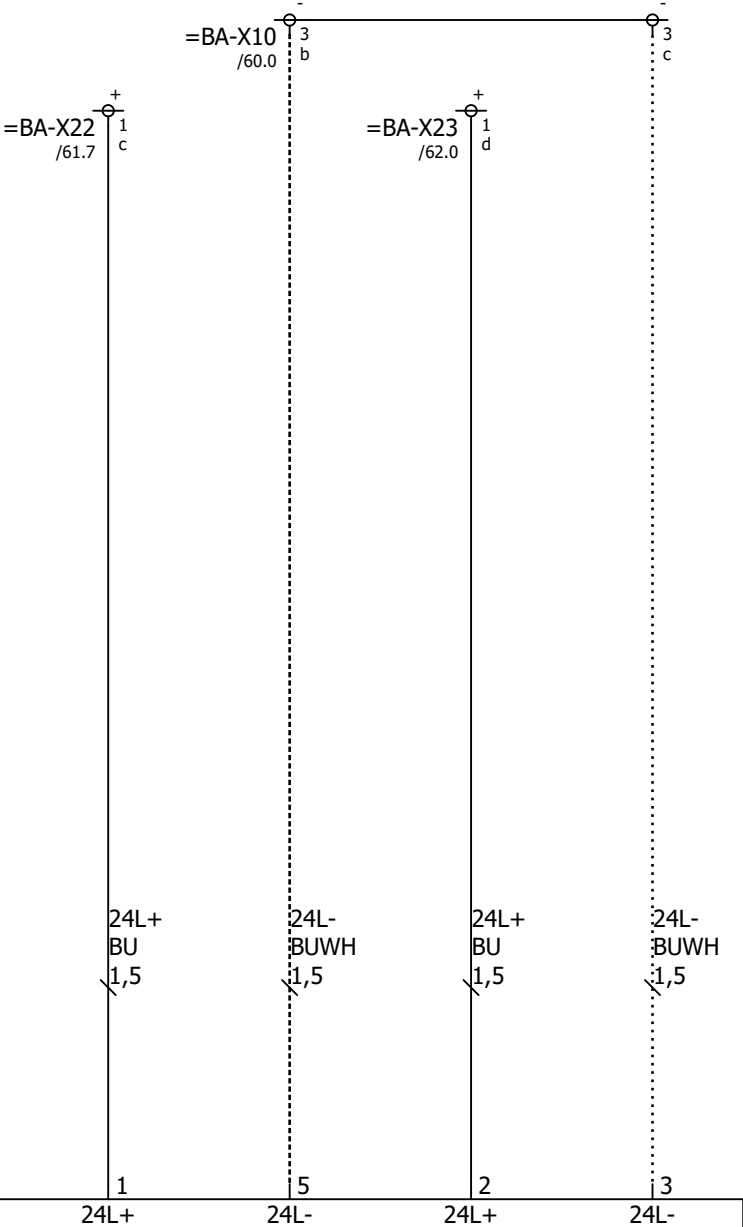
G13000079

bus terminal

-K6 power supply terminal, E-bus refreshment

supply		
1	24L+ EtherCAT operating supply	/105.5
power contacts		
2	24L+ EtherCAT powerbus	/105.6
3	24L- =	/105.7
4	PE	
supply		
5	24L- EtherCAT operating supply	/105.6
power contacts		
6	24L+	
7	24L-	
8	PE	

Safety Address: HEX -<>/ Dec: <>



-K6 / 105

bus terminal

G13000079

power supply terminal, E-bus refresh

[⚡]

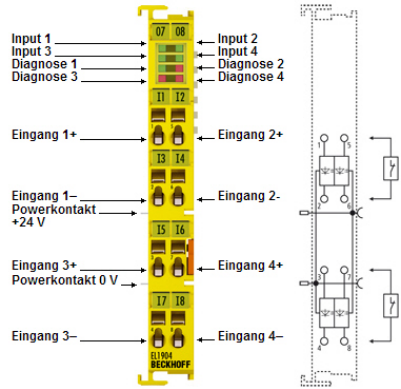


power supply terminal
EtherCAT

EtherCAT
operating supply

EtherCAT
powerbus

Digital-Eingang EL1904



G10001248

bus terminal

-K2 Safety, input, 4 channel, 24 V DC

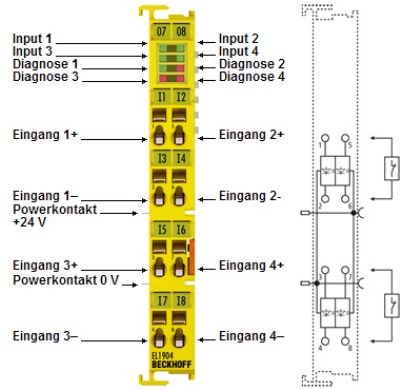
IN 01		
1	IN 1+	emergency stop =EA+E/88.0
2	IN 1 -	= =EA+E/88.1
IN 02		
5	IN 2+	emergency stop =EA+E/88.1
6	IN 2 -	= =EA+E/88.2
IN 03		
3	IN 3+	emergency stop =EAB+G/89.0
4	IN 3 -	= =EAB+G/89.1
IN 04		
7	IN 4+	emergency stop =EAB+G/89.1
8	IN 4 -	= =EAB+G/89.2

Safety Address: HEX -/ Dec: <11>

ON

X	X	X	X				
		X	X	X	X	X	X
1	2	3	4	5	6	7	8

Digital-Eingang EL1904



G10001248

bus terminal

-K3 Safety, input, 4 channel, 24 V DC

IN 01		
1	IN 1+	
2	IN 1 -	
IN 02		
5	IN 2+	
6	IN 2 -	
IN 03		
3	IN 3+	process switch-off return =EP/97.5
4	IN 3 -	=EP/97.5
IN 04		
7	IN 4+	
8	IN 4 -	

Safety Address: HEX -<d>/ Dec: <13>

ON

X	X	X	X				
	X		X	X	X	X	X
1	2	3	4	5	6	7	8

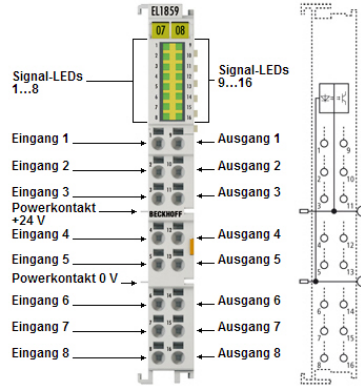


G10001248



G10001248

Digital-Eingang EL1859



G10000441

bus terminal

-K4 digital, E/A, 2 x 8 channel, 24 VDC

I1	
1	I1 potential distribution limiting =DL1+L1/85.4
I2	
2	I2 locator limit Y- =DL1B+L1/87.4
I3	
3	I3 control voltage turn on =OP+G/125.6
I4	
4	I4 machine start =OP+G/125.7
I5	
5	I5 machine Stop =OP+G/125.8
I6	
6	I6
I7	
7	I7
I8	
8	I8
O1	
9	O1
O2	
10	O2
O3	
11	O3 operator action request =OK+HOK/121.7
O4	
12	O4 signal pointer =OK+HOK/121.9

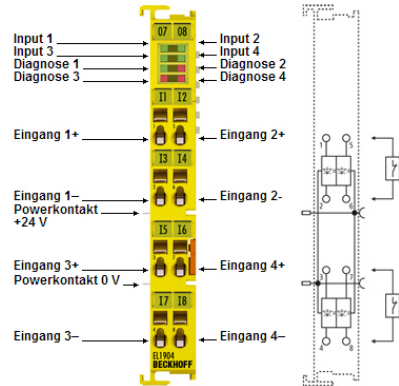


G10000441

O5	
13	O5 machine in start =OP+G/126.3
O6	
14	O6 machine in Stop =OP+G/126.2
O7	
15	O7 =OP+G/126.1
O8	
16	O8

Safety Address: HEX -<>/ Dec: <>

Digital-Eingang EL1904



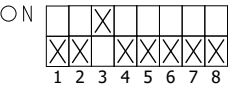
G10001248

bus terminal

-K5 Safety, input, 4 channel, 24 VDC

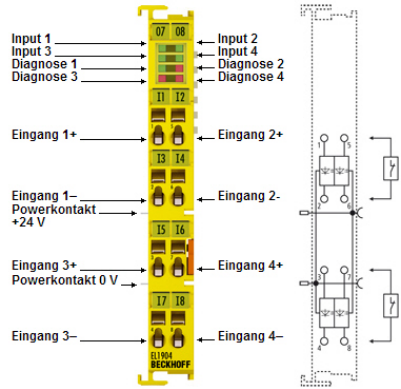
IN 01	
1	IN 1+ light barrier OSSD =EL+E/94.3
2	IN 1 - =EL+E/94.4
IN 02	
5	IN 2+ light barrier OSSD =EL+E/94.4
6	IN 2 - =EL+E/94.5
IN 03	
3	IN 3+ light barrier OSSD =ELB+E/95.3
4	IN 3 - =ELB+E/95.4
IN 04	
7	IN 4+ light barrier OSSD =ELB+E/95.4
8	IN 4 - =ELB+E/95.5

Safety Address: HEX -<4>/ Dec: <4>



G10001248

Digital-Eingang EL1904



G10001248

bus terminal

-K7

Safety, input, 4 channel, 24 V
DC

IN 01

1	IN 1+	light barrier fade out	=OK+HOK/121.4
2	IN 1 -	=	=OK+HOK/121.2

IN 02

5	IN 2+	light barrier fade out	=OK+HOK/121.3
6	IN 2 -	=	=OK+HOK/121.3

IN 03

3	IN 3+		
4	IN 3 -		

IN 04

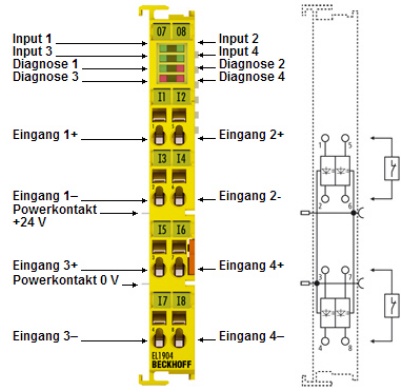
7	IN 4+		
8	IN 4 -		

Safety Address: HEX -<25>/ Dec: <37>

ON

☒	☒	☒	☒	☒	☒	☒	☒
1	2	3	4	5	6	7	8

Digital-Eingang EL1904



G10001248

bus terminal

-K8

Safety, input, 4 channel, 24 V
DC

IN 01

1	IN 1+	machine reset	=OP+G/125.3
2	IN 1 -		=OP+G/125.4

IN 02

5	IN 2+	panel Safety channel	=OP+G/125.1
6	IN 2 -	=	=OP+G/125.1

IN 03

3	IN 3+	panel Safety channel	=OP+G/125.2
4	IN 3 -	=	=OP+G/125.2

IN 04

7	IN 4+	monitoring safety circuit activate	=OK+HOK/121.6
8	IN 4 -	=	=OK+HOK/121.7

Safety Address: HEX -<24>/ Dec: <36>

ON

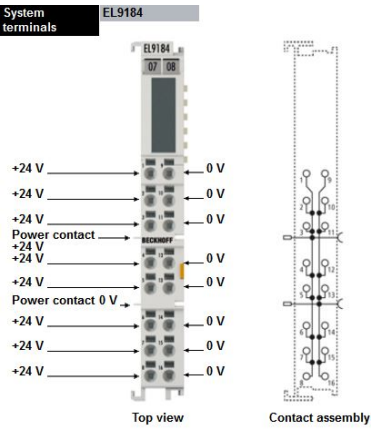
☒	☒	☒	☒	☒	☒	☒	☒
1	2	3	4	5	6	7	8



G10001248

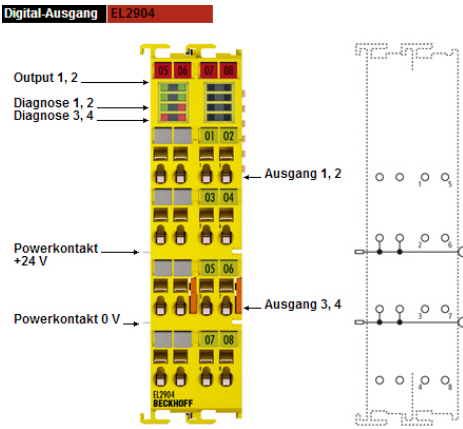


G10001248



	supply-13
13	-
	supply-14
14	-
	supply-15
15	-
	supply-16
16	-

Safety Address: HEX -<>/ Dec: <>



G10001255

bus terminal

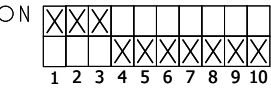
-K10

Safety, output, 4 channel, 24 V DC

	supply-1
1	+
	supply-2
2	+
	supply-3
3	+
	supply-4
4	+
	supply-5
5	+
	supply-6
6	+
	supply-7
7	+
	supply-8
8	+
	supply-9
9	- signal pointer =OK+HOK/121.9
	supply-10
10	- operator action request =OK+HOK/121.7
	supply-11
11	-
	supply-12
12	-

	Out 03
4'	3/24L- Safety process switch-off =EP/97.4
	Out 01
1'	1/24L+
2'	1/24L-
	Out 02
5'	2/24L+
6'	2/24L-
	Out 03
3'	3/24L+ Safety process switch-off =EP/97.3
	Out 04
7'	4/24L+
8'	4/24L-
	Power
2	24L+
3	24L-
6	24L+
7	24L-

Safety Address: HEX -<7>/ Dec: <7>

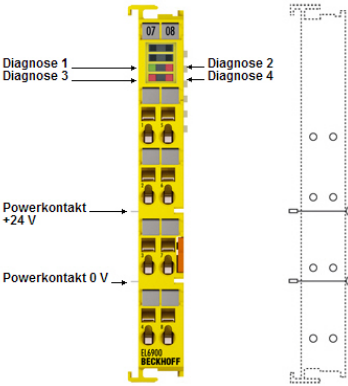


G10000699



G10001255

Kommunikation EL6900



G10001262

bus terminal

-K11 Safety, TwinSAFE-logic-terminal

1	n.c.
2	n.c.
3	n.c.
4	n.c.
5	n.c.
6	n.c.
7	n.c.
8	n.c.

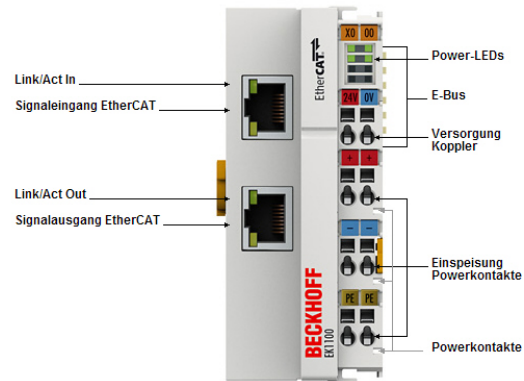
Safety Address: HEX -<2>/ Dec: <2>

ON

X									
X	X	X	X	X	X	X	X	X	X
1	2	3	4	5	6	7	8	9	10



EtherCAT EK1100



G13000078

bus terminal

-K1

bus coupler

supply

1 24L+ EtherCAT coupler operating supply /111.5

power contacts

2 + EtherCAT coupler powerbus /111.6

3 - = /111.7

4 PE

supply

5 24L- EtherCAT coupler operating supply /111.6

power contacts

6 +

7 -

8 PE

X1

Bus

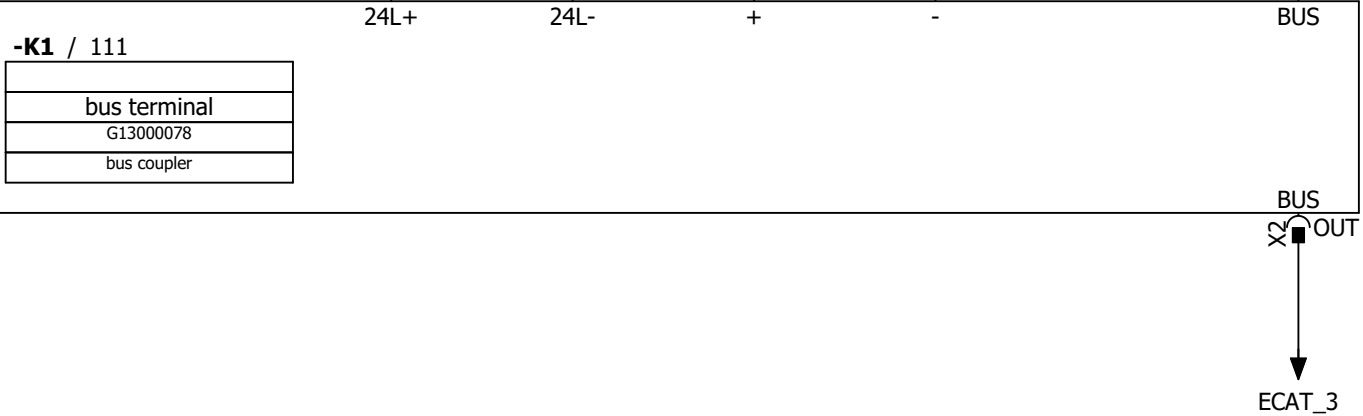
IN BUS EtherCAT coupler Bus /111.8

X2

OUT

BUS EtherCAT coupler Bus /111.8

Safety Address: HEX -<>/ Dec: <>



EtherCAT coupler

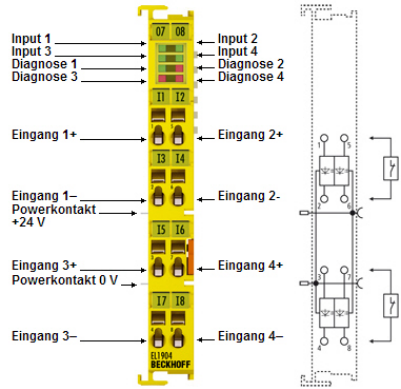


EtherCAT coupler
operating supply

EtherCAT coupler
powerbus

EtherCAT coupler
Bus

Digital-Eingang EL1904



G10001248

bus terminal

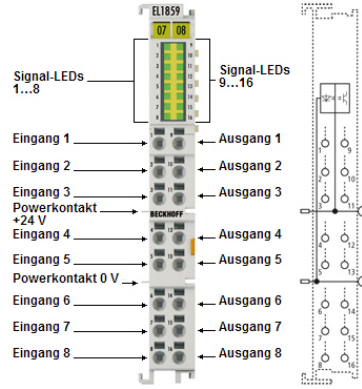
-K2 Safety, input, 4 channel, 24 V DC

IN 01	
1	IN 1+
2	IN 1 -
IN 02	
5	IN 2+
6	IN 2 -
IN 03	
3	IN 3+ Safety interface input =EI1/93.2
4	IN 3 - =EI1/93.3
IN 04	
7	IN 4+
8	IN 4 -

Safety Address: HEX -<50>/ Dec: <80>

ON				X	X		
	X	X	X	X	X	X	X
	1	2	3	4	5	6	7
							8

Digital-Eingang EL1859



G10000441

bus terminal

-K3 digital, E/A, 2 x 8 channel, 24 VDC

I1	
1	I1 exhaustion active =SP+P/138.3
I2	
2	I2
I3	
3	I3
I4	
4	I4
I5	
5	I5
I6	
6	I6
I7	
7	I7
I8	
8	I8
O1	
9	O1
O2	
10	O2
O3	
11	O3
O4	
12	O4

O5

13 O5

O6

14 O6

O7

15 O7

O8

16 O8

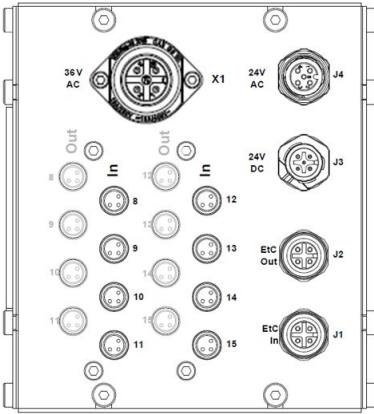
Safety Address: HEX -<>/ Dec: <>



G10001248



G10000441



72163210

lifter

-K1

SmartLifter 200, plasma/ARKI/PAN,
2 x 4 DI, 2 x 4 DO

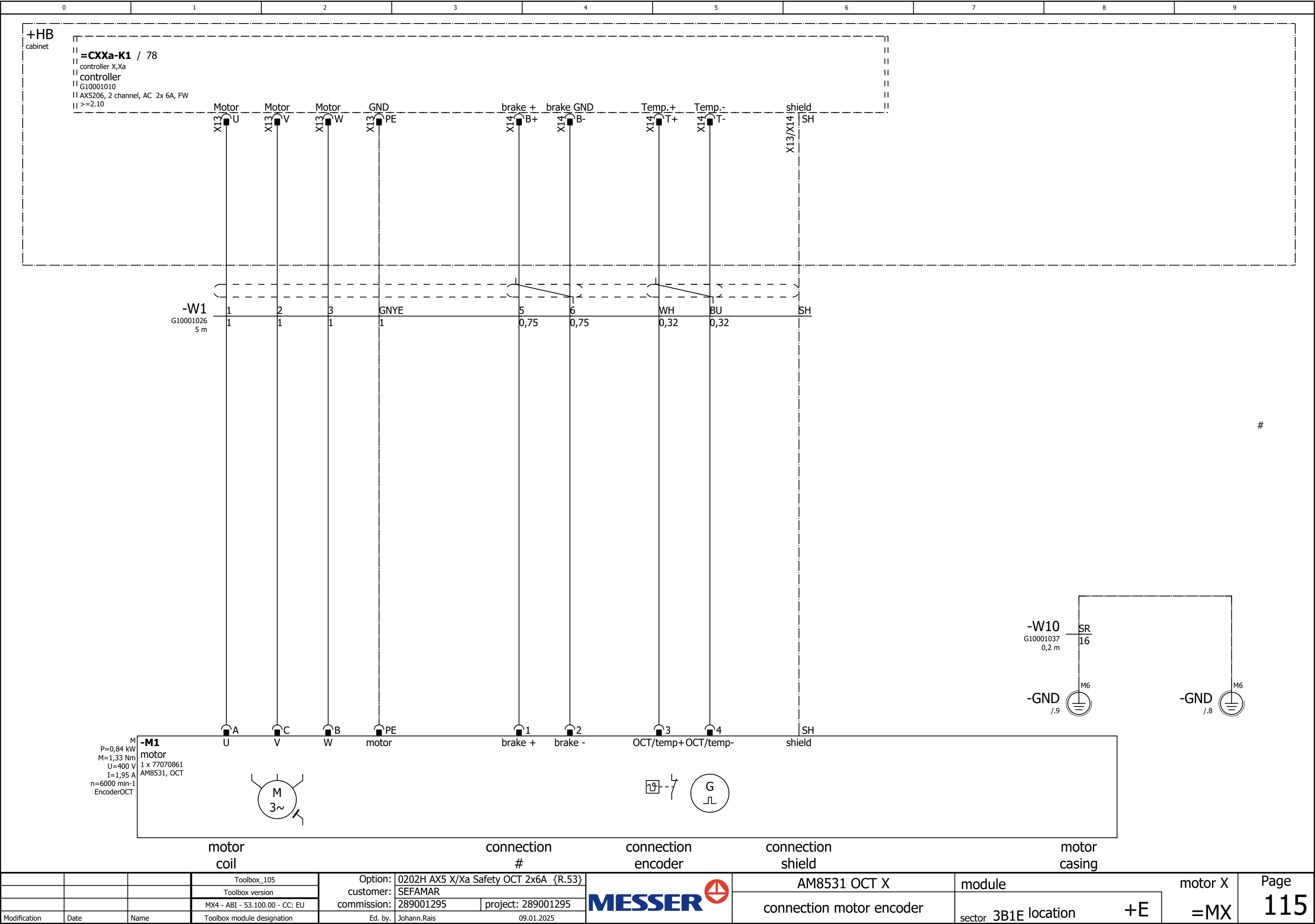
J5		PwrDrive		
3	36Ls		/114.3	
J1		Bus		
IN	BUS	SmartLifter EtherCAT	/114.9	
J2		OUT	BUS	SmartLifter EtherCAT /114.9
J3		PwrDC		
1	Us	SmartLifter DC operating voltage	/114.5	
2	Up	SmartLifter DC control voltage	/114.6	
3	GND	SmartLifter DC operating voltage	/114.7	
4	n.c	=	/114.7	
5	n.c		/114.8	
SH				
J4		PwrALFA		
1	24L			
2	nc			
3	24N			
4	nc			
5	nc			
SH				

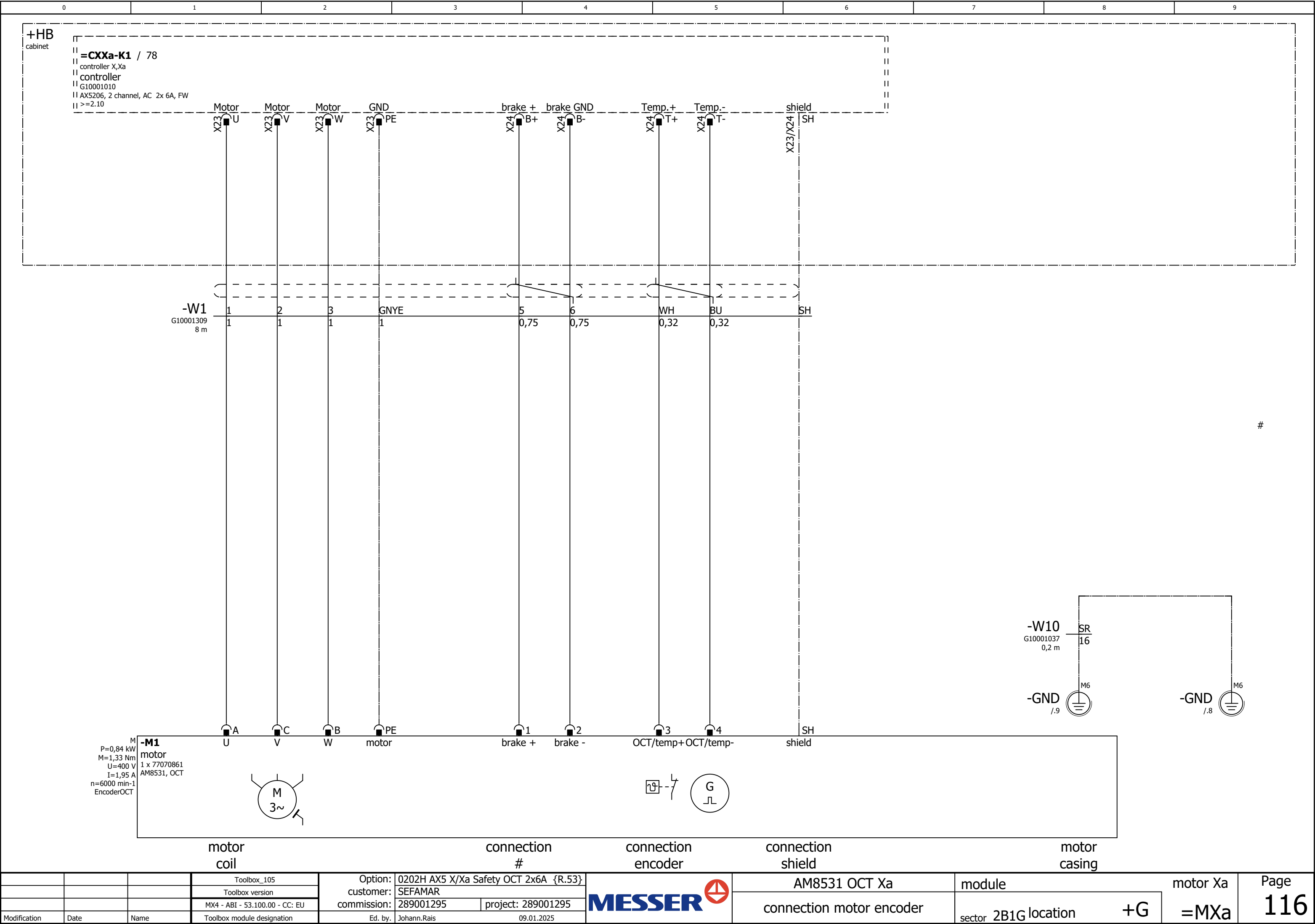
J5		PwrDrive		
1	36Lp	SmartLifter AC power supply	/114.3	
2	36N	=	/114.2	
4	PE	=	/114.1	
SH			/114.3	
J7_I8		I8		
1	24L+	magnet collision torch	=DC1/82.2	
3	24L-	=	=DC1/82.2	
4	I8	=	=DC1/82.2	
J7_I9		I9		
1	24L+	tactile switch	=DC1B/83.2	
3	24L-	=	=DC1B/83.2	
4	I9	=	=DC1B/83.2	
J7_I10		I10		
1	24L+			
3	24L-			
4	I10			
J7_I11		I11		
1	24L+			
3	24L-			
4	I11			
J7_I12		I12		
1	24L+			
3	24L-			
4	I12			
J7_I13		I13		
1	24L+			
3	24L-			
4	I13			
J7_I14		I14		
1	24L+			
3	24L-			

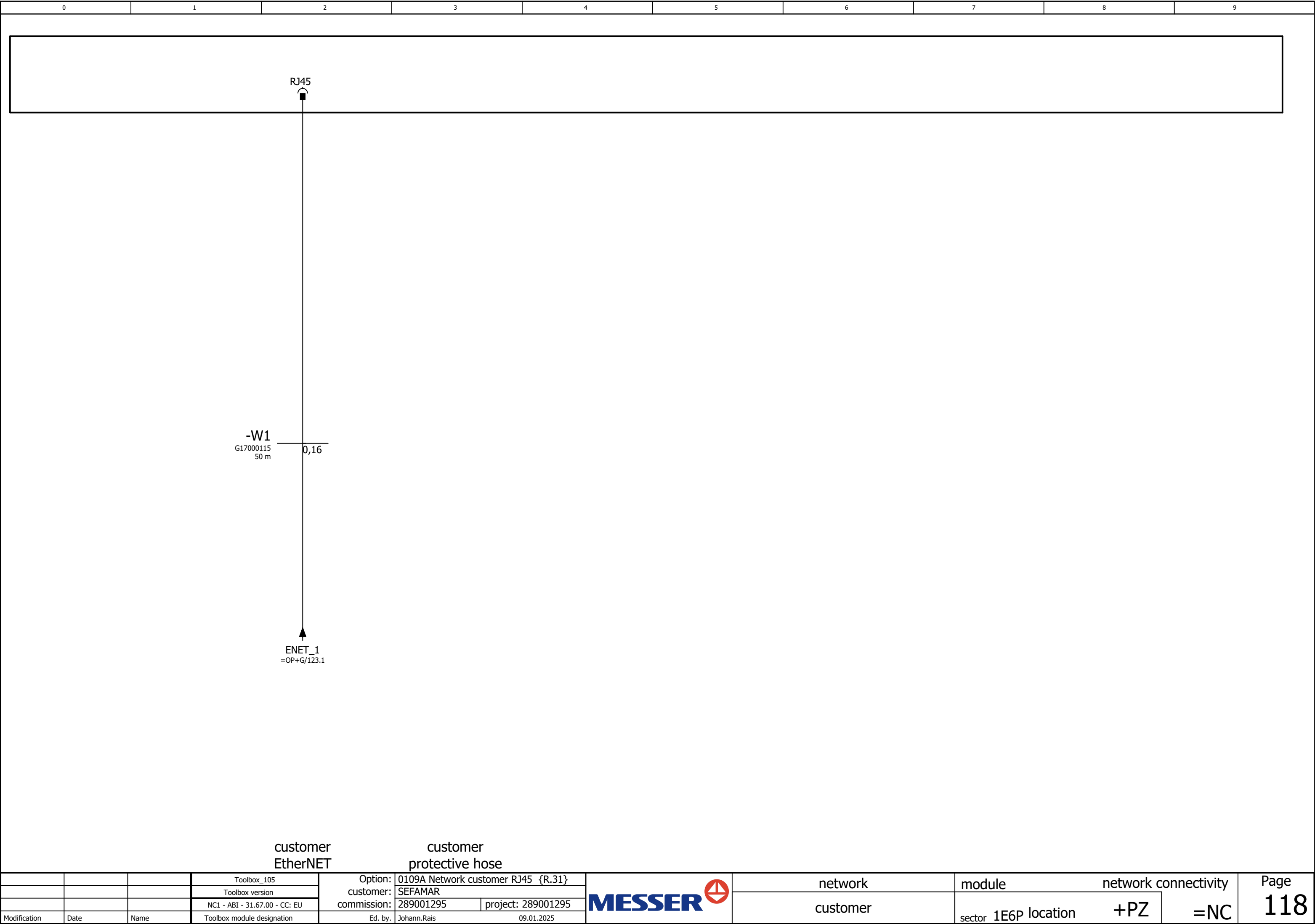
J7_I14		I14		
4	I14			
J7_I15		I15		
1	24L+			
3	24L-			
4	I15			
J7_O8		O8		
1	24L+	=UP1/141.2		
3	24L-	laser diode =UP1/141.2		
4	O8	=	=UP1/141.2	
J7_O9		O9		
1	24L+			
3	24L-			
4	O9			
J7_O10		O10		
1	24L+			
3	24L-			
4	O10			
J7_O11		O11		
1	24L+			
3	24L-			
4	O11			
J7_O12		O12		
1	24L+			
3	24L-			
4	O12			
J7_O13		O13		
1	24L+			
3	24L-			
4	O13			
J7_O14		O14		
1	24L+			

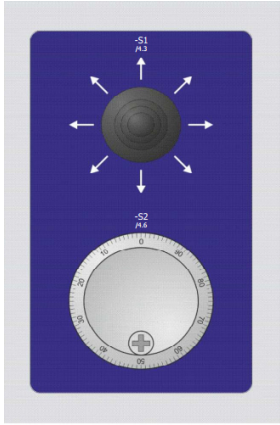
J7_O14		O14		
3	24L-			
4	O14			
J7_O15		O15		
1	24L+			
3	24L-			
4	O15			
J10_PAN		PAN		
1	+/-10V			
2	GND Signal			
7	24L-			
9	+/-1V			
11	collision			
15	24L+			
SH				
J11_ALFA		ALFA		
1				
2				
3				
4				
5				
6				
7				
9				
10				
11				
12				
13				
14				
15				
SH				

\$-FOS-F19-PLC









G10000038
control panel accessor
-S1 Joystick, hand wheel extension

400	
Versorgung	
1	+24V_In
2	GND_In

401	
1	+24V_Out hand wheel Joystick /120.0
2	GND_Out /120.1

K-Bus/Out	
K-Bus	
902	CON K-bus /120.7

PE	
grounding	
M6	grounding hand wheel, Joystick groundi /120.9

S2	
Encoder	
1	+5V hand wheel /120.4
5	A /120.5
6	A- /120.5
7	B /120.5
8	B- /120.6
10	0V /120.5

X01	
Verteiler	
1	+24V
2	+24V
3	PE

X01	
4	GND
5	GND
X3	
Joystick	
5	NO1 /120.2
7	NO2 /120.3
8	NO3 /120.3
9	NO4 /120.3
10	COM Joystick /120.2

X4	
Encoder	
1	+5V hand wheel /120.4
5	A /120.5
6	A- /120.5
7	B /120.5
8	B- /120.6
10	0V /120.5

X5	
K-Bus	
403	CON K-bus /120.7

X7	
Versorgung	
1	+24V_In hand wheel Joystick /120.0
2	GND_In /120.1

X101	
Bus	
IN	BUS hand wheel, Joystick EtherC /120.8

X102	
OUT	BUS hand wheel, Joystick EtherC /120.8

X103	
Interface-in	
1	panel Safety channel =OP/125.1
2	= =OP/125.1
3	= =OP/125.2
4	= =OP/125.2
5	=OP/125.4
6	GND operation panel supply =OP/124.4
7	=OP/126.1
8	operation panel battery pack =OP/124.1

X103	
9	machine in Stop =OP/126.2
10	machine in start =OP/126.3
11	operation panel battery pack =OP/124.1
12	PE operation panel supply =OP/124.3
13	PC turn off =OP/124.6
14	USV turn off =OP/124.7
15	machine reset =OP/125.3
16	machine start =OP/125.7
17	machine Stop =OP/125.8
18	control voltage turn on =OP/125.6
19	+24V operation panel supply =OP/124.3
Shield	=OP/124.8

X104	
Interface-out	
1	panel Safety channel =OP/125.1
2	= =OP/125.1
3	= =OP/125.2
4	= =OP/125.2
5	=OP/125.4
6	GND operation panel supply =OP/124.4
7	=OP/126.1
8	operation panel battery pack =OP/124.1
9	machine in Stop =OP/126.2
10	machine in start =OP/126.3
11	operation panel battery pack =OP/124.1
12	PE =OP/124.3
13	PC turn off =OP/124.6
14	USV turn off =OP/124.7
15	machine reset =OP/125.3
16	machine start =OP/125.7
17	machine Stop =OP/125.8
18	control voltage turn on =OP/125.6
19	+24V operation panel supply =OP/124.3
Shield	=OP/124.8



G37000289

PC operation panel

-P1 display 15", GCS export, SSD, touch display, Win10

XS01

shield	
SH	/124.8
supply	
8 Power Status USV turn off	/124.7

401

1 24L+ operation panel supply	/124.3
2 24L- =	/124.4
3 24L+Akku operation panel battery pack	/124.1
4 24L-Akku =	/124.1
7 PCON PC turn off	/124.6

PE

grounding	
M6 grounding panel grounding	/124.9
supply	
5 PE operation panel supply	/124.3

S1_A

ON/OFF	
1 XA	/126.1
2 XA	/126.1
3 XA	/125.4
4 XA	/125.4
5 XA	/125.6
6 XA control voltage turn on	/125.6

S1_B

1 XB	/125.1
------	--------

S1_B

2 XB panel Safety channel	/125.1
3 XB	/125.2
4 XB panel Safety channel	/125.2
5 XB	/125.5
6 XB	/125.5

S2

Stop	
1 E1 machine in Stop	/126.2
2 E1 =	/126.2
3 C1 #	/125.9
4 C1 =	/125.9
5 C2 machine Stop	/125.8
6 C2 =	/125.8

S3

Start	
1 E1 machine in start	/126.3
2 E1 =	/126.3
3 C1 machine start	/125.7
4 C1 =	/125.7
5 C2	/126.4
6 C2	/126.4

S4

Emergency Stop	
3 C1	/125.1
4 C1	/125.1
5 C2	/125.2
6 C2	/125.2

X104

Bus	
OUT 1 BUS panel EtherCAT	/123.9

X105

OUT 2 BUS panel EtherNET	/123.3
--------------------------	--------

X106

OUT 3 BUS panel EtherNET	/123.1
--------------------------	--------

XS01

Interface	
1 panel Safety channel	/125.1
2 =	/125.1
3 =	/125.2

XS01

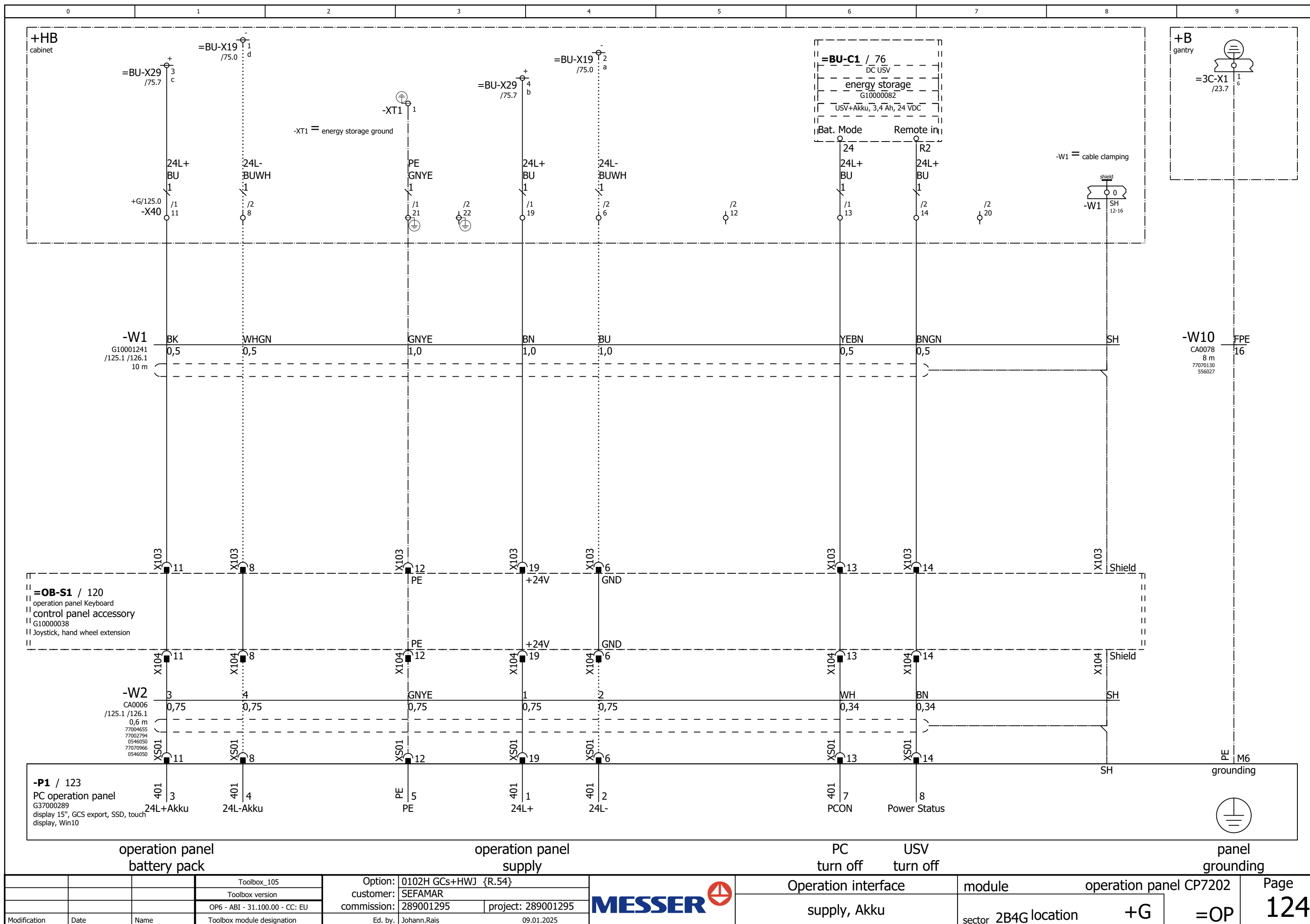
4 panel Safety channel	/125.2
5	/125.4
6 operation panel supply	/124.4
7	/126.1
8 operation panel battery pack	/124.1
9 machine in Stop	/126.2
10 machine in start	/126.3
11 operation panel battery pack	/124.1
12 operation panel supply	/124.3
13 PC turn off	/124.6
14 USV turn off	/124.7
15 machine reset	/125.3
16 machine start	/125.7
17 machine Stop	/125.8
18 control voltage turn on	/125.6
19 operation panel supply	/124.3

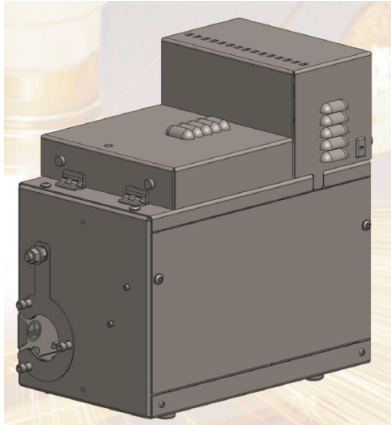
i24L+

supply intern	
1 i24L+	/125.5
2 i24L+	/125.6
3 i24L+ machine start	/125.7
4 i24L+ #	/125.9

i24L-

1 i24L-	/125.5
2 i24L-	/126.1
3 i24L- machine in start	/126.3
4 i24L- machine in Stop	/126.2
5 i24L-	





G72100310

plasma connection unit torch

-E1

KJB, Q

Contact

Power Module		
eEF	Position contact 2X	torch touch =PT1/137.5

GND

Grounding		
M8	ground	torch connection console ground /129.9

Gas fitting

Gas console		
G1/8"	Plasma gas1	plasma gas 1 /129.6
G1/8"	Plasma gas2	plasma gas 2 /129.7
G1/8"	Swirl gas 1	swirl gas /129.8
G1/8"L	Plasma gas3	

IN

Plasma Source		
BL	Coolant inlet	/129.2

M12

-	Cathode line	cathode line /129.4
-	Cathode line	

OUT

SW	Coolant outlet	return coolant /129.3
----	----------------	-----------------------

Torch

Torch		
1-8	Torch connect	control cable =PT1/137.7
BL	Plasma gas	plasma gas 2 =PT1/137.1
BL	Coolant supply	flow of coolant =PT1/137.3
GN	Swirl gas1	swirl gas =PT1/137.2

Torch

M8	ground torch	protective sleeve grounding	=PT1/137.9
NW14	Coolant return	return coolant	=PT1/137.4
PI	Pilot cable	pilot arc	=PT1/137.6
SW	Ignition gas	plasma gas 1	=PT1/137.1
YW	Exhaust		=PT1/137.8

X314

Plasma Source		
1-3	Pilot cable	pilot arc /129.2

X315

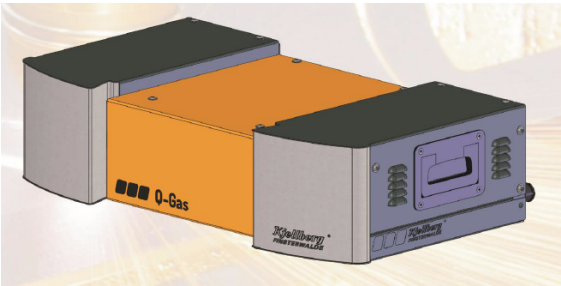
1-5	Power cable	supply /129.0
-----	-------------	---------------

X341

Gas console		
1-5	CAN	
Gas console		
1-5	CAN	CAN bus /129.5

X343

Plasma Source		
1-8	EtherNet	EtherNet /129.1



G72100309

plasma connection unit cutting gas console

-E1

Air Fitting

Gas Input		
G1/4"	WG2-Air	gas air /131.8
G3/8" LH	WG2-N2-H2	
Q-Port		
G1/8"	WG2-Air_N2-H2	
M12	WG1-N2_O2	swirl gas =PB1+L1/129.8

Cutting Gas

G1/4" LH	PG2-Air_N2_O2	plasma gas 2 =PB1+L1/129.7
G3/8" LH	PG3-H2	

Gas Fitting

Gas Input		
G1/4"	N2	/132.2
G1/4"	O2	/132.4
G1/4"	Ar	gas AR /132.6
G1/4"	Air	/132.8

M8

Grounding		
PE	GND	gas console grounding /131.9

Pilot Gas

Q-Port		
G1/4"	PG1-Air_Ar_N2	plasma gas 1 =PB1+L1/129.6

X310

Power Source		
1-5	24V DC	gas console supply /131.0

X340

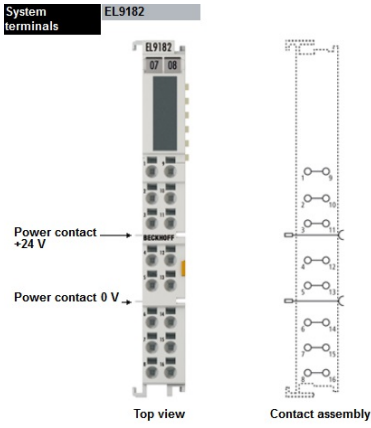
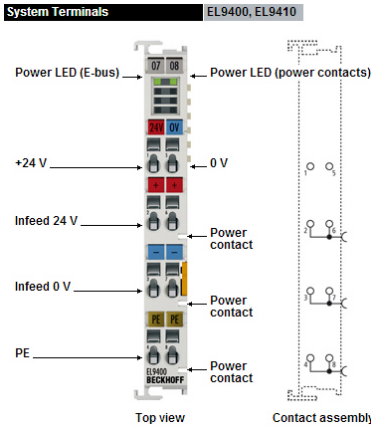
1-5	CAN Q-Gas	CAN bus /131.1
-----	-----------	----------------

X341

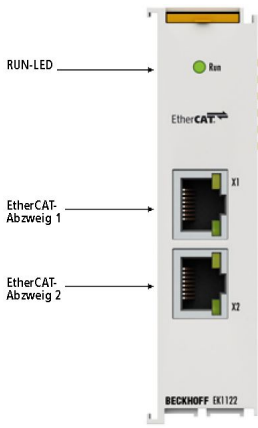
Q-Port		
1-5	CAN Q-Port	CAN bus =PB1+L1/129.5

X342

Power Source		
1-8	EtherNet	EtherNet /131.2



16 8-16



G13000079

bus terminal

-K1 power supply terminal, E-bus refreshment

power contacts

2	24L+	plasma interface supply	+P/136.0
3	24L-	=	+P/136.1
4	PE		
6	24L+	plasma interface supply	+P/136.0
7	24L-	=	+P/136.1
8	PE		

supply

1	24L+	plasma interface supply	+P/136.0
5	24L-	=	+P/136.1

G10000910

bus terminal

-K2 feed-through terminal 8x

Terminal-1-9

1	1-9	plasma emergency stop	+P/136.4
9	1-9	=	+P/136.4

Terminal-2-10

2	2-10	plasma emergency stop	+P/136.6
10	2-10	=	+P/136.6

Terminal-3-11

3	3-11	plasma emergency stop	+P/136.6
11	3-11	=	+P/136.6

Terminal-4-12

4	4-12	plasma power source standb	+P/136.3
12	4-12	=	+P/136.3

Terminal-5-13

5	5-13	plasma power source standb	+P/136.2
13	5-13	=	+P/136.2

Terminal-6-14

6	6-14		
14	6-14		

Terminal-7-15

7	7-15		
15	7-15		

Terminal-8-16

8	8-16		
---	------	--	--

G10000781

bus terminal

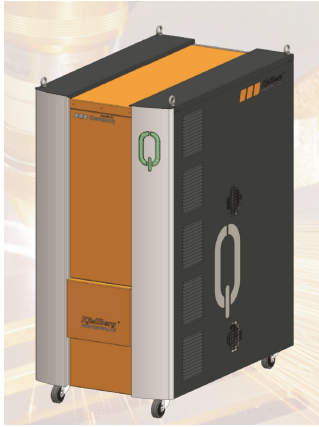
-K3 EK1122 EtherCAT port junction module, 2x RJ45, 100MBit/s

X1

Bus	
OUT1	Bus plasma power source EtherC +P/136.7

X2

OUT2	Bus
------	-----



G72100001
plasma power source
-E1 KJB, Q1500, 400V, 50/60 Hz, 37 kVA

Dia 12	
Q-Port	
BL	Coolant supply Dia12
Dia 14	
RD	Coolant return Dia 14 return coolant
M8	
Table	
PE	GND plasma power source functional ground
M10	
+	Work lead
M12	
Q-Port	
-	Negative lead cathode line
-	Negative lead
X302	
CNC Digital	
1	24L+ plasma emergency stop
2	COM-Emergency switch
3	NC-Emergency switch
4	activate device
5	activate current
6	IN-spare
7	IN-spare
8	IN-spare
9	24L+ extern
10	ready to cut

=PB1+L1/129.2

=PB1+L1/129.3

/135.6

/135.8

=PB1+L1/129.4

/136.6

X302	
11	benchmark
12	ready to move
13	OUT-spare
14	OUT-spare
15	main relays
16	Stand by plasma power source standb
17	Ground

/136.3

= /136.2

X304	
CNC Analog	
2	analog voltage
3	workpiece
4	Cathode voltage
5	Nozzle voltage
6	GND

X306	
Q-Desk	
1-8	Q-Desk

X307	
EtherNET	
IN	EtherNet plasma power source EtherN

/136.9

X308	
Bus	
IN	Bus plasma power source EtherC

/136.7

X309	
OUT	Bus

X310	
Q-Gas	
1-5	24V DC Q-Gas gas console supply

=PG1+G/131.0

X314	
Q-Port	
1-3	Pilot cable pilot arc

=PB1+L1/129.2

X315	
1-5	24V DC Q-Port supply

=PB1+L1/129.0

X340	
Q-Gas	
1-5	CAN CAN bus

=PG1+G/131.1

X342	
1-8	EtherNet Q-Gas EtherNet

=PG1+G/131.2

X343	
Q-Port	
1-8	EtherNet Q-port EtherNet
x	
mains	
PE	PE plasma power source power supply
U	L1
V	L2
W	L3

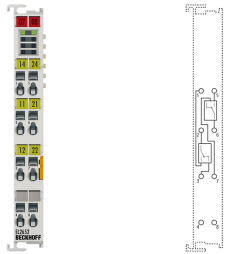
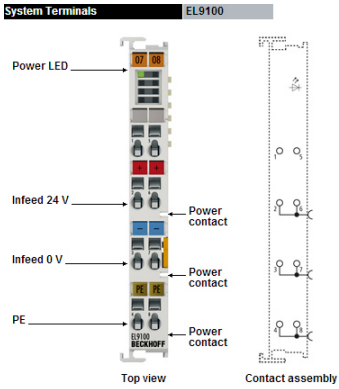
=PB1+L1/129.1

=PA1/127.1

=PA1/127.2

=PA1/127.3

=PA1/127.3



EL9100 | Potential supply terminal, 24 V DC

G10001266

bus terminal

-K1

potential supply terminal, 24 V DC

G10001372

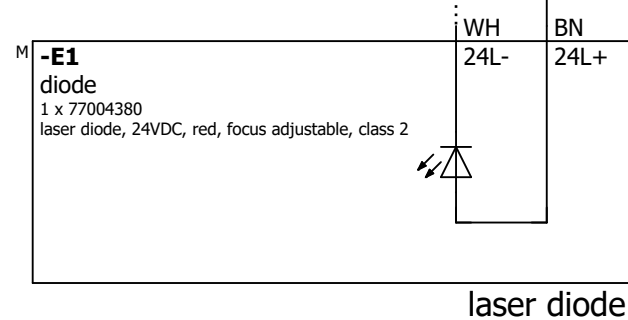
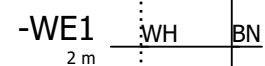
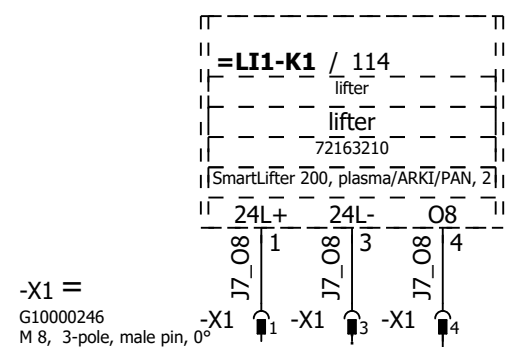
bus terminal

-K2

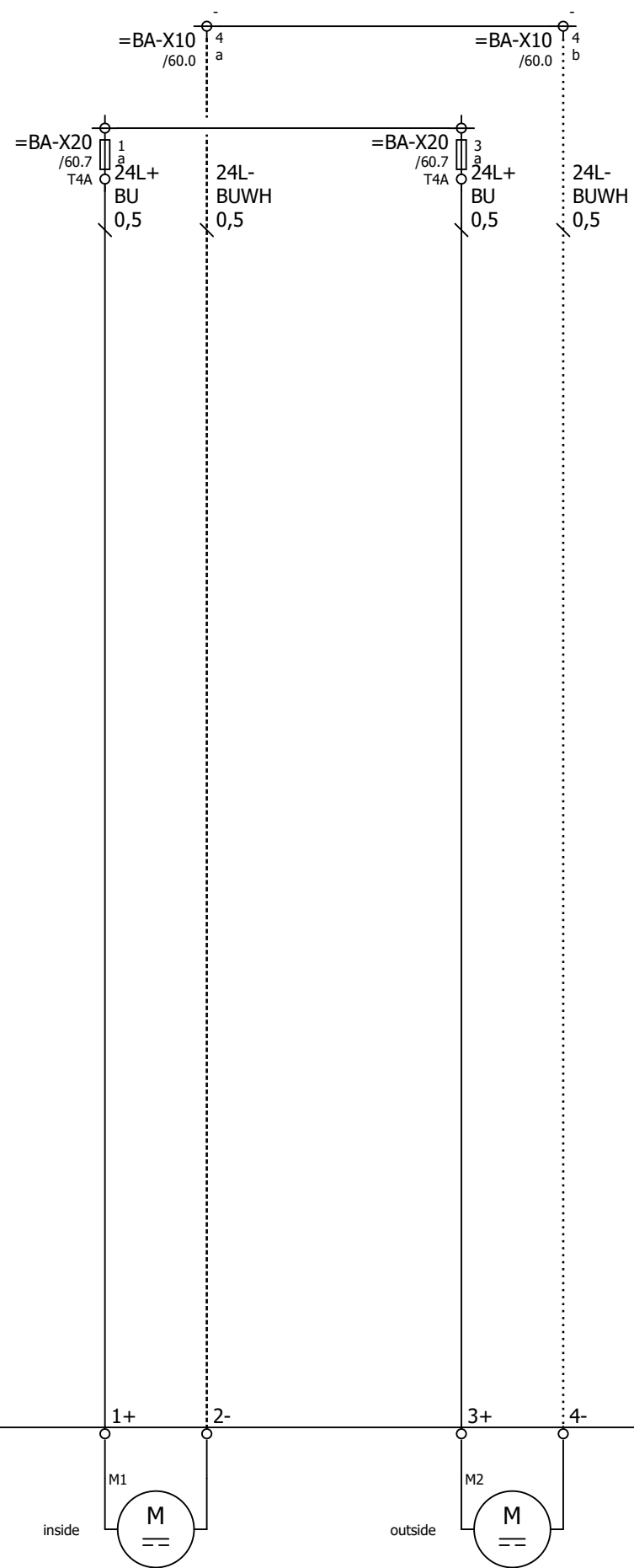
digital, relay, 2 changeover contact, 230 VAC, 1A

PE		
4	PE	exhaust grounding +P/140.6
8	PE	= +P/140.6
n.c.		
1	n.c	
5	n.c	
power contacts		
2	24L+	
3	24L-	
supply 24L+		
6	24L+	
supply 24L-		
7	24L-	

K1		
2	COM	filter fan turn on +P/140.2
7	NC	filter fan enabling +P/140.5
K1		
1	NO	exhaust turn on +P/140.2
3	NC	filter fan turn on +P/140.3
K2		
5	NO	exhaust turn off +P/140.4
6	COM	filter fan enabling +P/140.3



			Toolbox_105	Option:	1001A L1-Laser Pointer {R.31}			laser diode	module	unit pointer	Page 141
			Toolbox version	customer:	SEFAMAR			connection	sector CA2L location +L1	=UP1	
			UP1 - ABI - 31.99.00 - CC: EU	commission:	289001295	project: 289001295					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	09.01.2025					



CA0078

+HB=3B-W10

MCS* Cable Production

9 m - 16 mm²

77003786 // Kabeldurchführungsstülle, 1x11-12mm

M10 - female

+B=3C-X1

c

Page: 15

Kabel // FPE, Ring, OE-M10, 16mm²

OE -

+HB=3B-X0FPE

a

Erdung Schaltschrank // Schaltschrank

Cable Plan Formular: =Z6B++111+A/152

CA0078

+HP1=3B1-W10

MCS* Cable Production

11 m - 16 mm²

77070529 //

OE -

+HP1=3B1-X0FPE

a

Page: 18

Kabel // FPE, Ring, OE-M10, 16mm²

M10 - female

+T=3CB-X1

c

Erdung Schaltschrank // Schaltschrank

Cable Plan Formular: =Z6B++111+A/153

CA0078

+HB=3BB-W10

MCS* Cable Production

9 m - 16 mm²

M10 - female

+B=3C-X1

d

Page: 21

Kabel // FPE, Ring, OE-M10, 16mm²

OE -

+HB=3BB-X0FPE

a

Erdung Schaltschrank // Schaltschrank

Cable Plan Formular: =Z6B++111+A/154

CA0122

+B=3C-W1

MCS* Cable Production

0,2 m - 25 mm²

M10 - female

+B=3C-X1

a

Page: 24

Kabel // Erdung, PE, Ring, M8-M10, 25mm²

M8 - female

+B=3C-GND

Erdungsschiene // Erdungsschiene

Cable Plan Formular: =Z6B++111+A/155

CA0060

+B=3C-W2

MCS* Cable Production

35 m - 70 mm²

M10 - female

+B=3C-X1

b

Page: 24

Kabel // Erdung, PE, Ring, M10-M10, 70mm²

M10 - female

+T=3CB-X1

d

Erdungsschiene // Erdungsschiene

Cable Plan Formular: =Z6B++111+A/156

CA0065

+T=3CB-W1

MCS* Cable Production

15 m - 70 mm²

M10 - female

+T=3CB-X1

b

Page: 26

Kabel // Erdung, PE, Ring, OE-M10, 70mm²

OE -

+PZ=CUSTOMERGND

Erdungsschiene // Erdungsschiene

Cable Plan Formular: =Z6B++111+A/157

CA0060

+T=3CB-W12

MCS* Cable Production

0,2 m - 70 mm²

M10 - female

+T=3CB-X1

a

Page: 26

Kabel // Erdung, PE, Ring, M10-M10, 70mm²

M10 - female

+T=3CB-GND

Erdungsschiene // Erdungsschiene

Cable Plan Formular: =Z6B++111+A/158

G10001038

+B=3D-W1

0,1 m - 16 mm²

M6 - female

+B=3D-GND

Page: 27

Masseband // M6-M6, 0,1m

M6 - female

+B=3D-GND

Erdung Maschine // Erdung Maschine

G10001039

+E=3D-W1

0,3 m - 16 mm²

M6 - female

+E=3D-GND

Page: 27

Masseband // M6-M6, 0,3m

M6 - female

+E=3D-GND

Erdung Maschine // Erdung Maschine

G10001039

+G=3D-W1

0,3 m - 16 mm²

M6 - female

+G=3D-GND

Page: 27

Masseband // M6-M6, 0,3m

M6 - female

+G=3D-GND

Erdung Maschine // Erdung Maschine

G10001108

+PRE=3DB-W0

0,3 m - 70 mm²

Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108

+PRG=3DB-W0

0,3 m - 70 mm²

Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRE=3DB-W1

0,3 m - 70 mm²



Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRG=3DB-W1

0,3 m - 70 mm²



Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRE=3DB-W2

0,3 m - 70 mm²



Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRG=3DB-W2

0,3 m - 70 mm²



Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRE=3DB-W3

0,3 m - 70 mm²



Page: 28

Kabel // Einzelader

Erdung Maschine // Erdung Maschine

G10001108
+PRG=3DB-W3

0,3 m - 70 mm²



Page: 28

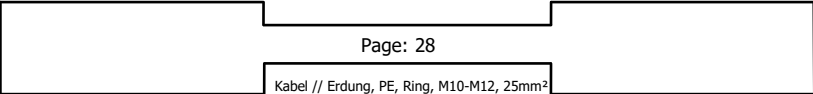
Kabel // Einzelader

Erdung Maschine // Erdung Maschine

CA0212
+T=3DB-W10

MCS* Cable Production

9 m - 25 mm²



Page: 28

Kabel // Erdung, PE, Ring, M10-M12, 25mm²

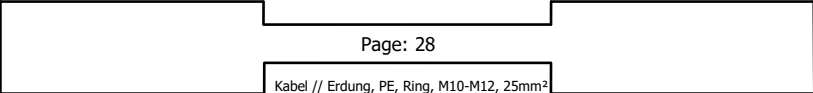
Erdung Maschine // Erdung Maschine

Cable Plan Formular: =Z6B++111+A/159

CA0212
+T=3DB-W20

MCS* Cable Production

9 m - 25 mm²



Page: 28

Kabel // Erdung, PE, Ring, M10-M12, 25mm²

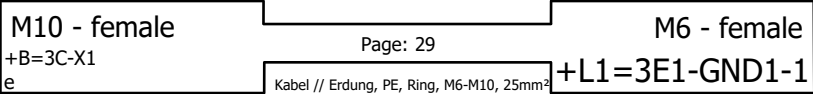
Erdung Maschine // Erdung Maschine

Cable Plan Formular: =Z6B++111+A/160

CA0126
+L1=3E1-W1

MCS* Cable Production

8 m - 25 mm²



M10 - female
+B=3C-X1
e

Page: 29

M6 - female
+L1=3E1-GND1-1

Kabel // Erdung, PE, Ring, M6-M10, 25mm²

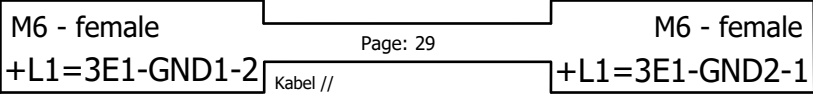
Erdung Elemente // Erdung Lifter

Cable Plan Formular: =Z6B++111+A/161

CA0135
+L1=3E1-W2

MCS* Cable Production

2,2 m - 16 mm²



M6 - female
+L1=3E1-GND1-2

Page: 29

M6 - female
+L1=3E1-GND2-1

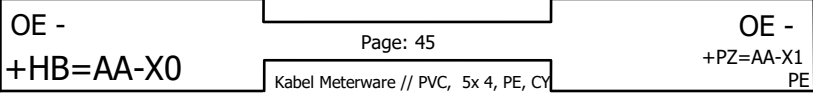
Kabel //

Erdung Elemente // Erdung Lifter

Cable Plan Formular: =Z6B++111+A/162

G10000911
+HB=AA-W0

25 m - 4 mm²



OE -
+HB=AA-X0

Page: 45

OE -
+PZ=AA-X1
PE

Kabel Meterware // PVC, 5x 4, PE, CY

AC Versorgung Schaltschrank // AC Betriebsspannung

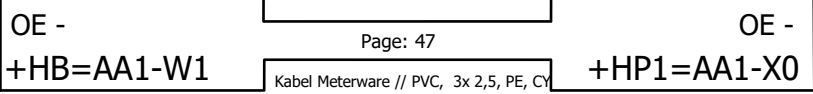
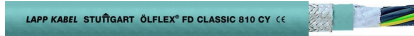
Cable Plan Formular: =Z6B++111+A/163

G10000797
+HP1=AA1-W1

34 m - 2,5 mm²



G10000423 // Schirmklammer, Zugentlastung, Hutschienen, 8-11



OE -
+HB=AA1-W1

Page: 47

OE -
+HP1=AA1-X0

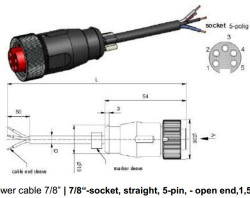
Kabel Meterware // PVC, 3x 2,5, PE, CY

AC Versorgung Schaltschrank // Einspeisung

Cable Plan Formular: =Z6B++111+A/164

G10001295
+B=BI-W1

10 m - 2,5 mm²



OE - 7/8 - female
+HB=BI-X1 Kabel // OE-7/8", Buchse, 5 x 2,5, 10m +B=BI-K1 X1

DC Versorgung Schnittstelle // EP Power - Verteiler

Cable Plan Formular: =Z6B++111+A/165

G10001070
+B=BI-W9

7,5 m - 0,34 mm²

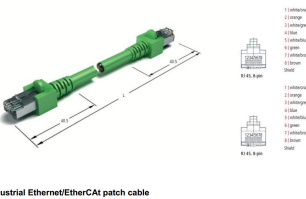


RJ45 - male M8 - male
+HB=IC-K1 X8 Kabel Netzwerk // EtherCAT, M8-RJ45, 7,5m +B=BI-K1 L/A

DC Versorgung Schnittstelle // EP Power - Verteiler

77004561
+HB=CXXa-W9

10 m - 0,16 mm²

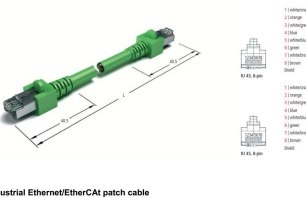


RJ45 - male RJ45 - male
+HB=CXXa-K1 X04 Kabel Netzwerk // EtherCAT, RJ45-RJ45, 10m +G=OB-X29

Regler X,Xa // AX5206 X,Xa

G10000209
+HB=CY1-W9

0,25 m - 0,14 mm²



RJ45 - male RJ45 - male
+HB=CY1-K1 X04 Kabel Netzwerk // EtherCAT, RJ45-RJ45, 0,25m +HB=CXXa-K1 X05

Regler Y1 // AX5103 Y1

+L1=DC1-WB1

2 m



+L1=DC1-X1 Page: 82
J7_I8:1 vergossenes Leitungsende +L1=DC1-B1

Erkennung Kollision // Brenner Magnethalter

+L1=DC1B-WB1

2 m



+L1=DC1B-B1 Page: 83
vergossenes Leitungsende +L1=DC1B-X1

Erkennung Kollision // Brenner Aushebeschalter taktil

G10000458
+L1=DL1-W1

15 m - 0,34 mm²



OE - M12 - female
+HB=DL1-W1 Page: 85 Kabel // OE-M12, 5-polig, Buchse, 15m +L1=DL1-B1

Erkennung Begrenzung // Begrenzung Y+

Cable Plan Formular: =Z6B++111+A/166

G10000458
+L1=DL1B-W1

15 m - 0,34 mm²



OE - M12 - female
+HB=DL1B-W1 Page: 87 Kabel // OE-M12, 5-polig, Buchse, 15m +L1=DL1B-B1

Erkennung Begrenzung // Begrenzung Y-

Cable Plan Formular: =Z6B++111+A/167

G10000957
+E=EA-W1

10 m - 0,34 mm²



M12 - female OE -
+E=EA-S1 X1 Page: 88 Kabel // OE-M12, 5-polig, Buchse, 10m +HB=IT-K2

Not-Halt Zusätze // Not-Halt

Cable Plan Formular: =Z6B++111+A/168

G10000957
+G=EAB-W1

10 m - 0,34 mm²



M12 - female OE -
+G=EAB-S1 X1 Page: 89 Kabel // OE-M12, 5-polig, Buchse, 10m +HB=IT-K2

Not-Halt Zusätze // Not-Halt

Cable Plan Formular: =Z6B++111+A/169

G10000965
+E=EF-W1

5 m - 0,34 mm²



M12 - male M12 - female
+B=IBB-K1 01 Page: 90 Kabel // M12-M12, 5-polig, 5m +E=EF-S1

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+E=EF-W2

5 m - 0,34 mm²



M12 - male M12 - female
+B=IBB-K1 23 Page: 90 Kabel // M12-M12, 5-polig, 5m +E=EF-S2

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+E=EF-W3

5 m - 0,34 mm²



M12 - male
+B=IBB-K1 45

Page: 90
Kabel // M12-M12, 5-polig, 5m

M12 - female
+E=EF-S3

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+E=EF-W4

5 m - 0,34 mm²



M12 - male
+B=IBB-K1 67

Page: 90
Kabel // M12-M12, 5-polig, 5m

M12 - female
+E=EF-S4

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+G=EFB-W1

5 m - 0,34 mm²



M12 - male
+B=IB-K1 01

Page: 91
Kabel // M12-M12, 5-polig, 5m

M12 - female
+G=EFB-S1

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+G=EFB-W2

5 m - 0,34 mm²



M12 - male
+B=IB-K1 23

Page: 91
Kabel // M12-M12, 5-polig, 5m

M12 - female
+G=EFB-S2

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+G=EFB-W3

5 m - 0,34 mm²



M12 - male
+B=IB-K1 45

Page: 91
Kabel // M12-M12, 5-polig, 5m

M12 - female
+G=EFB-S3

Not-Halt Fusschutzsicherung // Fusschutzsicherung

G10000965
+G=EFB-W4

5 m - 0,34 mm²



M12 - male
+B=IB-K1 67

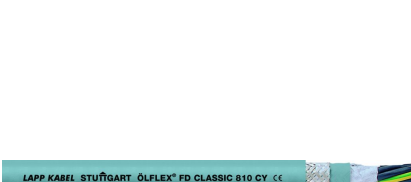
Page: 91
Kabel // M12-M12, 5-polig, 5m

M12 - female
+G=EFB-S4

Not-Halt Fusschutzsicherung // Fusschutzsicherung

77002347
+HP1=EI1-W1

34 m - 1 mm²



OE -
+HB=EI1-W1

Page: 93
Kabel Meterware // PVC, 5x 1, PE, CY

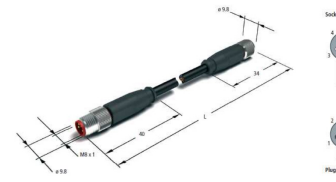
OE -
+HP1=EI1-X1

Not-Halt Schnittstelle // Schaltschrank extern

Cable Plan Formular: =Z6B++111+A/170

G10000124
+B=IB-W1

0,3 m - 0,34 mm²



M8 - male
+B=BI-K1 1

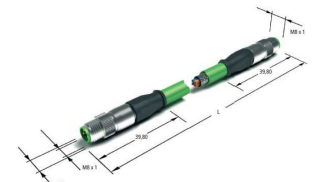
Page: 99
Kabel // Powerleitung, M8-M8, 0,3m

M8 - female
+B=IB-K1 24V IN

Schnittstelle Box // Feldbus Modul

G10001272
+B=IB-W9

0,3 m - 0,34 mm²



M8 - male
+B=IB-K1 L/A

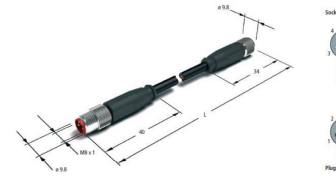
Page: 99
Kabel Netzwerk // EtherCAT, M8-M8, 0,3m

M8 - male
+B=BI-K1 L/A

Schnittstelle Box // Feldbus Modul

G10000127
+B=IBB-W1

10 m - 0,34 mm²



M8 - male
+B=IB-K1 24V OUT

Page: 101
Kabel // Powerleitung, M8-M8, 10m

M8 - female
+B=IBB-K1 24V IN

Schnittstelle Box // Feldbus Modul

G10001071
+B=IBB-W9

7,5 m - 0,34 mm²



M8 - male
+B=IBB-K1 L/A

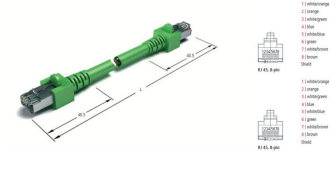
Page: 101
Kabel Netzwerk // EtherCAT, M8-M8, 7,5m

M8 - male
+B=IB-K1 L/A

Schnittstelle Box // Feldbus Modul

G10000252
+HB=IC-W9

1 m - 0,16 mm²



RJ45 - male
+HB=IT-K1 X2

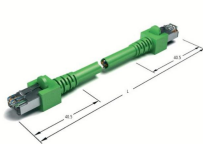
Page: 103
Kabel Netzwerk // EtherCAT, RJ45-RJ45, 1m

RJ45 - male
+HB=IC-K1 X1

Schnittstelle Steuerung // Sternverteiler

G10000253
+HB=IT-W9

2 m - 0,16 mm²



RJ45 - male
+HB=IT-K1 X1

Page: 104

RJ45 - male
+HB=CY1-K1 X05

Kabel Netzwerk // EtherCAT, RJ45-RJ45, 2m

Schnittstelle Klemme // Buskoppler

G17000114
+HP1=IT1-W9

40 m - 0,25 mm²



RJ45 - male
+HP1=IT1-K1 X1

Page: 111

RJ45 - male
+HB=IC-K1 X3

Kabel Netzwerk // EtherCAT, RJ45-RJ45, 40m

Schnittstelle Klemme // Buskoppler

CA0000
+L1=LI1-W1

MCS* Cable Production

11 m - 1 mm²



OE -
+HB=LI1-W1

Page: 114

M22 - female
+L1=LI1-K1 J5

Kabel // OE-M22, 4-adrig

Höhenverstellung // SL 200

Cable Plan Formular: =Z6B++111+A/171

CA0168
+L1=LI1-W2

MCS* Cable Production

10 m - 0,34 mm²



M8 - male
+B=BI-K1 2

Page: 114

M12 - female
+L1=LI1-K1 J3

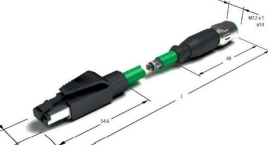
Kabel // M8-M12, 5-adrig, <=15m

Höhenverstellung // SL 200

Cable Plan Formular: =Z6B++111+A/172

G10000414
+L1=LI1-W9

12 m - 0,34 mm²



+HB=IC-K1 X2

Page: 114

M12 - male
+L1=LI1-K1 J1

Kabel Netzwerk // EtherCAT, M12-RJ45, 12m

Höhenverstellung // SL 200

G10001026
+E=MX-W1

5 m - 2,07 mm²



M23 - female
+E=MX-M1

Page: 115

X13+X14 - female
+HB=CXXa-K1 X13

Kabel Motor // Motor, AMxxxx, OCT, 1,0+0,75+0,32, 5m

Motor X // AM8531 OCT X

G10001037
+E=MX-W10

0,2 m - 16 mm²



M6 - female
+E=MX-GND

Page: 115

M6 - female
+E=MX-GND

Masseband // M6-M6, 0,2m

Motor X // AM8531 OCT X

G10001309
+G=MXa-W1

8 m - 2,07 mm²



M23 - female
+G=MXa-M1

Page: 116

X13+X14 - female
+HB=CXXa-K1 X23

Kabel Motor // Motor, AMxxxx, OCT, 1,0+0,75+0,32, 8m

Motor Xa // AM8531 OCT Xa

G10001037
+G=MXa-W10

0,2 m - 16 mm²



M6 - female
+G=MXa-GND

Page: 116

M6 - female
+G=MXa-GND

Masseband // M6-M6, 0,2m

Motor Xa // AM8531 OCT Xa

G10001310
+L1=MY1-W1

10 m - 1,0+0,75+0,32 mm²



X13+X14 - female
+HB=CY1-K1 X13

Page: 117

M23 - female
+L1=MY1-M1

Kabel Motor // Motor, AMxxxx, OCT, 1,0+0,75+0,32, 10m

Motor Y1 // AM8531 OCT Y1

G17000115
+PZ=NC-W1

50 m - 0,25 mm²



RJ45 - male
+G=OP-X17

Page: 118

RJ45 - male
+PZ=NC-A1

Kabel Netzwerk // EtherCAT, RJ45-RJ45, 50m

Netzwerkanbindung // Netzwerk

G10000074
+G=OB-W9

0,5 m - 0,16 mm²



RJPP - male
+G=OP-P1 X104

Page: 120

RJPP - male
+G=OB-S1 X101

Kabel Netzwerk // EtherCAT, RJPP-RJPP, 0,5m

Bedienfeld Tastatur // Handrad - Joystick

CA0167
+G=OB-W10

MCS* Cable Production

0,5 m - 4 mm²

M6 - female +G=OB-GND	Page: 120	M6 - female +G=OB-S1 PE
Kabel // Erdung, PE, Ring, M6-M6, 4mm²		

Bedienfeld Tastatur // Handrad - Joystick

Cable Plan Formular: =Z6B++111+A/173

77005783
+G=OB-W29

3 m - 0,14 mm²



RJ45 - male +G=OB-X29	Page: 120	RJPP - male +G=OB-S1 X102
Kabel Netzwerk // EtherCAT, RJPP-RJ45, 3m		

Bedienfeld Tastatur // Handrad - Joystick

77011240
+HOK=OK-W1

10 m - 0,14 mm²

77003783 //

OE - +HB=OK-W1	Page: 121	M12 - female +HOK=OK-X1 X1
Kabel // OE-M12, 12-polig, Buchse, 10m		

Bedienfeld Schlüsselschalter // Schlüsselschalter Kasten

Cable Plan Formular: =Z6B++111+A/174

G10001241
+G=OP-W1

10 m - 0,5+1,0 mm²



M23 - female +HB=OP-W1	Page: 124	OE - +G=OB-S1 X103
Kabel // OE-M23, 19-polig, Buchse, 10m		

Bedienfeld CP7202 // Bedienerchnittstelle

Cable Plan Formular: =Z6B++111+A/175

CA0006
+G=OP-W2

MCS* Cable Production

0,6 m - 1,09 mm²



M23 - male +G=OB-S1 X104	Page: 124	M23 - female +G=OP-P1 XS01
Kabel // M23-M23, 16+5-adrig		

Bedienfeld CP7202 // Bedienerchnittstelle

Cable Plan Formular: =Z6B++111+A/176

CA0078
+G=OP-W10

MCS* Cable Production

8 m - 16 mm²

M10 - female +B=3C-X1 f	Page: 124	OE - +G=OP-P1 PE
Kabel // FPE, Ring, OE-M10, 16mm²		

Bedienfeld CP7202 // Bedienerchnittstelle

Cable Plan Formular: =Z6B++111+A/177

77005783
+G=OP-W17

3 m - 0,14 mm²



RJ45 - male +G=OP-X17	Page: 123	RJPP - male +G=OP-P1 X106
Kabel Netzwerk // EtherCAT, RJPP-RJ45, 3m		

Bedienfeld CP7202 // Bedienerchnittstelle

77005783
+G=OP-W18

3 m - 0,14 mm²

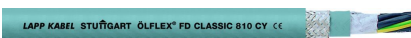


RJ45 - male +G=OP-X18	Page: 123	RJPP - male +G=OP-P1 X105
Kabel Netzwerk // EtherCAT, RJPP-RJ45, 3m		

Bedienfeld CP7202 // Bedienerchnittstelle

G10000662
+P=PA1-W0

6 m - 16 mm²



OE - +P=PS1-E1 x	Page: 127	OE - +PZ=PA1-F0
Kabel Meterware // PVC, 4x16, PE, CY		

Plasma Netzschalterbox // Q1500

Cable Plan Formular: =Z6B++111+A/178

G72100038
+L1=PB1-W1

35 m - 9999 mm²

M23 - female +L1=PB1-E1 X315	Page: 129	M23 - male +P=PS1-E1 X315
Plasma Verbindung // Versorgungsleitung, Q/Q-Gas, 35m		

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

G72100036
+L1=PB1-W2

35 m - 0,34 mm²

M12 - female +L1=PB1-E1 X343	Page: 129	M12 - male +P=PS1-E1 X343
Plasma Verbindung // KJB, Ethernetkabel, M12-M12, 35m		

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

G72100114
+L1=PB1-W3

35 m - 9999 mm²

M17 - female +L1=PB1-E1 X314	Page: 129	M17 - male +P=PS1-E1 X314
Plasma Verbindung // KJB, Versorgungsleitung, M17-M17, 35m		

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

G72100115

+L1=PB1-W4

35 m - 1/4 Zoll

\$QCH - female

+L1=PB1-E1 IN

Page: 129

\$QCH - female

+P=PS1-E1 Dia 12

Schlauch Gas // Schlauch, Kühlmittel, 35m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

G72100116

+L1=PB1-W5

35 m - 1/4 Zoll

\$QCH - female

+L1=PB1-E1 OUT

Page: 129

\$QCH - female

+P=PS1-E1 Dia 14

Schlauch Gas // Schlauch, Kühlmittel, 35m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

72423910

+L1=PB1-W6

35 m - 50 mm²

CatodePBA440 - female

+L1=PB1-E1 M12

Page: 129

CatodeHF600 - female

+P=PS1-E1 M12

Plasma // KJB, Katodenleitung, 50sqmm, 35m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

72423910

+L1=PB1-W7

35 m - 50 mm²

CatodePBA440 - female

+L1=PB1-E1 M12

Page: 129

CatodeHF600 - female

+P=PS1-E1 M12

Plasma // KJB, Katodenleitung, 50sqmm, 35m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

G72100022

+L1=PB1-W10

10 m - 0,34 mm²

M12 - male

+G=PG1-E1 X341

Page: 129

M12 - female

+L1=PB1-E1 X341

Plasma Verbindung // KJB, CAN Kabel, M12-M12, 10m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

72421722

+L1=PB1-W11

10 m - 1/8-1/4 Zoll

G1/4 - female

+G=PG1-E1 Pilot Gas

Page: 129

G1/8 - female

+L1=PB1-E1 Gas fitting

Schlauch Gas // Schlauch, Gas, PG1, PGV-PBA, 10m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

72421723

+L1=PB1-W12

10 m - 1/8-1/4 Zoll

G1/4L - female

+G=PG1-E1 Cutting Gas

Page: 129

G1/8 - female

+L1=PB1-E1 Gas fitting

Schlauch Gas // Schlauch, Gas, PG2, PGV-PBA, 10m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

72421725

+L1=PB1-W14

10 m - 1/8-M12x1 Zoll

M12x1 - female

+G=PG1-E1 Air Fitting

Page: 129

G1/8 - female

+L1=PB1-E1 Gas fitting

Schlauch Gas // Schlauch, Gas, WG, PGV-PBA, 10m

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

CA0122

+L1=PB1-W15

MCS* Cable Production

8 m - 25 mm²

M10 - female

+B=3C-X1

Page: 129

M8 - female

+L1=PB1-E1 GND

Kabel // Erdung, PE, Ring, M8-M10, 25mm²

Plasma Brenneranschlusseinheit // Typ Q-Port 4500

Cable Plan Formular: =Z6B++111+A/179

G72100035

+G=PG1-W1

30 m - 9999 mm²

M23 - female

+G=PG1-E1 X310

Page: 131

M23 - male

+P=PS1-E1 X310

Plasma Verbindung // Versorgungsleitung, Q/Q-Gas, 30m

Plasma Gaskonsole // Typ Q-Gas 4500

G72100034

+G=PG1-W2

30 m - 0,34 mm²

M12 - female

+G=PG1-E1 X340

Page: 131

M12 - male

+P=PS1-E1 X340

Plasma Verbindung // KJB, CAN Kabel, M12-M12, 30m

Plasma Gaskonsole // Typ Q-Gas 4500

G72100033

+G=PG1-W3

30 m - 0,34 mm²

M12 - female

+G=PG1-E1 X342

Page: 131

M12 - male

+P=PS1-E1 X342

Plasma Verbindung // KJB, Ethernetkabel, M12-M12, 30m

Plasma Gaskonsole // Typ Q-Gas 4500

CA0122
+G=PG1-W10

MCS* Cable Production

7 m - 25 mm²

M10 - female

+B=3C-X1

h

Page: 131

M8 - female

+G=PG1-E1 M8

Kabel // Erdung, PE, Ring, M8-M10, 25mm²

Plasma Gaskonsole // Typ Q-Gas 4500

Cable Plan Formular: =Z6B++111+A/180

G72100135
+P=PS1-W1

5 m - 0,5 mm²

OE -

+HP1=PS1-K2

Page: 136

M23 - male

+P=PS1-E1 X302

Plasma Verbindung // Versorgungsleitung M23 17pol 5m

Plasma Quelle // Q1500

Cable Plan Formular: =Z6B++111+A/181

G72100138
+P=PS1-W9

5 m - 0,34 mm²

RJ45 - male

+HP1=PS1-K3 X1

Page: 136

RJ45IN - male

+P=PS1-E1 X308

Plasma Verbindung // KJB, EtherCAT, RJ45-RJ45, 5m

Plasma Quelle // Q1500

G72100137
+P=PS1-W9.1

5 m - 0,34 mm²

RJ45 - male

+HP1=PS1-A1

Page: 136

RJ45 - male

+P=PS1-E1 X307

Plasma Verbindung // KJB, Ethernetkabel, RJ45-RJ45, 5m

Plasma Quelle // Q1500

CA0122
+P=PS1-W10

MCS* Cable Production

10 m - 25 mm²

M10 - female

+T=3CB-X1

g

Page: 135

M8 - female

+P=PS1-E1 M8

Kabel // Erdung, PE, Ring, M8-M10, 25mm²

Plasma Quelle // Q1500

Cable Plan Formular: =Z6B++111+A/182

G72100007
+P=PS1-W13

10 m - 70 mm²

M10 - female

+P=PS1-E1 M10

Page: 135

M10 - female

+T=3CB-GND

Plasma // Werkstückkabel Q3000, 70mm², 10m

Plasma Quelle // Q1500

G72100282
+L1=PT1-W1

2 m - 9999 mm²

\$QTorch - female

+L1=PB1-E1 Torch

Page: 137

\$QTorch - female

+L1=PT1-A1 Torch

Plasma Brenner // Schaft QT 4510, 2m

Plasma Brenner // Typ Q-Torch Brenner

G10001345
+P=SP-W1

6 m - 0,5 mm²

OE -

+HP1=SP-W1

Page: 138

OE -

+P=SP-X1

Kabel Meterware // PVC, 3x 0,5, PE, CY

Sensor Druck // Differenzdruckwächter

Cable Plan Formular: =Z6B++111+A/183

G10001341
+P=UE-W1

6 m - 0,75 mm²

OE -

+HP1=UE-W1

Page: 140

OE -

+P=UE-PE1

Kabel Meterware // PVC, 5x 0,75, PE, CY

Einheit Absaugung // externe Ansteuerung

Cable Plan Formular: =Z6B++111+A/184

+L1=UP1-WE1

2 m

+L1=UP1-X1
J7_O8:3

Page: 141

vergossenes Leitungsende

+L1=UP1-E1

Einheit Pointer // Laserdiode

G10000957
+G=UT-W1

10 m - 0,34 mm²

OE -

+HB=UT-X1

Page: 142

M12 - female

+G=UT-X1 -X1

Kabel // OE-M12, 5-polig, Buchse, 10m

Einheit Temperatur // Klimatisierung

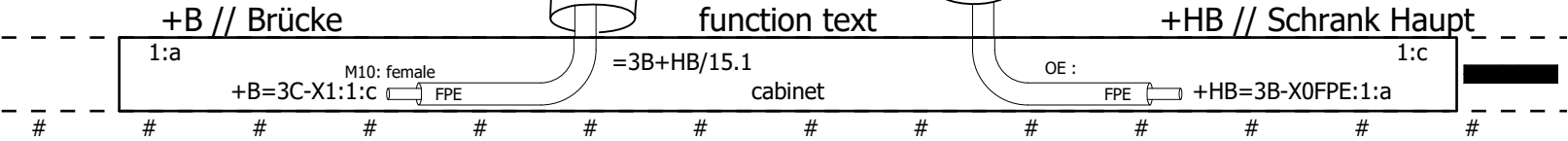
Cable Plan Formular: =Z6B++111+A/185

9 Meter

CA0078 Baugruppe

grounding cabinet
cable

+HB=3B-W10

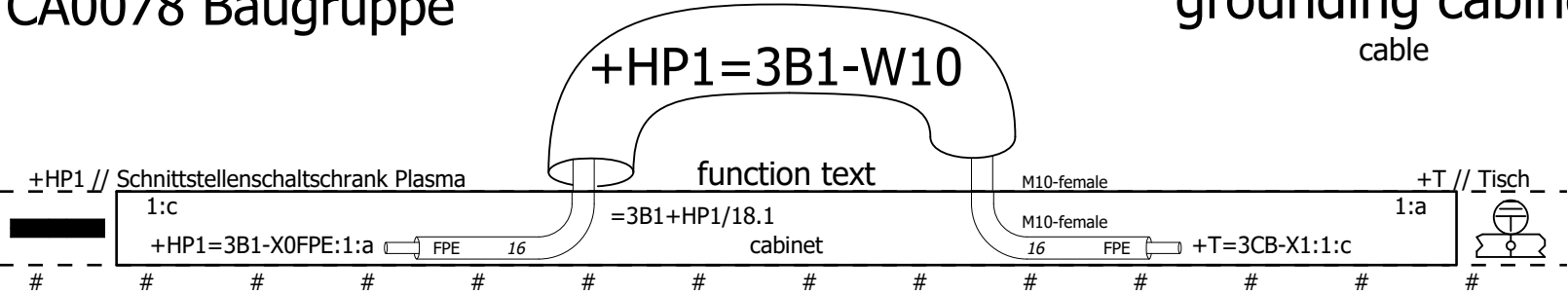


description: FPE, ring, OE-M10, 16mm²
number of cores:
cross section 16 mm²
length: 9 m
assembly Article: Kabel Zubehör: 77003786
Kabeldurchführungstülle, 1x11-12mm

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	
Stück 1	cable accessory cable lug, M10, 16mm ²	556027	A

CA0078 Baugruppe

grounding cabinet



description:

FPE, ring, OE-M10, 16mm²

number of cores:

cross section

16 mm²

length:

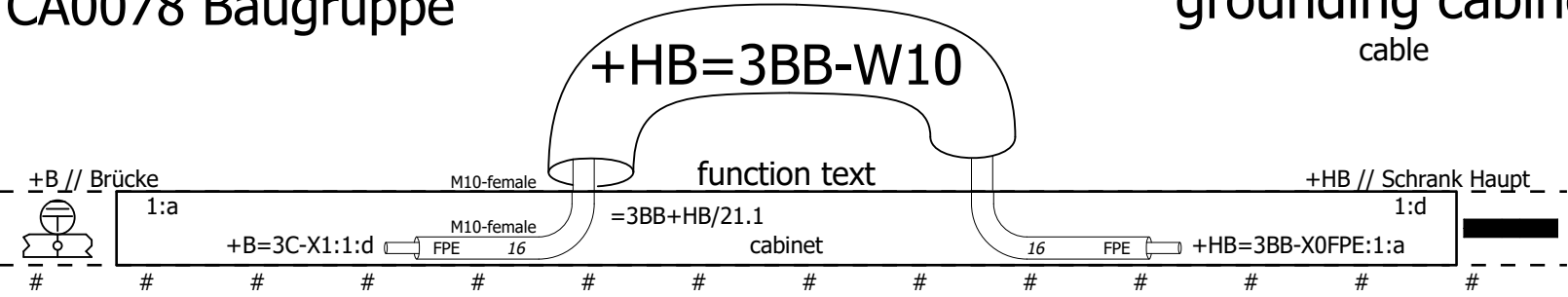
11 m

Add-on: Kabel Zubehör: 77070529

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	
Stück 1	cable accessory cable lug, M10, 16mm ²	556027	A

CA0078 Baugruppe

grounding cabinet



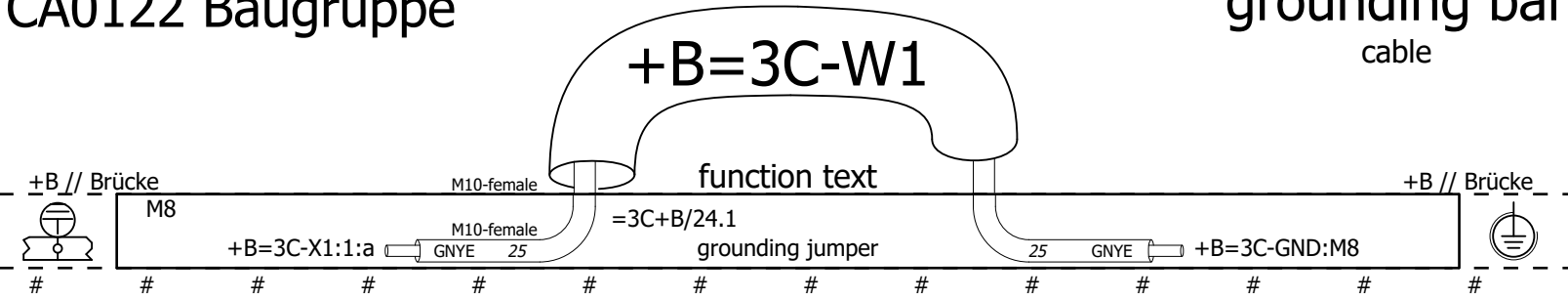
description: FPE, ring, OE-M10, 16mm²
number of cores:
cross section 16 mm²
length: 9 m

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	
Stück 1	cable accessory cable lug, M10, 16mm ²	556027	A

CA0122 Baugruppe

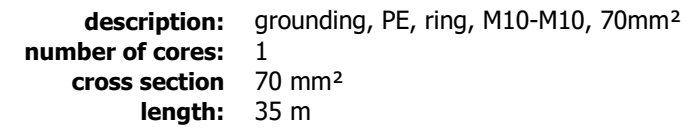
grounding bar
cable

description: grounding, PE, ring, M8-M10, 25mm²
number of cores:
cross section 25 mm²
length: 0,2 m



Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079	A B
		grounding jumper	
Stück 1	cable accessory cable lug, M8, 25mm ²	556077 =	
Stück 1	cable accessory cable lug, M10, 25mm ²	556091 =	

grounding bar
cable

function text

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x70, PE	0551080 grounding table	A B
Stück 1	cable accessory cable lug, M10, 70mm²	556084 =	
Stück 1	cable accessory cable lug, M10, 70mm²	556084 =	

grounding bar

```

cable
1x$NG-Dummy
Dummy// ...

```

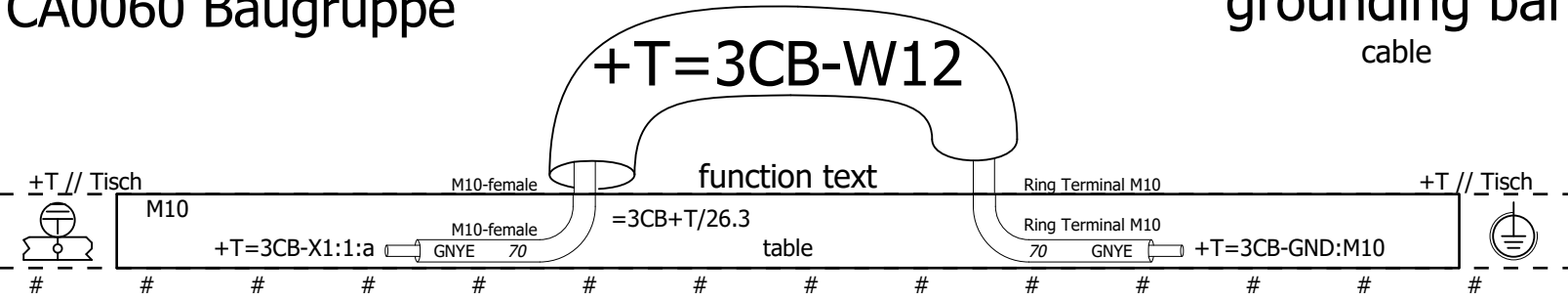
description:	grounding, PE, ring, OE-M10, 70mm ²
number of cores:	
cross section	70 mm ²
length:	15 m

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x70, PE	0551080 grounding customer	A
Stück 1	cable accessory cable lug, M10, 70mm²	556084 =	

CA0060 Baugruppe

grounding bar
cable

description: grounding, PE, ring, M10-M10, 70mm²
number of cores: 1
cross section 70 mm²
length: 0,2 m



Quantity	Designation	Part number	
	cable sold by meter PVC, 1x70, PE	0551080	A B
Stück 1	cable accessory cable lug, M10, 70mm ²	556084 =	
Stück 1	cable accessory cable lug, M10, 70mm ²	556084 =	

CA0212 Baugruppe

grounding machine

cable

description:

grounding, PE, ring, M10-M12, 25mm²

number of cores:

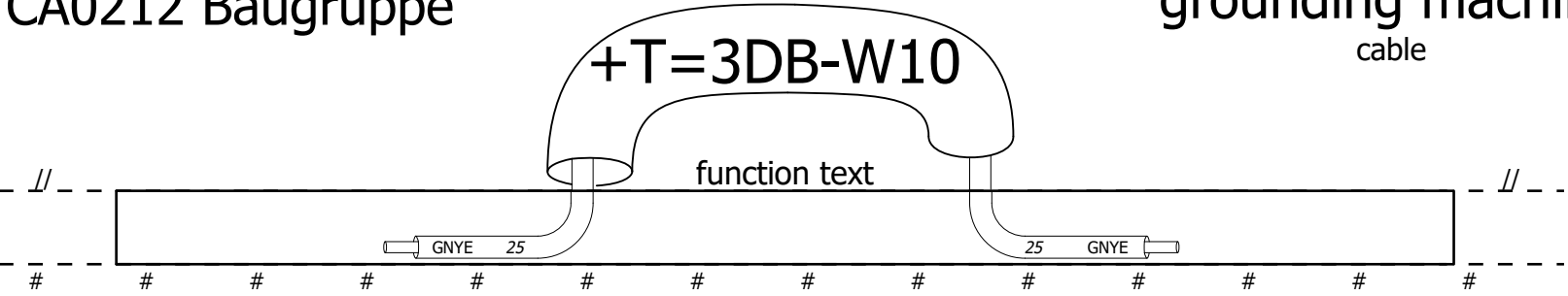
1

cross section

25 mm²

length:

9 m

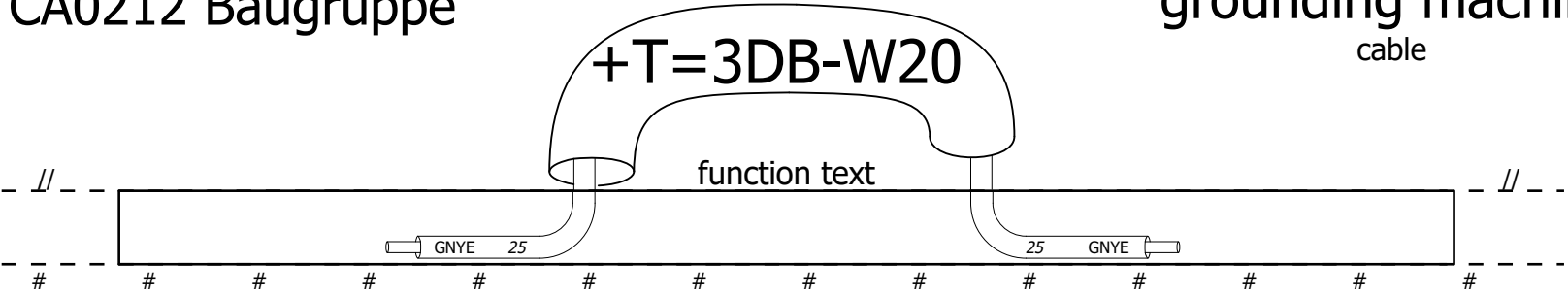


Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079 track 2	A B
Stück 1	cable accessory cable lug, M10, 25mm ²	556091 =	
Stück 1	cable accessory cable lug, M12, 25mm ²	G10001087 =	

CA0212 Baugruppe

grounding machine
cable

description: grounding, PE, ring, M10-M12, 25mm²
number of cores: 1
cross section 25 mm²
length: 9 m



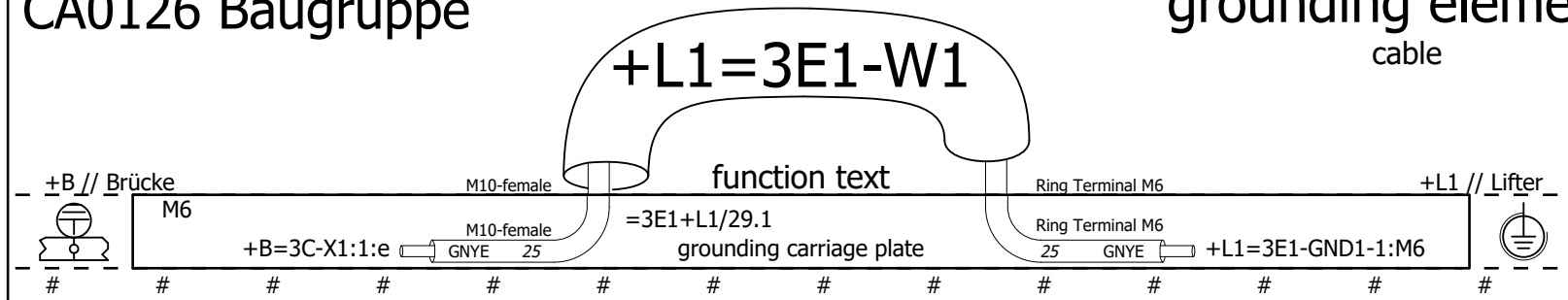
Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079 track 2	A B
Stück 1	cable accessory cable lug, M10, 25mm ²	556091 =	
Stück 1	cable accessory cable lug, M12, 25mm ²	G10001087 =	

CA0126 Baugruppe

grounding elements

cable

description:	grounding, PE, ring, M6-M10, 25mm ²
number of cores:	
cross section	25 mm ²
length:	8 m



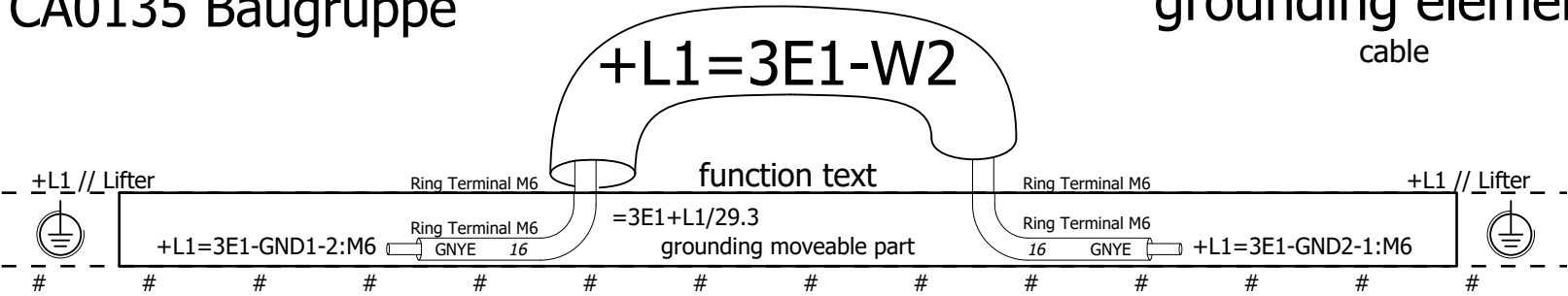
Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079 grounding carriage plate	A B
Stück 1	cable accessory cable lug, M6, 25mm²	556087 =	
Stück 1	cable accessory cable lug, M10, 25mm²	556091 =	

			Toolbox_105	Option:					module	Page 161
			Toolbox version	customer:	SEFAMAR			cable connection plan	sector 111 location +A =Z6B	
			automated	commission:	289001295	project: 289001295				
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	09.01.2025				

CA0135 Baugruppe

grounding elements
cable

description: Erdung, PE, Ring, M6-M6, 16mm²
number of cores: 1
cross section 16 mm²
length: 2,2 m

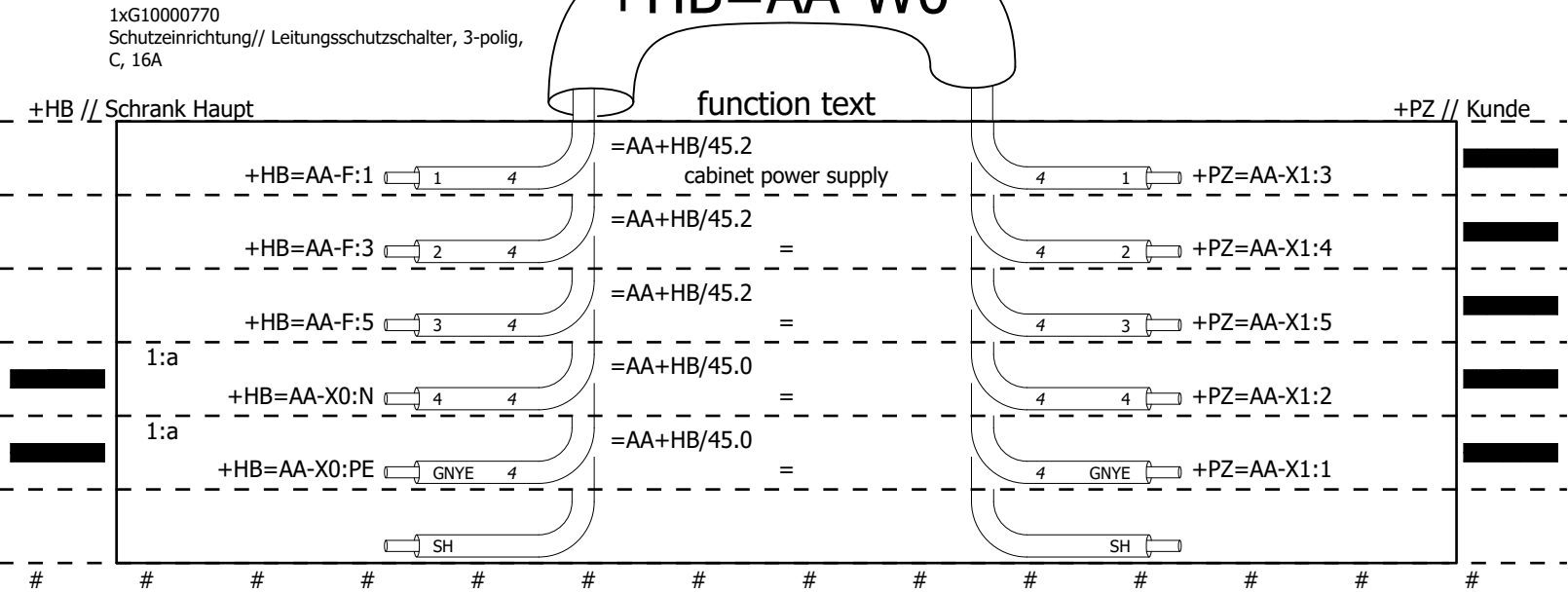


Quantity	Designation	Part number	
	cable sold by meter PVC, 1x16, PE	G10000669	
		grounding moveable part	
Stück 1	cable accessory cable lug, M6, 16mm²	556018	A B
Stück 1	cable accessory cable lug, M6, 16mm²	556018	

G10000911 Einzelteil

AC power supply cabinet
cable sold by meter

description: PVC, 5x 4, PE, CY
number of cores: 5
cross section: 4 mm²
length: 25 m



G10000797 Einzelteil

AC power supply cabinet

cable sold by meter

description:

PVC, 3x 2,5, PE, CY

number of cores:

3

cross section

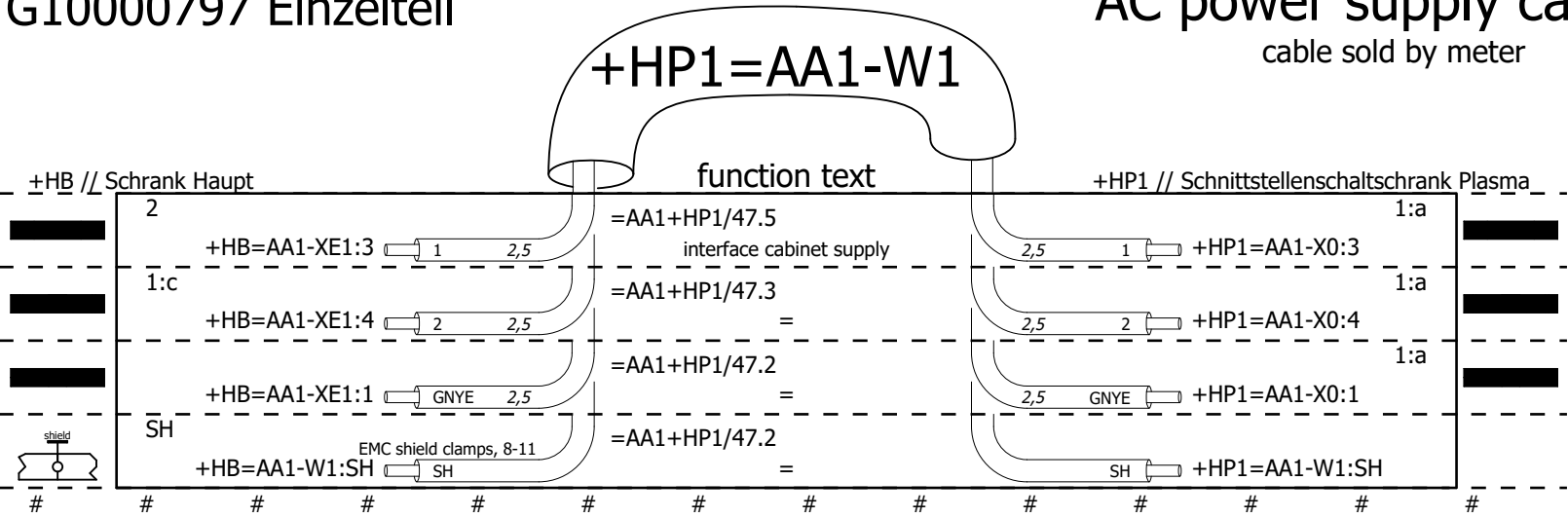
2,5 mm²

length:

34 m

Add-on: Kabel Zubehör: G10000423

Schirmklammer, Zugentlastung, Hutschienen, 8-11



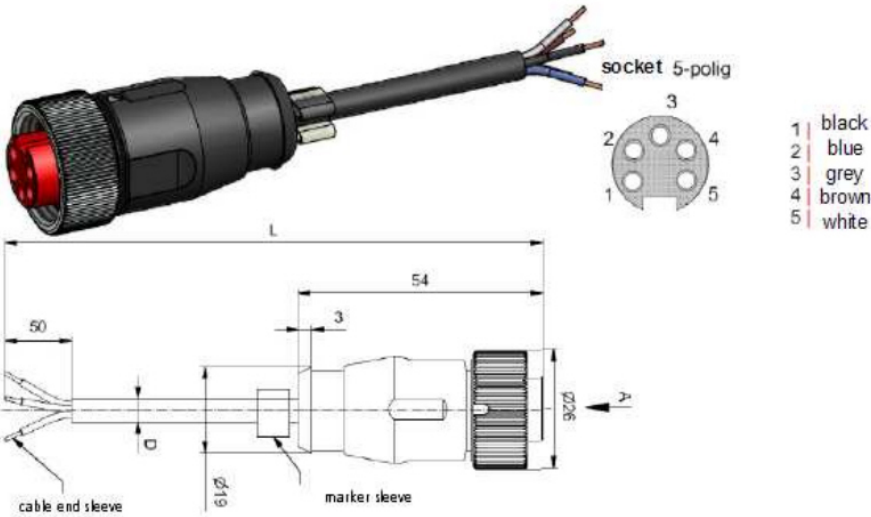
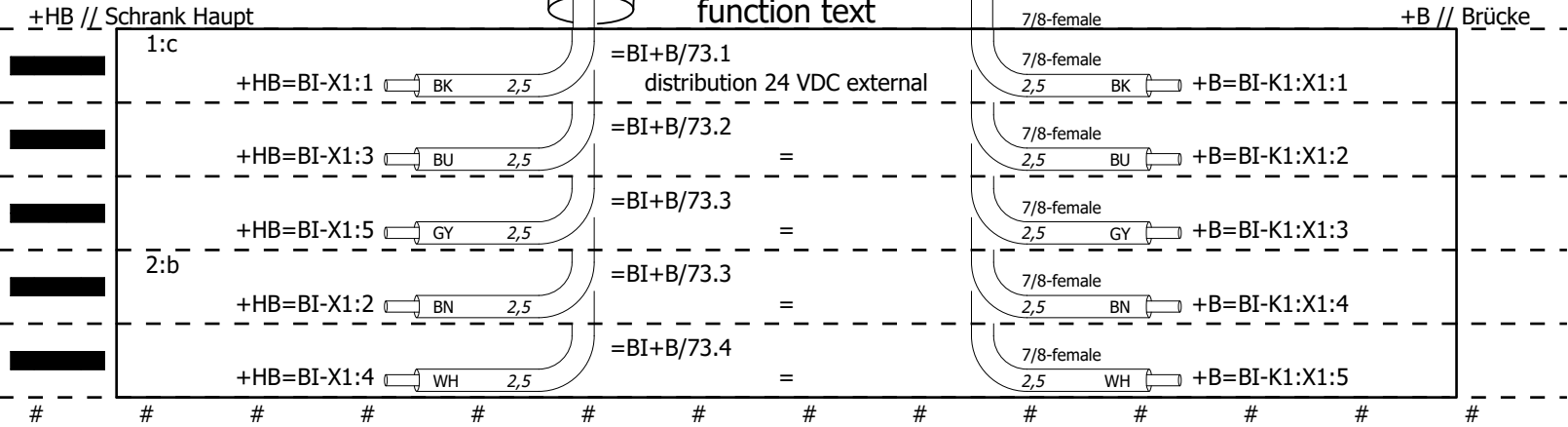
G10001295 Einzelteil

DC power supply interface

+B=BI-W1

1xG10000103
Verteiler Power// EtherCAT-Box, Powerverteilung,
M8, 4 x 4A

description: OE-7/8", socket, 5 x 2,5, 10m
number of cores: 5
cross section: 2,5 mm²
length: 10 m



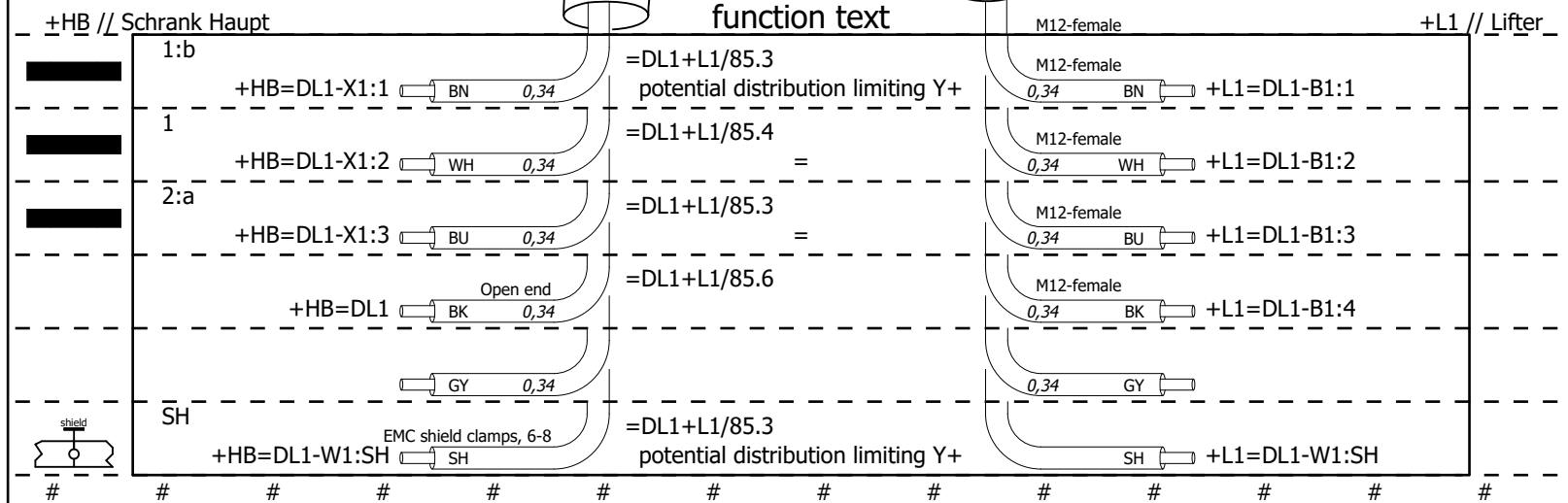
Power cable 7/8" | 7/8"-socket, straight, 5-pin, - open end,1,5 mm²

detection limiting

cable

Sensor// induktiv, Zylinder M18, M12, 1x NC

description:	OE-M12, 5-pole, socket, 15m
number of cores:	5
cross section	0,34 mm ²
length:	15 m

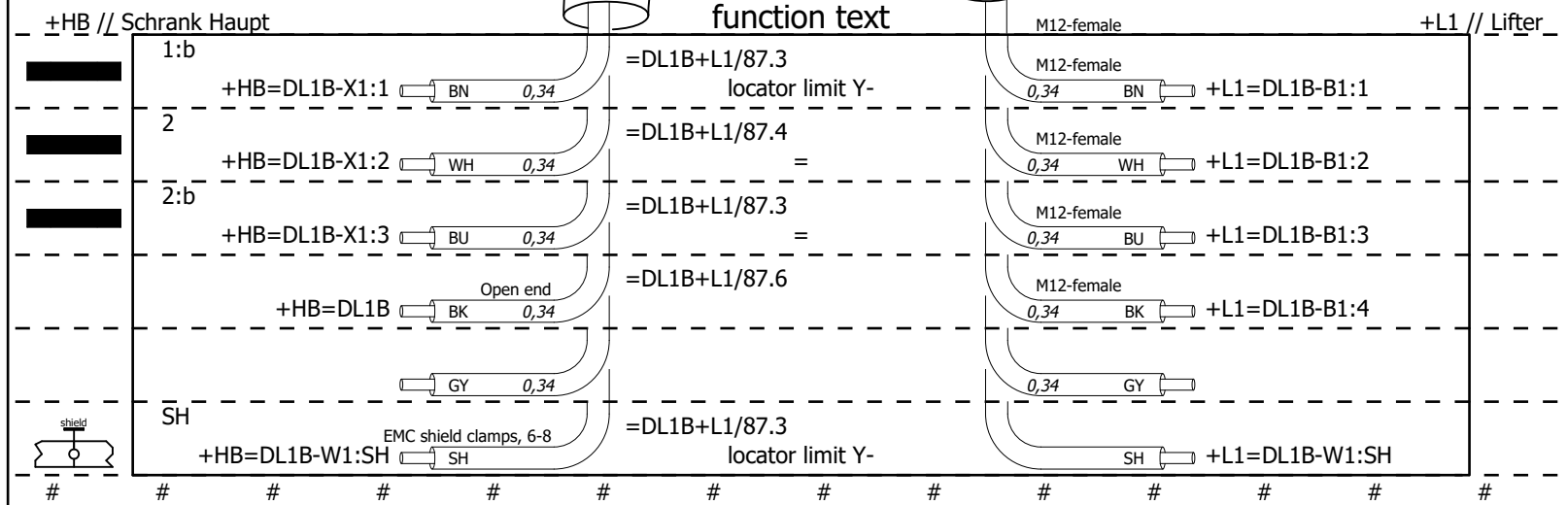


detection limiting

cable

Sensor// induktiv, Zylinder M18, M12, 1x NC

description: OE-M12, 5-pole, socket, 15m
number of cores: 5
cross section 0,34 mm²
length: 15 m



G10000957 Einzelteil

1xG10001013

Schalter Zubehör// Not-Aus, M12, 2x NC

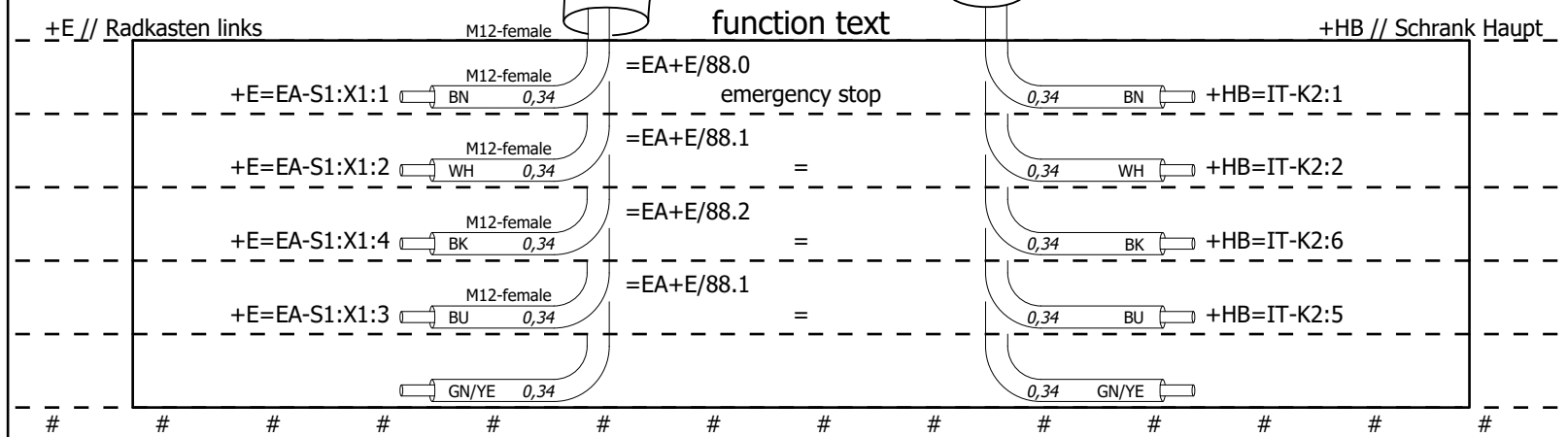
emergency stop additions

cable

1xG10001248

Busklemme// Safety, Eingang, 4 Kanal, 24 V DC

description:	OE-M12, 5-pole, socket, 10m
number of cores:	5
cross section	0,34 mm ²
length:	10 m



			Toolbox_105	Option: customer: commission:					module	cable connection plan	sector 111 location +A =Z6B	Page 168
			Toolbox version		SEFAMAR							
			automated		289001295	project: 289001295						
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	09.01.2025						

emergency stop additions

1xG10001248
Busklemme// Safety, Eingang, 4 Kanal, 24 V DC

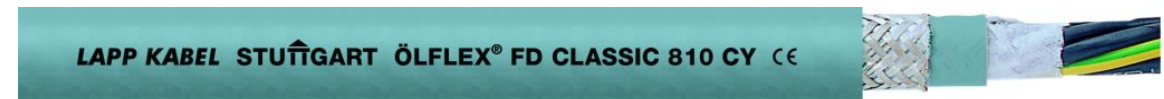
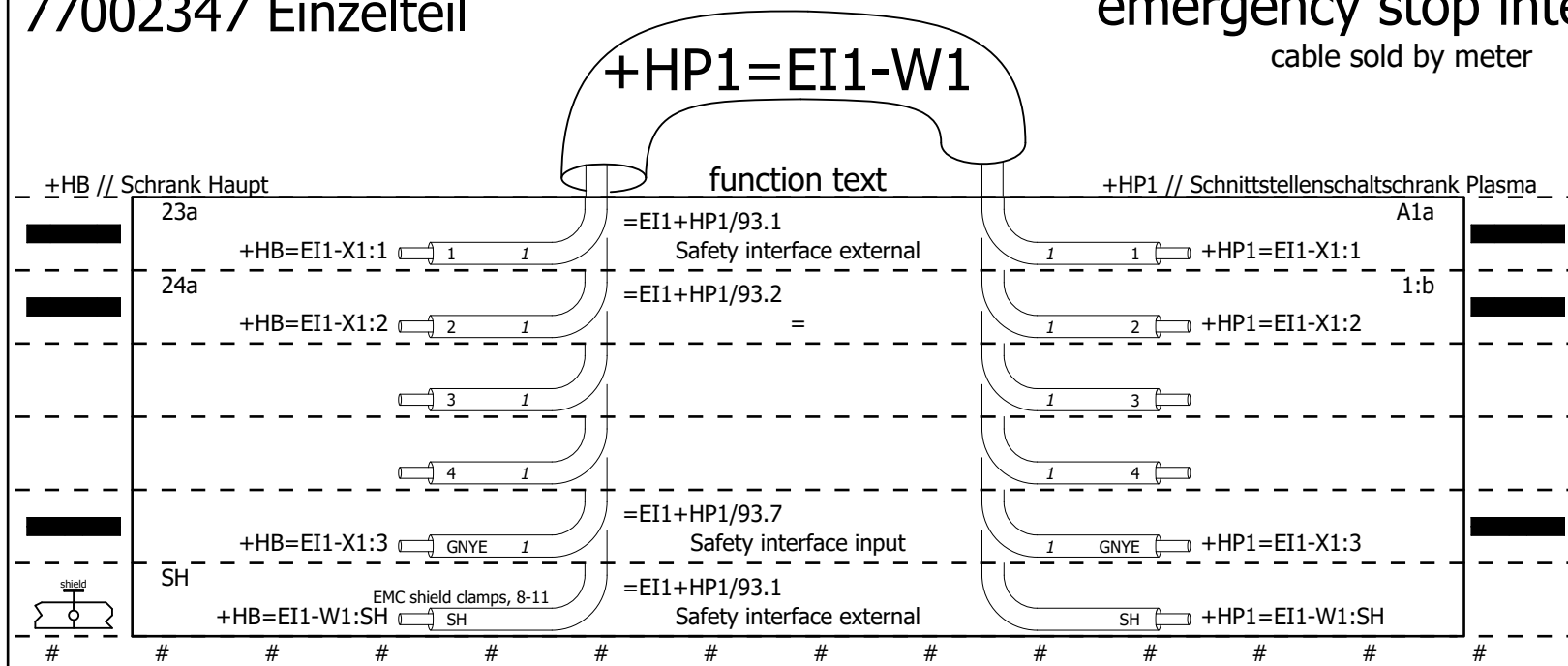
description:	OE-M12, 5-pole, socket, 10m
number of cores:	5
cross section	0,34 mm ²
length:	10 m



			Toolbox_105	Option: customer: commission:					module			Page 169	
			Toolbox version		SEFAMAR								
			automated		289001295	project: 289001295							
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	09.01.2025		cable connection plan			sector 111 location +A =Z6B		

emergency stop interface
cable sold by meter

description: PVC, 5x 1, PE, CY
number of cores: 5
cross section 1 mm²
length: 34 m



			Toolbox_105	Option: customer: commission:					module			Page 170	
			Toolbox version		SEFAMAR								
			automated		289001295	project: 289001295							
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	09.01.2025		cable connection plan	sector 111	location	+A	=Z6B	

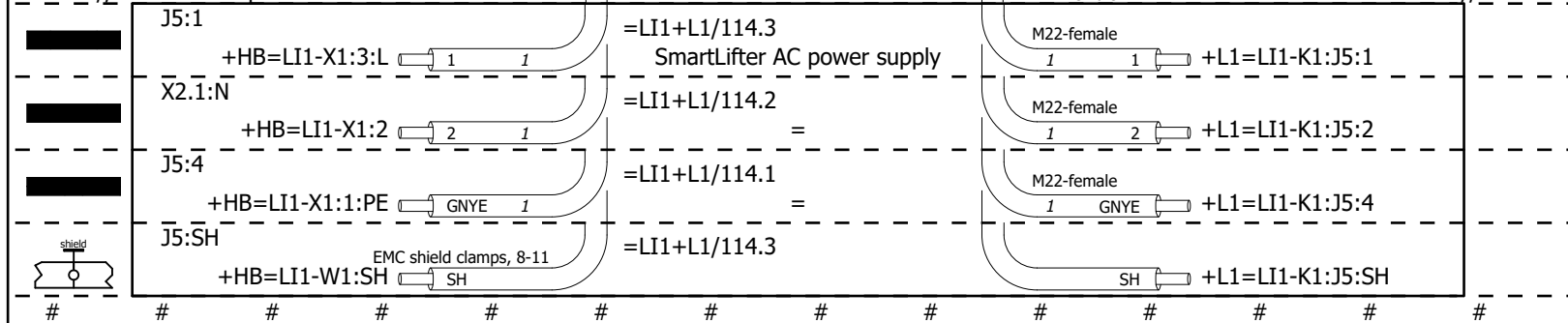
lifter

1x72163210

$$+L1=LI1-W1$$

M22-female

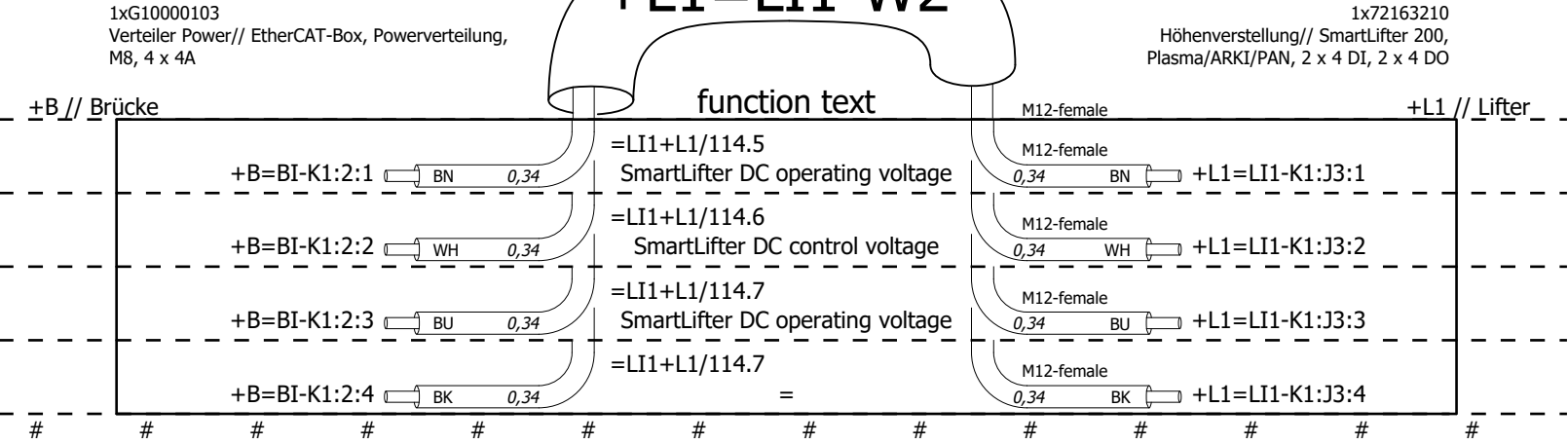
+HB // Schrank Haupt



Quantity	Designation	Part number	
	cable sold by meter PVC, 3x 1, PE, CY	0551042 SmartLifter AC power supply	A
Stück 1	plug M22, 4-pole, socket, 0°	77070191 =	

CA0168 Baugruppe

lifter

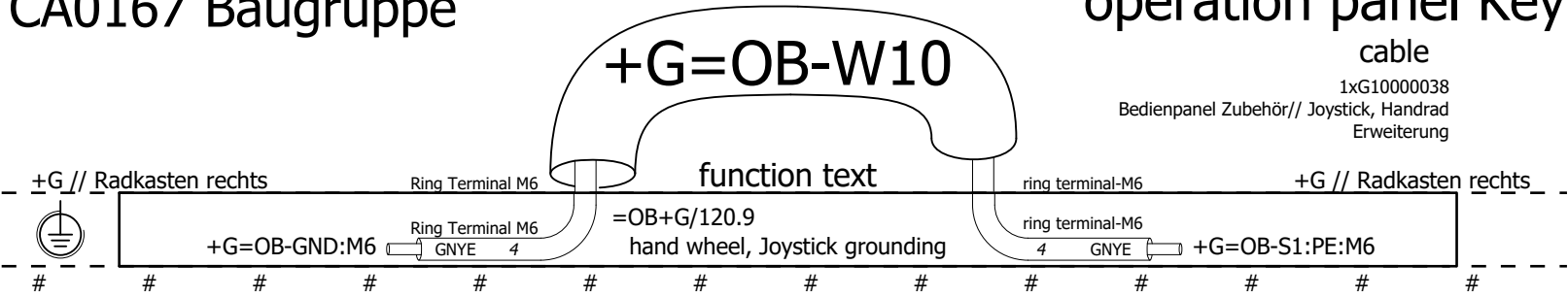


description: M8-M12, 5-core, <=15m
number of cores:
cross section 0,34 mm²
length: 10 m

Quantity	Designation	Part number	
Stück 1	cable power line, M8-M8, 15m	G10001293 SmartLifter DC operating voltage	B
Stück 1	plug M12, 4-pole, socket, 0°	77005053 =	

CA0167 Baugruppe

operation panel Keyboard



description:
number of cores:
cross section 4 mm²
length: 0,5 m

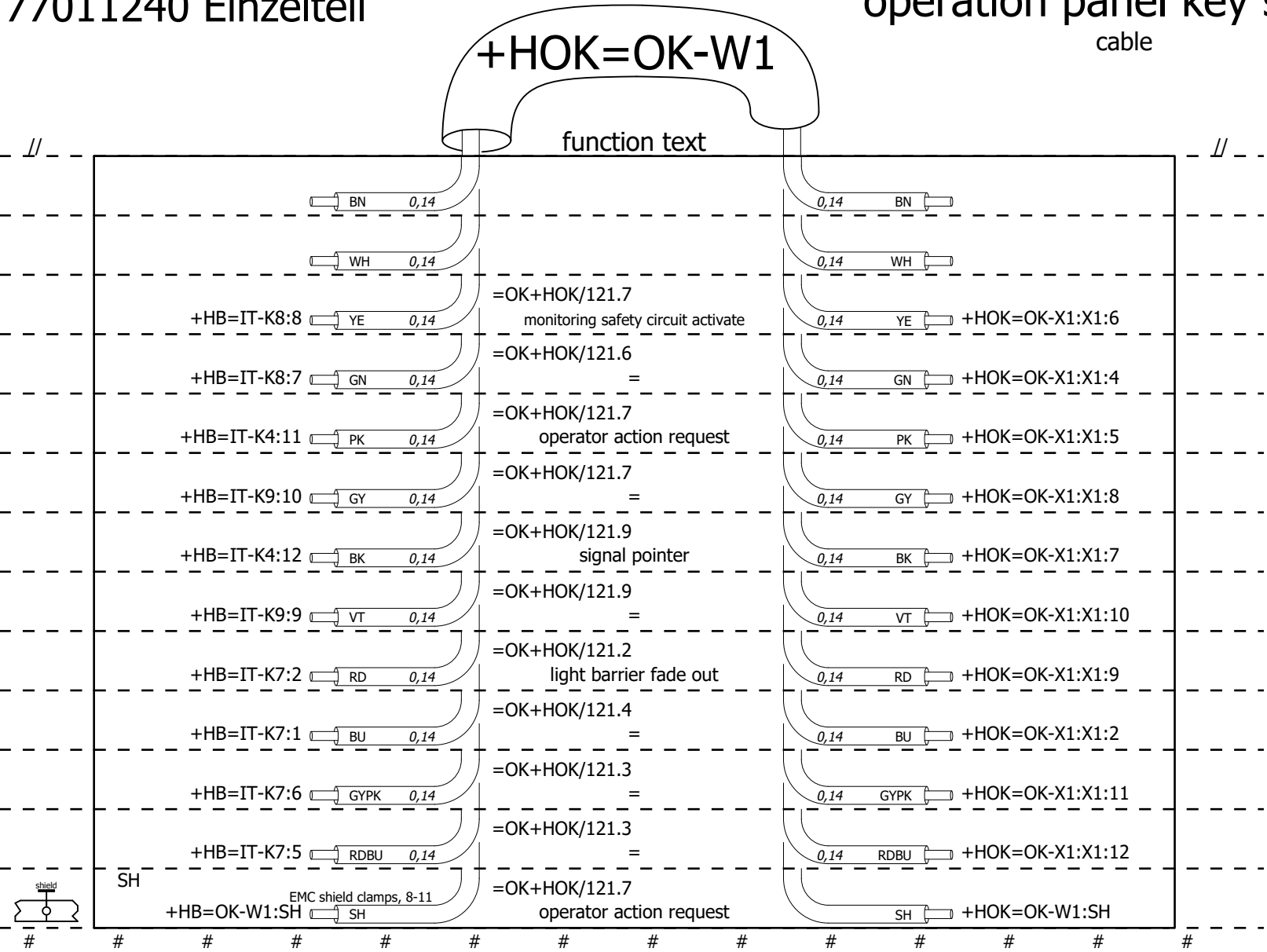
Quantity	Designation	Part number	
	cable sold by meter PVC, 1x 4	570130	A B
		hand wheel, Joystick grounding	
Stück 1	cable accessory cable lug, M6, 16mm ²	556018 =	
Stück 1	cable accessory cable lug, M6, 16mm ²	556018 =	

77011240 Einzelteil

operation panel key switch

cable

description: OE-M12, 12-pole, socket, 10m
number of cores: 12
cross section 0,14 mm²
length: 10 m
Add-on: Kabel Zubehör: 77003783



G10001241 Einzelteil

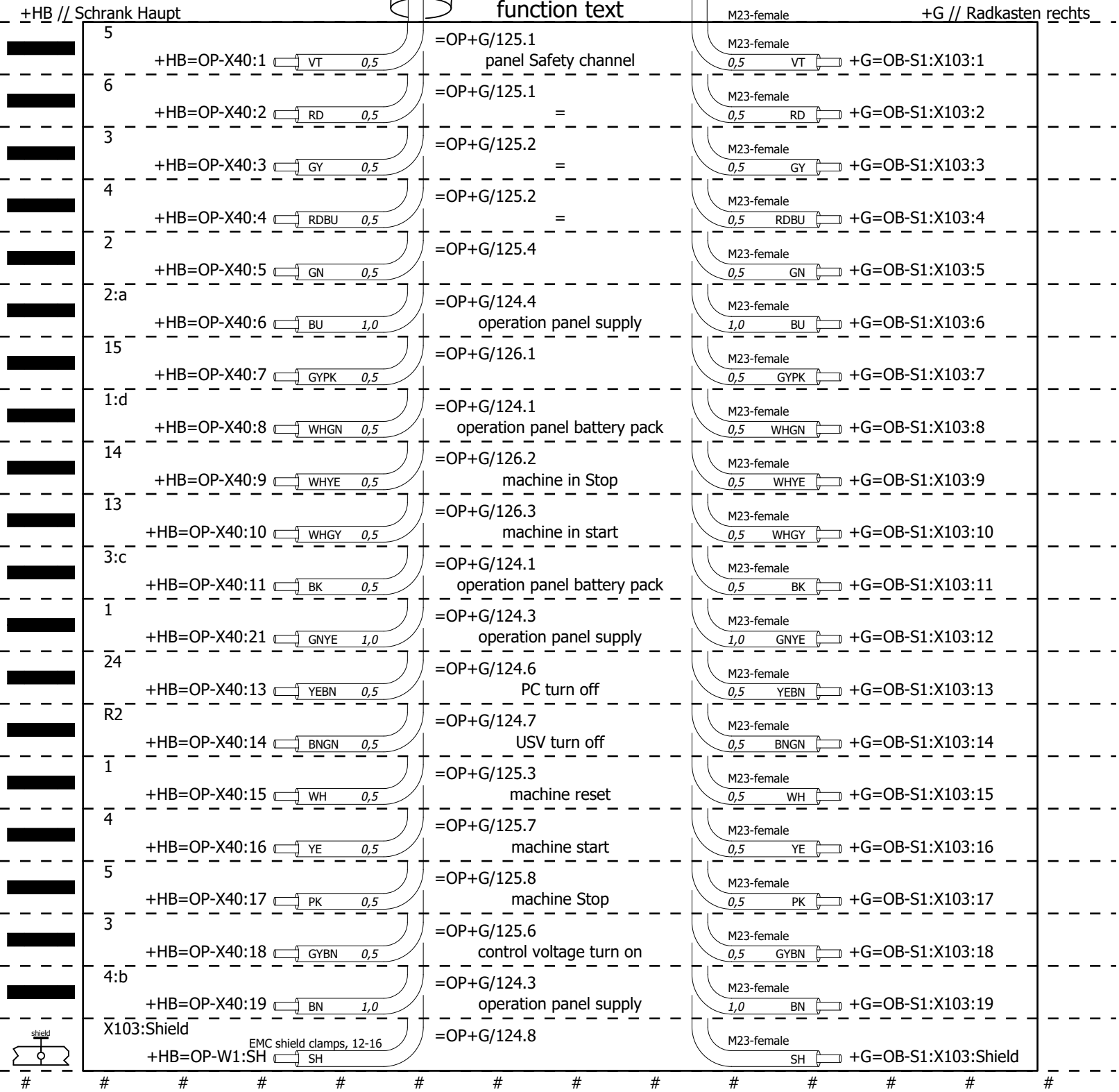
operation panel CP7202

description: OE-M23, 19-pole, socket (plug female), 10m
number of cores: 16+3
cross section: 0,5+1,0 mm²
length: 10 m

+G=OP-W1

cable

1xG10000038
Bedienpanel Zubehör// Joystick, Handrad
Erweiterung



CA0006 Baugruppe

operation panel CP7202

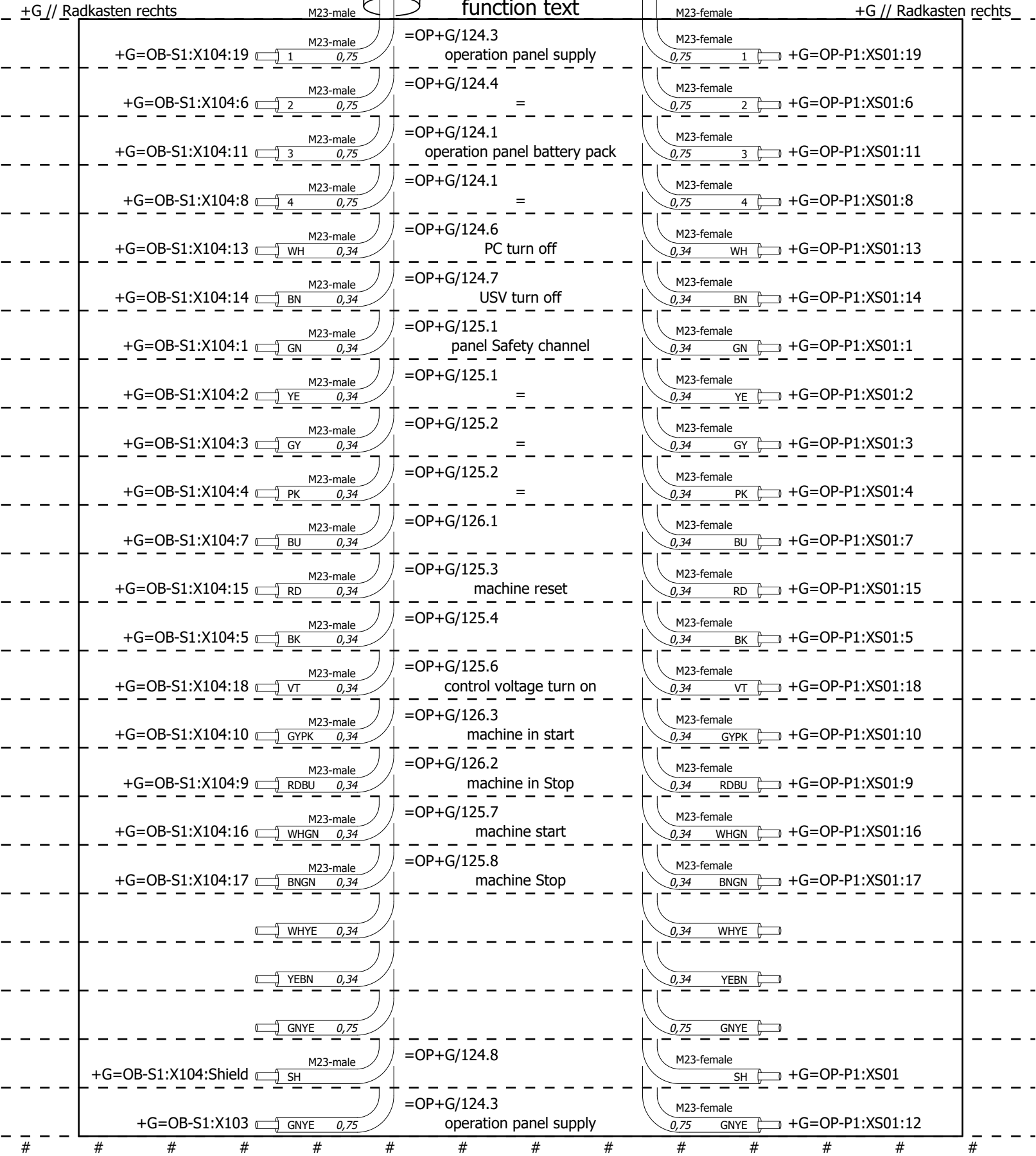
1xG10000038
Bedienpanel Zubehör// Joystick, Handrad
Erweiterung

1xG37000289
PC Bedienpanel// Display 15", GCS Export, SSD,
Touchdisplay, Win10

+G=OP-W2

cable

description: M23-M23, 16+5-core
number of cores:
cross section 1,09 mm²
length: 0,6 m

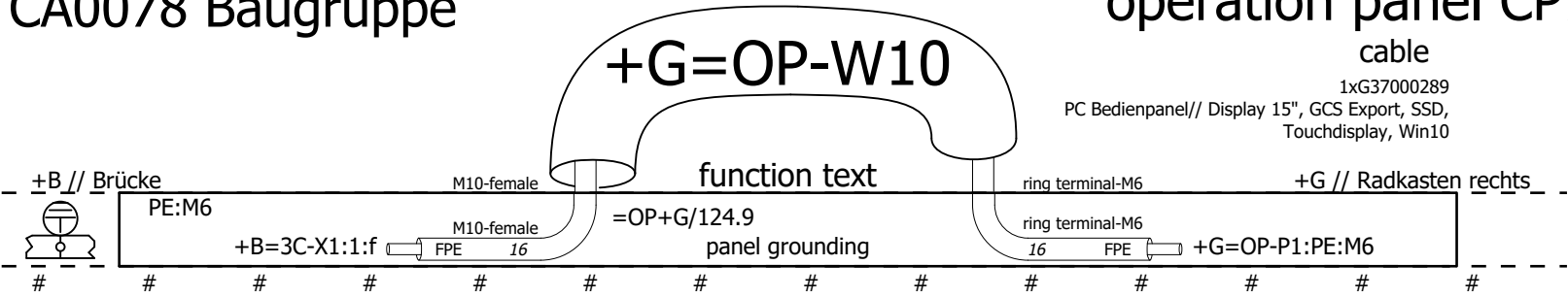


Quantity	Designation	Part number	
	cable sold by meter PVC, 16x0,34+5G0,75, PE, CY	77004655 operation panel battery pack	
Stück 1	plug accessory M23, 16+3-pole, plug, 0°	77002794 =	A
Stück 1	cable accessory cable gland, PG13,5, EMV	0546050 =	A
Stück 1	plug accessory M23, 16+3-pole, socket, 90°	77070966 =	B
Stück 1	cable accessory cable gland, PG13,5, EMV	0546050 =	B



CA0078 Baugruppe

operation panel CP7202



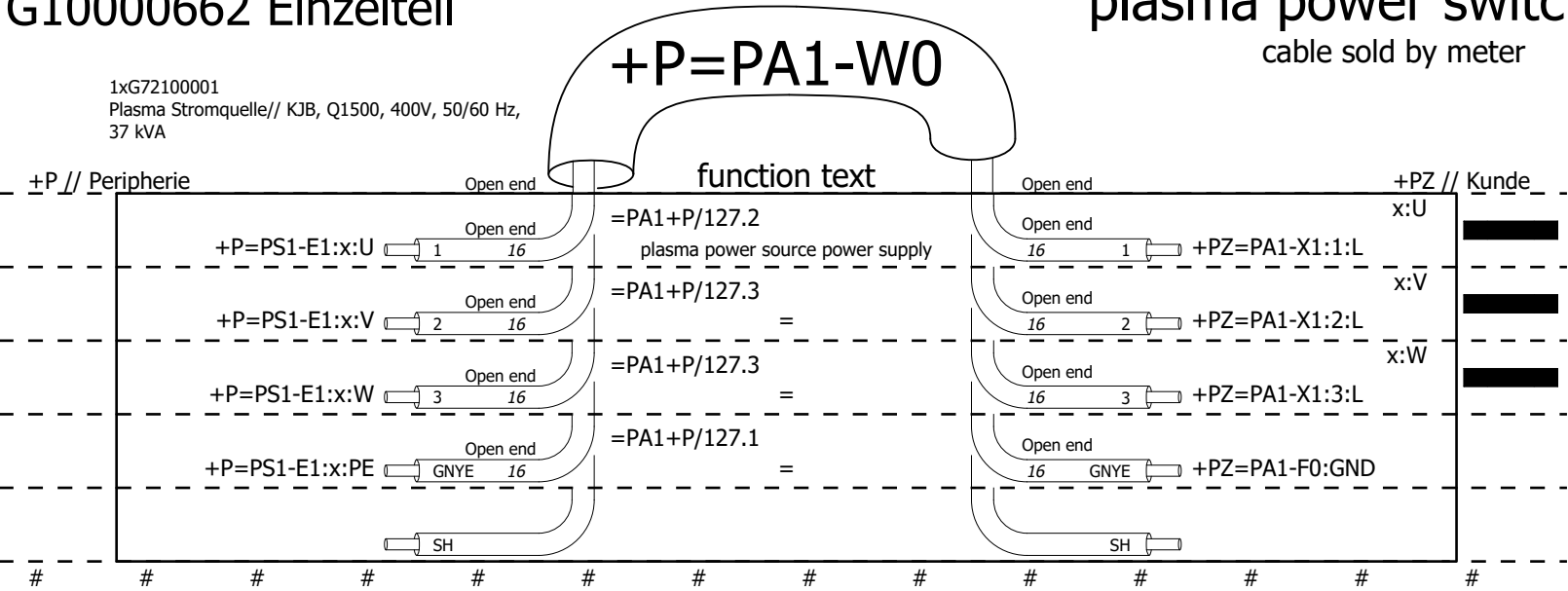
description: FPE, ring, OE-M10, 16mm²
number of cores:
cross section 16 mm²
length: 8 m

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130 panel grounding	
Stück 1	cable accessory cable lug, M10, 16mm ²	556027 =	A

G10000662 Einzelteil

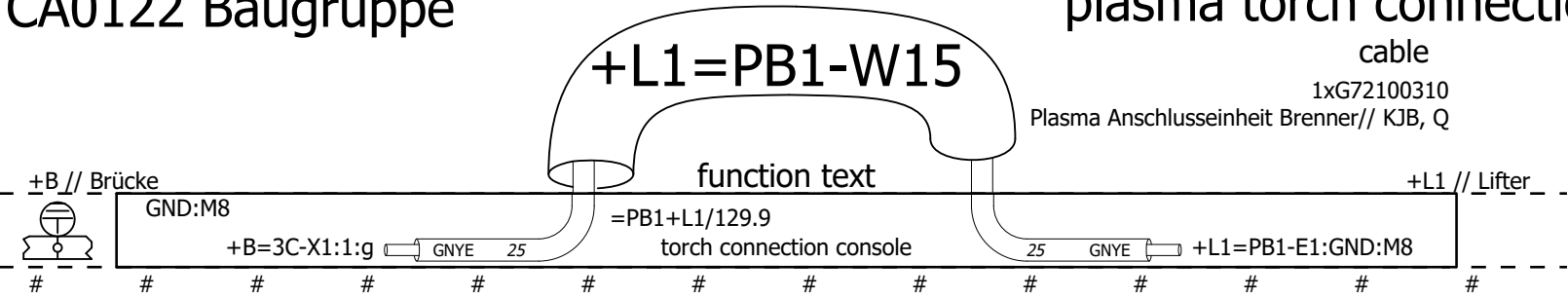
plasma power switch box
cable sold by meter

description: PVC, 4x16, PE, CY
number of cores: 4
cross section: 16 mm²
length: 6 m



CA0122 Baugruppe

plasma torch connection unit

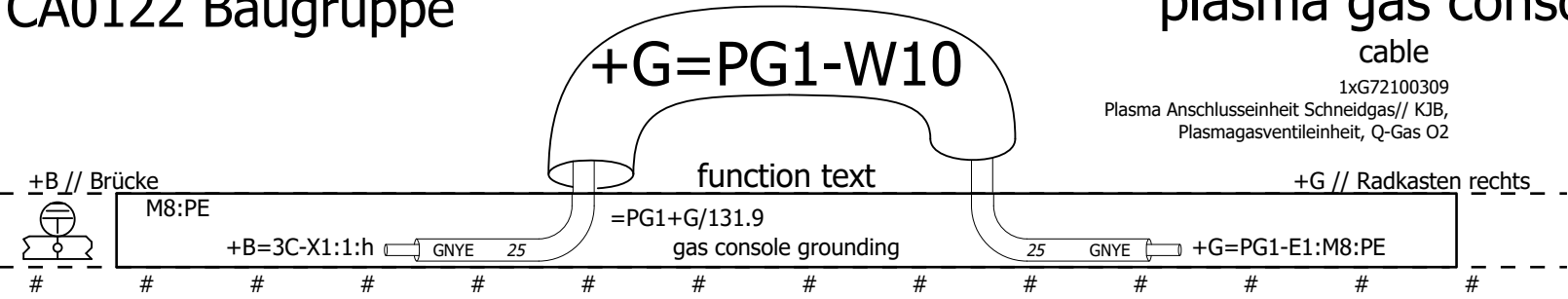


description: grounding, PE, ring, M8-M10, 25mm²
number of cores:
cross section 25 mm²
length: 8 m

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079	A B
		torch connection console grounding	
Stück 1	cable accessory cable lug, M8, 25mm ²	556077 =	
Stück 1	cable accessory cable lug, M10, 25mm ²	556091 =	

CA0122 Baugruppe

plasma gas console

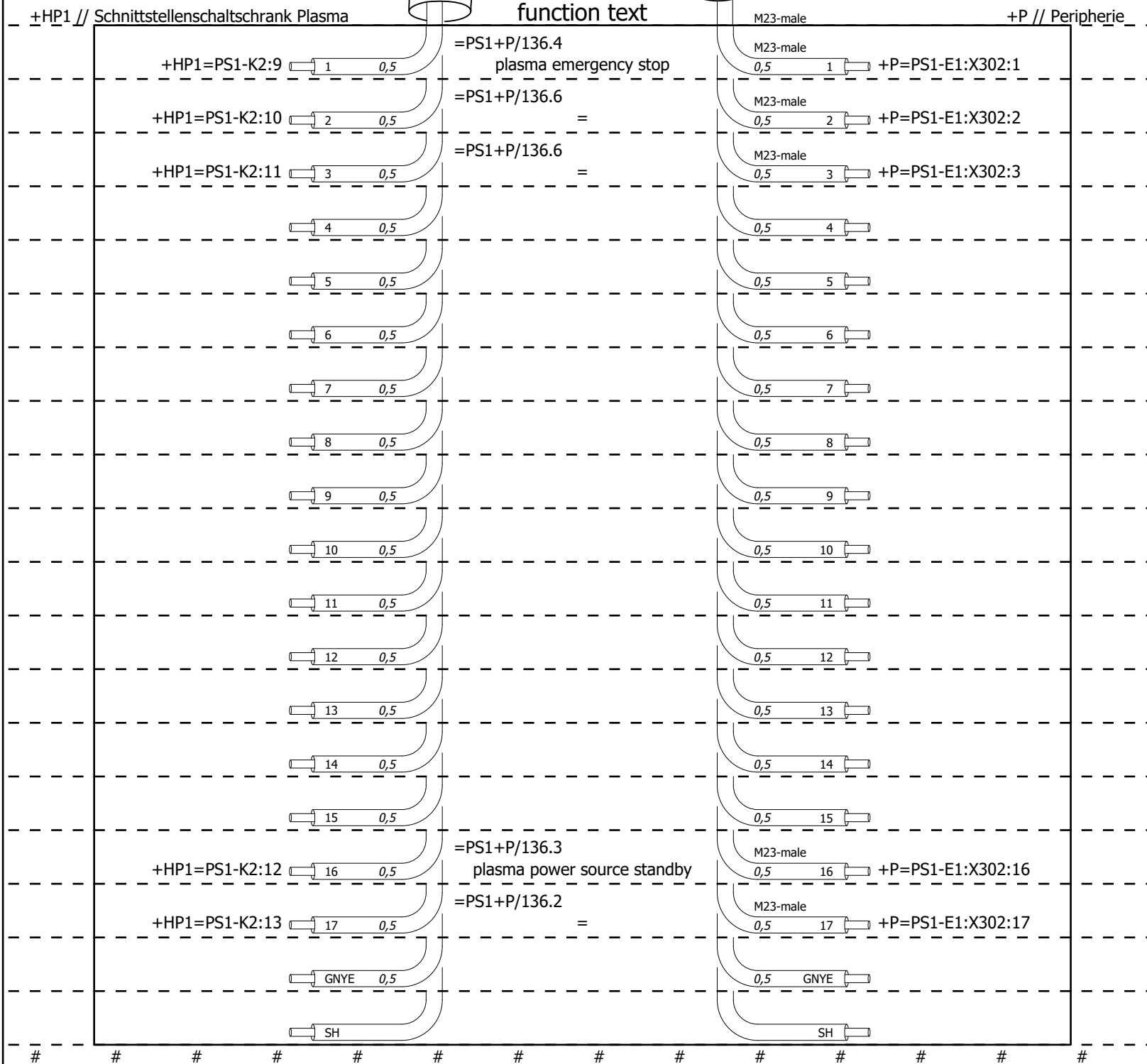


description: grounding, PE, ring, M8-M10, 25mm²
number of cores:
cross section 25 mm²
length: 7 m

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079	A B
		gas console grounding	
Stück 1	cable accessory cable lug, M8, 25mm ²	556077 =	
Stück 1	cable accessory cable lug, M10, 25mm ²	556091 =	

plasma source

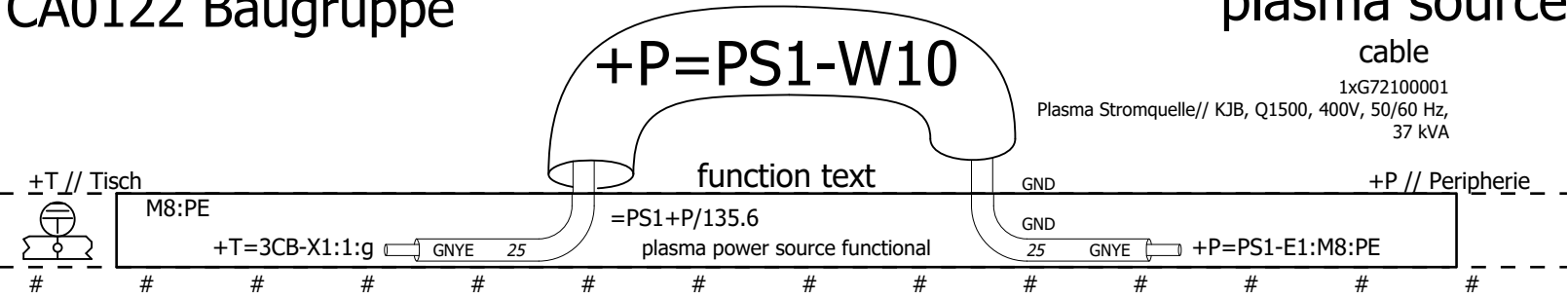
description: supply line M23 17pol 5m
number of cores: 17
cross section 0,5 mm²
length: 5 m



			Toolbox_105	Option: customer: commission:				module	Page 181	
			Toolbox version		SEFAMAR					
			automated		289001295		project: 289001295			
Modification	Date	Name	evaluation has been generated	Ed. by: Johann.Rais	09.01.2025			cable connection plan		sector 111 location +A =Z6B

CA0122 Baugruppe

plasma source

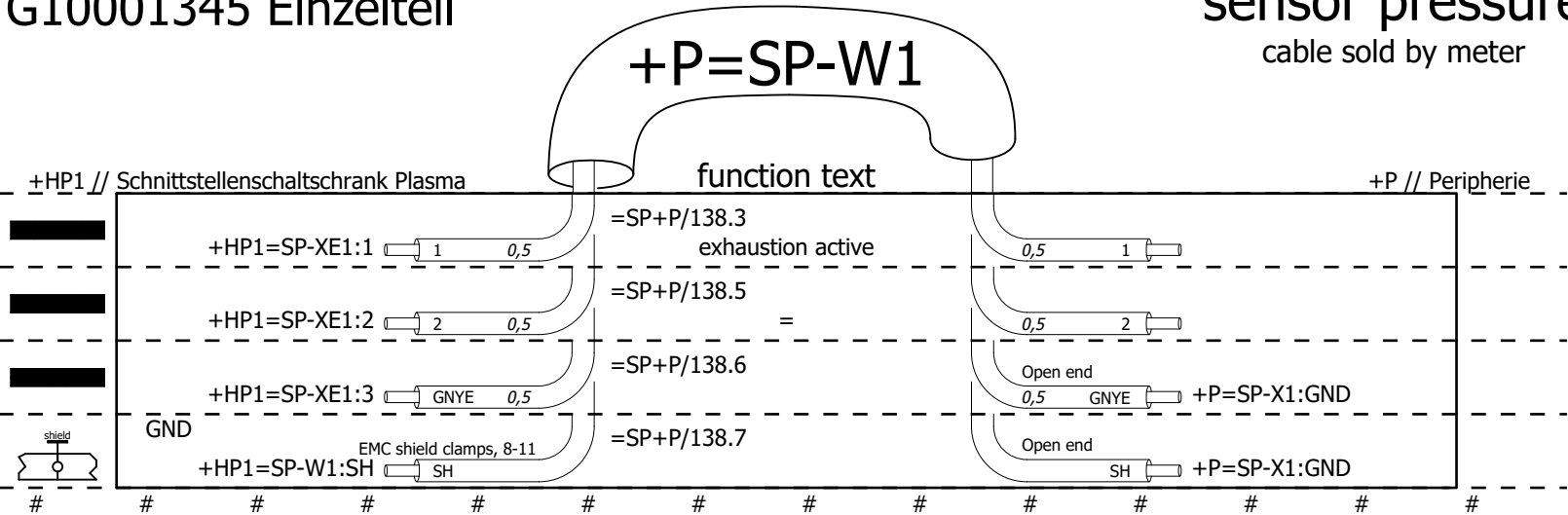


Quantity	Designation	Part number	
	cable sold by meter PVC, 1x25, PE	0551079	A B
		plasma power source functional ground	
Stück 1	cable accessory cable lug, M8, 25mm ²	556077	
		=	
Stück 1	cable accessory cable lug, M10, 25mm ²	556091	B
		=	

G10001345 Einzelteil

sensor pressure
cable sold by meter

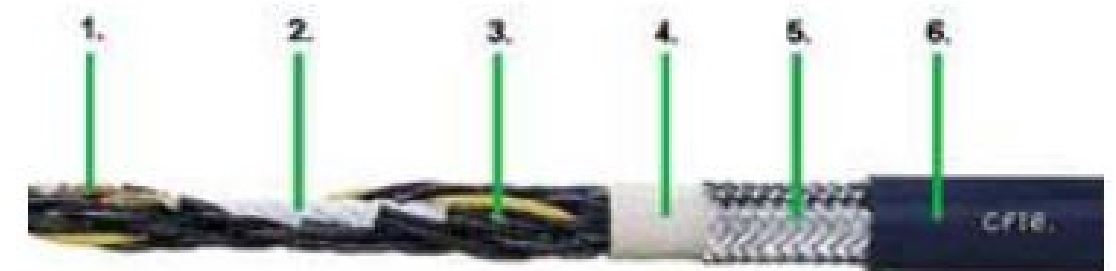
description: PVC, 3x 0,5, PE, CY
number of cores: 3
cross section 0,5 mm²
length: 6 m



#



unit exhaustion
cable sold by meter

$$+P=UE-W1$$


G10000957 Einzelteil

unit temperature

+G=UT-W1

cable

1x77004724

Stecker// M12, 4-polig, Flansch, Stift, 0°

description: OE-M12, 5-pole, socket, 10m
number of cores: 5
cross section: 0,34 mm²
length: 10 m

